

SuperMoTo

The Super Monitoring Tool

Monitoring Matrix, Measurement & Speaker Correction

User and Technical Manual

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Work in progress...

Contents

1	Overview	5
1.1	What is SuperMoto?	5
1.2	Why calibrate a monitoring system?	5
1.2.1	The room's low-frequency modes	5
1.2.2	Boundary and early reflections	6
1.2.3	Imperfect loudspeakers	6
1.2.4	Subwoofers and the crossover region	6
1.2.5	What SuperMoTo can and cannot do	6
1.2.6	Treat the room first	7
1.3	Signal flow at a glance	7
1.4	Plugin format and channel layout	7
2	Host (DAW) compatibility	9
2.1	Recommended and likely-compatible hosts	9
2.2	Hosts likely to be awkward	9
2.3	Suggested REAPER setup (reference)	9
2.4	What the host project saves	9
3	The monitoring matrix	12
3.1	Concepts	12
3.1.1	What a frame contains	13
3.1.2	What an output (loudspeaker) contains	13
3.1.3	Configurations A-F	14
3.2	Interacting with the matrix	14
3.3	Per-output FIR correction and latency compensation	14
3.4	The global spectrum analyzer	15
3.5	Host-automatable parameters	16
3.6	Compact (collapsed) view	16
4	The monitor target curve	17
4.1	Why a flat in-room response is the wrong target	17
4.2	Choosing how much tilt	17
4.3	Curve parameters	18
4.4	Usage	19
4.5	The curve bank	19
5	The Configuration tool	21
5.1	Ambisonics decoding	21

6	Measurement (Calibration view)	25
6.1	The three measurement modes	25
6.1.1	The raw loudspeaker (Dry)	26
6.1.2	Verifying the correction (FIR)	26
6.1.3	The complete monitored chain (System)	26
6.2	The Calibration pane, control by control	27
6.2.1	Microphone input and calibration	27
6.2.2	Measurement type and channel selection	27
6.2.3	The Sub switch	27
6.2.4	Stimulus, duration and level	27
6.2.5	The measurement folder	27
6.2.6	Comments	28
6.2.7	Running the measurement	28
6.2.8	The SPL meter	28
6.3	Microphone calibration	28
6.3.1	Why calibrate	28
6.3.2	The calibration file format	29
6.3.3	Where the calibration is used	29
6.4	What a measurement writes to disk	29
6.4.1	The capture files	30
6.4.2	The "sent" channel convention	30
6.4.3	The two manifests	30
6.4.4	What the manifest records	30
6.5	Recommendations	31
6.5.1	Choosing and orienting the microphone	31
6.5.2	How many positions, and how far apart	32
6.5.3	Choosing the stimulus	32
7	Analysis & correction design	34
7.1	Loading and estimating the response	34
7.1.1	Choosing the estimator	35
7.1.2	Microphone calibration	35
7.1.3	The Welch window	35
7.1.4	Delay removal and spatial averaging	36
7.1.5	The swept-sine estimator	36
7.2	The plot component	36
7.2.1	Level reference	37
7.2.2	Frequency response or impulse response	37
7.2.3	The raw impulse view, and the reverberation time	37
7.3	Fractional-octave smoothing	38
7.3.1	Frequency-dependent smoothing	38
7.3.2	Why smoothing matters, and a recommendation	38
7.4	Designing the correction	39
7.5	Subwoofer phase alignment	39
7.5.1	Setting it up	40
7.5.2	The Mains delay control	40
7.6	Phase type and latency	41
7.7	Exporting	41

8	Group analysis and multi-speaker alignment	43
8.1	Overview	43
8.2	Loading measurements	43
8.3	Loading a whole measurement folder in one step	44
8.3.1	Choosing the measurement runs	44
8.4	One set of correction settings for the whole group	45
8.5	Computing the alignment	45
8.6	Level matching between speakers	45
8.7	The subwoofer gets no correction filter	46
8.8	Applying and exporting	46
8.8.1	The export prefix	47
8.8.2	Figures in the report	47
8.9	Background processing	47
9	Building from source	48
9.1	License	48
A	Worked examples	49
A.1	Three things to know first	49
A.2	Example 1: a corrected stereo pair	50
A.3	Example 2: adding a subwoofer	51
A.4	Example 3: several speakers at once	52
A.5	Example 4: two monitor pairs, a subwoofer and headphones	52
A.6	Example 5: a 5.1 system switchable to 2.1, 5.0 and 2.0	53
A.7	Example 6: an Ambisonic rig with eight monitors and a subwoofer	54
B	Analysis and correction theory	56
B.1	Welch estimation	56
B.1.1	The H_1 estimator	56
B.1.2	Windowing, overlap and resolution	56
B.2	Delay removal and complex averaging	57
B.3	Synchronized swept-sine deconvolution	59
B.3.1	The synchronized sweep	59
B.3.2	Deconvolution by the analytic inverse	59
B.3.3	Why the harmonic orders separate	60
B.3.4	Extracting the linear and harmonic responses	60
B.4	Fractional-octave smoothing	60
B.5	Correction design	61
B.6	Subwoofer phase alignment	61
B.7	The complete correction filter	62
B.8	Limiting the phase to the crossover region	64
B.9	Rendering the FIR	65
B.9.1	Linear- and mixed-phase rendering	65
B.9.2	Minimum-phase rendering	66
C	Minimum phase, excess phase and invertibility	67
C.1	Physically	67
C.2	Mathematically	67
C.3	What it means for the correction	68

D	Spatial averaging and microphone positions	69
D.1	Spacing and the correlation length	69
D.2	How many positions are enough	70
D.3	Why a subwoofer argues for a tight cluster	70
E	Why the sweep runs to Nyquist	71
E.1	The inverse filter and its group delay	71
E.2	Where the ringing can come from	71
E.3	How the mechanisms are separated	72
E.4	Results	73
E.5	Why the sweep cannot be cut at a zero crossing	74
E.6	The band and fades SuperMoTo uses	75
E.7	What the figures mean for a real measurement	75
F	Delay estimation and subwoofer alignment	77
F.1	Two estimators	77
F.2	Why the two disagree	78
F.3	Which estimator for which problem	78
F.4	The alignment condition	79
F.5	Group alignment and the subwoofer trim	80
G	Ambisonics primer	81
G.1	Encoding the field and decoding to the rig	81
G.2	Spherical harmonics	81
G.3	Ambisonic order	81
G.4	The AmbiX convention (ACN and SN3D)	82
G.5	Decoding	82
G.6	The max- r_E weighting	82
H	Glossary	84

Chapter 1

Overview

This document, as well as the software it describes, are work in progress. The author is open to remarks and questions. Anything that can improve the tool and its documentation is welcome.

1.1 What is SuperMoto?

SuperMoTo is an audio plugin (VST3, AU and Standalone) for the monitoring section of a studio. It is the big brother of [MoTo](#). From a single instance it lets you:

- route up to 32 inputs to 32 outputs (8 by default) through a full routing matrix, with per-crosspoint gain, a 2-band EQ and phase;
- apply one [FIR](#) correction filter per output (zero-latency [WDL convolution](#)) to flatten each loudspeaker;
- measure the loudspeakers in the room with built-in stimuli;
- analyse those measurements and design the FIR correction curves (magnitude and phase), including phase-coherent subwoofer integration;
- align and correct a whole rig at once (several speakers plus a shared subwoofer) in the "Group analysis" mode.

The plugin is organised in four functional parts that share one engine: the **Monitoring matrix** (Chapter [3](#)), the **Measurement / Calibration** view (Chapter [6](#)), the **Analysis & correction design** view (Chapter [7](#)) and the **Group analysis** view (Chapter [8](#)).

1.2 Why calibrate a monitoring system?

What you hear at the listening position is never the loudspeaker alone: it is the loudspeaker convolved with the room, sampled by your ears (or a microphone) at one place. Several independent mechanisms each leave their mark on that combined response, and they behave very differently, which is exactly why some of them can be corrected electronically and some cannot.

1.2.1 The room's low-frequency modes

Below roughly the Schroeder frequency (typically 150 Hz to 300 Hz in a domestic-sized room), the room behaves as a small set of discrete resonances. Each room mode stores energy and releases it slowly, producing large peaks and dips in the magnitude response (often 10 dB or more) together with audible ringing (slow modal decay) in the time domain. Modes are strongly position dependent: a dip at one position can be a peak half a metre away.

1.2.2 Boundary and early reflections

Sound radiated towards nearby surfaces returns to the listener delayed and attenuated. Reflections off the desk, floor, ceiling and side walls combine with the direct sound to produce **comb filtering** (regularly spaced peaks and notches) across the midrange and treble, and the speaker-boundary interference (**SBIR**) cancellation places a characteristic dip in the upper bass whose frequency is set by the speaker-to-wall distance.

1.2.3 Imperfect loudspeakers

Even in an anechoic chamber no speaker is perfectly flat: drivers have their own magnitude ripple and roll-offs, which are essentially **minimum-phase**, and the **crossover** between drivers adds a phase rotation which is not. That excess phase is **all-pass** rather than a cancellation: it has unit magnitude, so unlike a reflection notch it still has a stable inverse, and a **linear-phase** filter can pre-compensate it at the cost of latency (Appendix C, Section 7.6). This part of the error is a property of the box and is the same at every listening position, which is what makes it the most cleanly correctable: one filter fixes it everywhere in the listening area, whereas a room correction is only valid where it was measured.

1.2.4 Subwoofers and the crossover region

Adding a subwoofer introduces a second source, in a different place, with its own roll-off and delay. Where the sub and the mains overlap their outputs add as complex pressures: unless they are in phase through the crossover the sum shows a dip or, worse, a deep cancellation [20]. Getting a smooth, coherent hand-off is as much a phase / timing problem as a level problem.

1.2.5 What SuperMoTo can and cannot do

Which errors a filter can undo depend on a single property of the response: whether it is minimum-phase, meaning its energy arrives as early as its magnitude allows, so that its inverse is itself causal and stable. A single loudspeaker resonance, a driver roll-off and an isolated low-frequency **room mode** are essentially minimum-phase, and are invertible. A pure propagation delay, a discrete reflection arriving after the direct sound, and the deep cancellation notch where two out-of-phase sources sum are not: they carry **excess phase**, lag beyond that minimum which no stable causal filter can remove. Those three are not equally hopeless, though: excess phase that is merely all-pass, such as a delay or a crossover's phase rotation, still has a stable inverse and can be pre-compensated by a linear-phase filter at the cost of latency, whereas a true cancellation null cannot be undone at all. Appendix C develops both pictures and what they imply for the correction filter.

Correction is a linear filter applied to the signal before the speaker. That makes it powerful for some of the problems above and powerless for others:

- **Can do well.** Flatten the loudspeaker's own magnitude and phase response; tame the broad *minimum-phase* part of the room's low-frequency response that is common to the listening area; apply a defensible target/house curve; and time- and phase-align a subwoofer with the mains so they sum coherently through the crossover.
- **Cannot do.** It cannot remove energy that arrives later than the direct sound: a reflection is a separate event in time, so "correcting" for it only pre-distorts the direct sound and the dip usually returns elsewhere. Deep cancellation notches are non-minimum-phase and effectively non-invertible: boosting into them wastes headroom and excites the mode further rather than filling the null [17, 16]. And because the measurement is taken around one reference area, a correction is only valid there: it cannot be flat everywhere at once [19].

This is why SuperMoTo, like other tools aiming at room correction, derives its correction from a **spatial average** of several microphone positions (so it follows the response that is consistent

across the listening area rather than chasing one position’s interference notches), limits its boost with a soft ceiling, and smooths the target more coarsely as frequency rises (Chapter 7).

1.2.6 Treat the room first

Electronic correction should not be considered as a substitute for acoustics. It is the last step of an audio system optimization. The errors it handles worst (reflections and modal ringing) are precisely the ones that respond to physical treatment, so the most effective gains come upstream, before any filter is designed:

- **Placement** of speakers and listening position relative to the walls (avoiding strong **SBIR** dips and the worst modal positions).
- **Symmetry** of the left/right setup, so the stereo image is not skewed by the room.
- **Bass trapping** in the corners to shorten modal decay. This reduces ringing, which no magnitude filter can.
- **Broadband absorption** at the first reflection points to reduce comb filtering.

A well-treated room makes the residual error smoother, more minimum-phase and more uniform across positions, which is exactly the kind of error a filter can fix. Room treatment and software correction are complementary, in that order.

1.3 Signal flow at a glance

Figure 1.1 shows the per-block processing performed by the matrix engine (see chapter 3). Audio enters on the discrete input channels, is summed through the active matrix configuration(s), passes the per-output trim and **FIR** correction, is time-aligned across outputs, and finally the master section (Level / Mute / Dim / Mono) is applied before the discrete outputs. The plugin’s bus is a fixed 32-in/32-out discrete layout (no host negotiation involved); the active matrix size (2 in / 8 out by default, up to 32 each) is a separate, independent setting made with the Inputs/Outputs selectors, and channels beyond it are simply left unrouted.

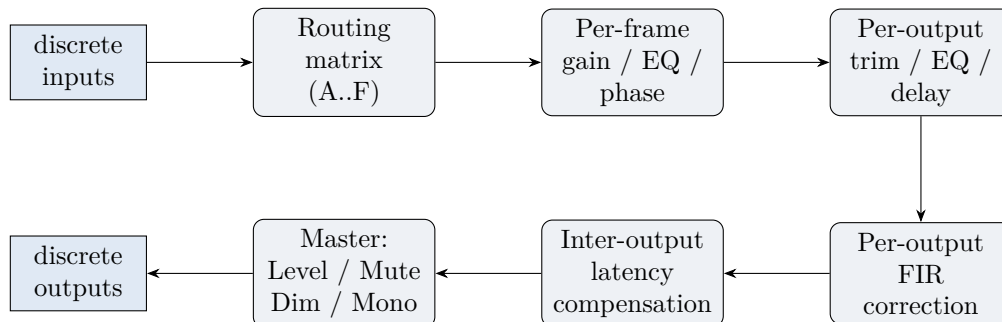


Figure 1.1: Per-block signal flow of the matrix engine. Each crosspoint (“frame”) of the matrix has its own gain, phase inversion and a small 2-band EQ; each output then has its own trim, a further 2-band EQ, a time-alignment delay and an optional FIR correction filter. Outputs are then re-aligned so loudspeakers with different FIR lengths stay time-coherent.

1.4 Plugin format and channel layout

- Formats: **VST3**, **AU**, **Standalone**.
- Bus layout: one input bus and one output bus, each 32 discrete channels. The plugin only reports support for the exact 32-in / 32-out layout: no other channel count is negotiated.
- Plugin type: audio effect (not a synth, no MIDI).

The bus is fixed rather than negotiated on purpose: JUCE’s per-host discrete channel-count discovery is inconsistent (the AU wrapper only probes up to 16 channels; VST3’s channel-set/**SpeakerArrangement** mapping has no defined bit for more than one unnamed discrete channel), which left some hosts stuck offering only 8 channels when the layout was negotiable. A single fixed 32-channel bus sidesteps that entirely: unused channels are left disconnected in the host, and the active matrix size (independent of the bus, Section 3.1.2) is what actually limits how many are routed. This fixed 32×32 discrete layout is the key constraint for host compatibility (see Chapter 2).

Chapter 2

Host (DAW) compatibility

*SuperMoTo is primarily designed and tested in **REAPER**. The informations given here were not known by the author and were gathered with the help of an AI assistant.*

To host it usefully a DAW must satisfy two requirements:

1. **Flexible channel routing**: the ability to feed the plugin's discrete inputs from arbitrary sources and to send its discrete outputs to arbitrary hardware outputs / busses.
2. **A monitoring (control-room) insert point**: a place to put the plugin on the path between the mix and the physical speaker outputs, after the program material is mixed but before it leaves for the monitors, ideally not printed into renders/bounces.

2.1 Recommended and likely-compatible hosts

2.2 Hosts likely to be awkward

- **Pro Tools**: plugin channel formats are constrained to named surround layouts; an arbitrary high-count discrete insert is not the native model.
- **Logic Pro**: surround formats are fixed and discrete 32-channel routing through a monitor insert is not straightforward.
- **Studio One / FL Studio / Live**: usable as generic effect hosts, but they lack a true control-room monitor insert, so the plugin would have to live on the master bus and be manually bypassed for exports.

2.3 Suggested REAPER setup (reference)

1. Open View → Monitoring FX and insert SuperMoTo there so it processes the hardware output path and is excluded from renders.
2. In the plugin's pin connector / channel mapping, map your mix busses to inputs 1..N and the plugin outputs 1..M to the physical speaker outputs.
3. Build your routing in the matrix (or with the Configuration tool) and store rigs in configurations A..F.

2.4 What the host project saves

The plugin state, saved with the host project and in presets, holds what SuperMoTo needs to monitor. That is the six matrix configurations, the matrix size, each output's processing (trim, polarity, EQ, delay and **FIR** correction, the impulses included) and the host parameters.

The work in progress in the other views lasts only as long as the plugin instance. That covers the measurements loaded in Analysis and Group analysis with their settings and results, the

Host	Fit	Notes
REAPER	Primary	Per-track/track-channel routing up to 64 channels, and a dedicated <i>Monitoring FX</i> chain (View → Monitoring FX) that sits on the hardware output path and is excluded from renders. The reference environment.
Ardour	Strong	Arbitrary bus channel counts and flexible port routing; a dedicated <i>Monitor Section</i> on the master/monitor path. A natural fit on Linux.
Steinberg Nuendo / Cubase	Strong	The <i>Control Room</i> provides a monitor insert chain with flexible downmix and multi-channel busses, ideal for a post-production monitoring matrix. Confirm the Control Room insert accepts a 32-channel plugin layout.
Bitwig Studio	Possible	Very flexible routing and per-device chains; place the plugin on the master or on a hardware-output chain. No formal “monitor section”, so the monitoring insert must be built manually and bypassed before export.
Standalone (built in)	Always	The Standalone build talks straight to the audio device (up to 32 in / 32 out if the interface provides them); useful for a dedicated monitor-controller machine with no DAW.

Table 2.1: Hosts that can drive SuperMoTo. "Fit" is the author's assessment based on the routing / monitor-section requirements; only REAPER is formally tested.

Calibration and Config tool controls, and the editor's page, window size and matrix selection. Closing and reopening the editor loses none of it, but reopening the project (or removing and adding the plugin again) starts these views afresh. What an analysis has written onto the outputs is part of the plugin state and comes back with the project, and the measurement folders, exported impulses and reports stay on disk to be loaded again. The microphone and SPL calibrations are machine-wide settings, remembered across projects (Section 6.3).

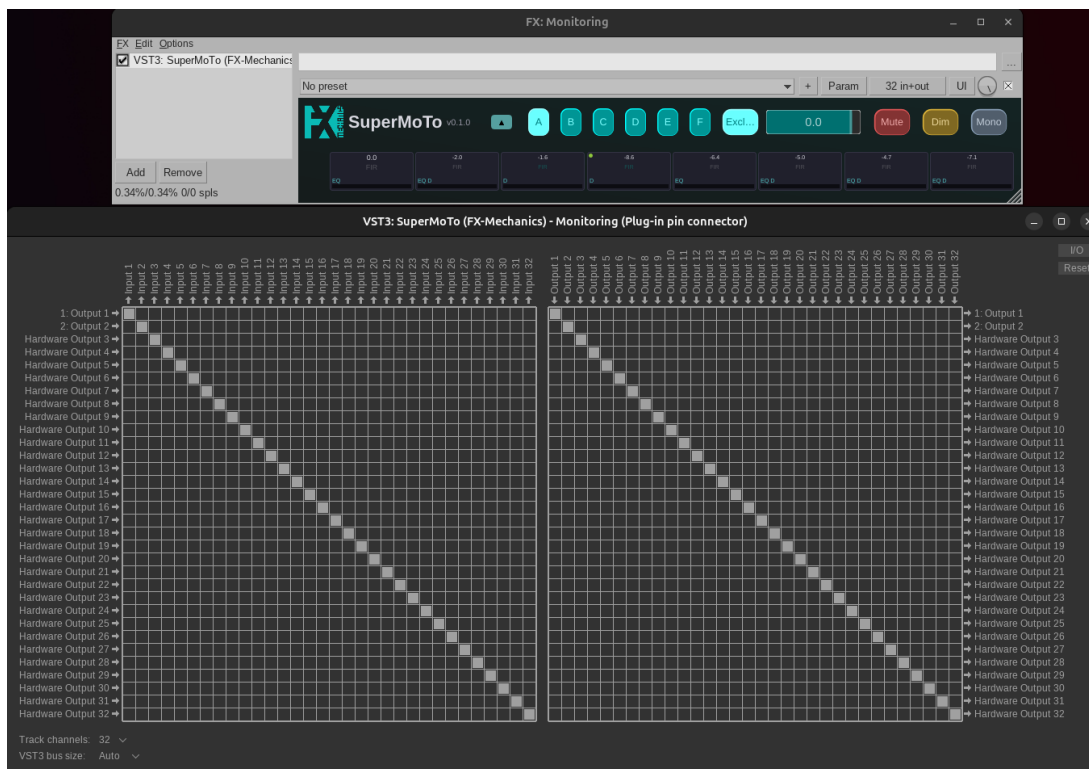


Figure 2.1: REAPER “Monitoring FX” window with SuperMoTo inserted, showing the fixed 32-in/32-out pin/channel mapping to the hardware outputs.

Chapter 3

The monitoring matrix



Figure 3.1: The Matrix view: top bar with configuration buttons A–F, Exclusive toggle, master Level and Mute/Dim/Mono; the matrix grid of frames (here in a 2nd order ambisonic configuration 10×14, resizable up to 32×32) each with its vu-meter; the per-output strip; the global spectrum analyzer; and the frame editor panel (bottom-right).

3.1 Concepts

The matrix is a grid of up to 32×32. The cell at row "in", column "out" is a frame (crosspoint): mainly a routing cell (how much of input "in" is sent to output "out"), but it also carries a small EQ of its own for a config-specific tonal tweak of crossover tuning on just that one route. Everything that describes the physical speaker (trim, polarity, its own EQ, delay and the FIR correction) lives on the output instead, since output settings are global and therefore shared by all configurations (A–F) that route into it.

A fresh instance starts at 2 in / 8 out, set with the Inputs and Outputs selectors in the bottom bar and saved with the session. Two inputs is the stereo mixing case, one stereo source feeding several monitoring destinations, which is what the outputs are for. Raise the input count

when the source itself is multichannel:

- **Surround.** 6 inputs for 5.1, 8 for 7.1, so every bed channel reaches its own speaker instead of being folded down.
- **Immersive / Dolby Atmos.** 10 inputs for a 7.1.2 layout and 12 for 7.1.4, counting the height layer. SuperMoTo monitors the rendered bed; it is not a renderer itself.
- **Ambisonics**, where the count depends on which side of the decode SuperMoTo sits.
 - **SuperMoTo decodes.** Feed it the B-format signal in the **AmbiX** convention, that is ACN channel order with SN3D normalisation, and set the input count to $(N + 1)^2$ for order N : 4 for first order, 9 for second, 16 for third. The Config tool writes the decode matrix itself, either its own max- r_E sampling decoder or an imported IEM AllRAD design (Section 5.1).
 - **Another decoder decodes.** If an upstream plugin already produces the speaker feeds for your exact rig, set the input count to the number of speakers instead and leave the matrix as plain diagonal routing. SuperMoTo then contributes the per-output correction, delay and level, and the A-F switching, without touching the decode.

The Config tool (Chapter 5) sets both counts for you: applying a rig raises them to whatever that layout needs, and never lowers them. Channels past the active size still exist on the plugin's fixed 32-channel bus, they are simply not routed (Section 1.4).

3.1.1 What a frame contains

- **Active** flag (is this crosspoint routed at all);
- **Gain** (overall frame level, dB);
- **Phase invert** (a frequency-independent polarity flip, useful for e.g. a differential decode coefficient);
- a **2-band EQ**: same band types and controls as the output's EQ below, applied only to this one route (before it sums into the output); useful for a config-specific voicing tweak without touching the shared output settings;
- a **vu-meter** and an **analyzer** checkbox to show the frame's signal on the spectrum analyzer.

3.1.2 What an output (loudspeaker) contains

Each output is the chain applied to the summed signal of all active frames feeding it. Click an output cell (top strip) to edit it:

- a **Description**: what this output drives, as you name it — "Genelec 8030 Left", "Surround Right". It is part of the preset, so it travels with the configuration it describes. Hovering the output in the matrix shows "Output 3: Genelec 8030 Left" above whatever the output has to say about its delay, the numbered channel squares in Measurement & calibration name themselves the same way, and a measurement run writes the name into the folder's manifest and readme (Section 6.4.1), so a set of captures says which loudspeaker it came from;
- **Trim** (output level, dB);
- **Phase invert**, the same flip as a frame's but applied to everything the output plays;
- a **2-band EQ**: each band can be lowpass, highpass, bandpass or peaking; lowpass/highpass are 2nd or 4th order; bands are cascaded in series. This is where the bass-management **crossover** lives (a highpass on a main, a lowpass on the sub);
- a **delay**, 0 ms to 100 ms, applied to the nearest whole sample (23 μ s, or 8 mm of path, at 44.1 kHz), for **time-aligning** the speaker;
- one **FIR correction** filter (Section 7.6), loaded from a wav **impulse response**;
- a **vu-meter** and an **analyzer** checkbox.

Frame or output polarity. The two switches multiply, so an inverted frame feeding an inverted output plays in normal polarity. A frame's switch belongs to one route of one configuration, which is where a decoder puts its negative coefficients. An output's switch belongs to the speaker as wired, whatever feeds it, and is the one to use for a subwoofer that sums better inverted against the mains. With an Ambisonics decoder in the frames, this keeps the subwoofer balance out of the decode (Section 5.1). Group analysis sets the subwoofer output's switch from "Invert sub" (Section 8.7). Measurements in "Dry" mode bypass the output's switch, while "FIR" and "System" include it (Section 6.1). Either switch fades through zero over 50 ms rather than clicking.

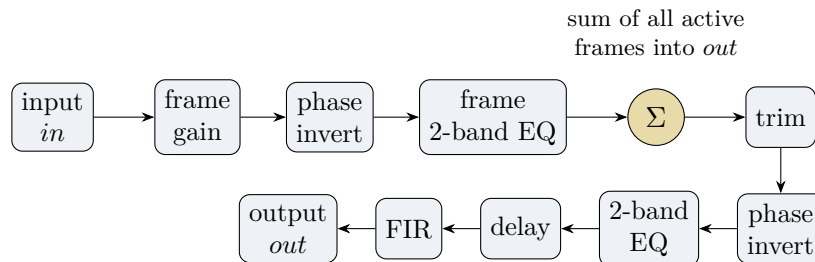


Figure 3.2: Routing frames (gain + phase + their own 2-band EQ) sum into each output, which then applies the per-speaker trim, polarity, its own 2-band EQ, delay and FIR correction.

3.1.3 Configurations A-F

There are 6 configurations (A..F), each a complete matrix at the active size (up to 32×32), stored per preset. They are engaged from the top-bar buttons:

- **Exclusive** mode (default): engaging one releases the others, an instant A/B between rigs.
- **Non-exclusive** mode: active configurations are summed, so you can stack layers.

The configuration you are editing is chosen with the "Edit: A..F" buttons above the analyzer, independently of which one is engaged for listening.

3.2 Interacting with the matrix

3.3 Per-output FIR correction and latency compensation

Each output can carry one **FIR** filter (loaded from a wav impulse response, convolved with the zero-latency WDL engine) to compensate the attached loudspeaker. Linear-phase correction IRs are rendered with their energy centred (see Chapter 7). In that case, a FIR of length L adds about $L/2$ samples of delay to that output.

SuperMoTo applies automatic **inter-output latency compensation**: among the outputs fed by the engaged configuration, the ones with shorter (or no) FIR are delayed so that all outputs stay time-aligned with the longest output FIR. This is what keeps a subwoofer aligned with a **linear-phase**-corrected main. A gold LED on an output cell indicates its alignment is automatic (hover to read the details).

Self-absorption. Before adding that compensation delay, an output that already carries its own manual delay first "spends" up to $L/2$ samples of it paying for its own FIR's latency, instead of that latency being added as extra delay somewhere else. Only the unabsorbed remainder, if any, still needs compensating on the other outputs, so a main that already has slack in its Delay control needs little or no compensation added to the sub at all. For example, a main delayed 25 ms to align with a farther sub, corrected with a 1024-sample linear-phase FIR ($L/2 \approx 11.6$ ms

Gesture	Action
Click a frame	Select it (opens it in the frame editor panel).
Double-click a frame	Activate / deactivate the crosspoint.
Vertical drag	Adjust the frame gain.
Alt+click	Toggle the frame's analyzer trace.
Right-click	Context menu for the frame.
Click an output	Select it (opens it in the output editor: trim, polarity, EQ, delay, FIR).
Output: double-click	Toggle that output's FIR on/off.
Output: vertical drag	Adjust the output trim.
Output: alt+click	Toggle the output-sum analyzer trace.
Output: right-click	Context menu for the output: FIR on/off, load / clear the impulse-response wav file, phase invert, analyzer trace, trim reset.

Table 3.1: Mouse interactions in the matrix and output strip. Every page also has an info button (i) summarising its controls and shortcuts.

at 44.1 kHz). Since $25 > 11.6$ ms, the main's delay control fully absorbs the FIR's latency (only $25 - 11.6 \approx 13.4$ ms of it is actually applied as a real delay) and the sub needs no extra compensation at all, instead of the ≈ 11.6 ms it would have received otherwise. The relative alignment between main and sub is identical either way: self-absorption only lowers the overall system latency, never changes what you hear. The delay control itself always keeps showing the value you set. Only how much of it becomes an actual delay line changes.

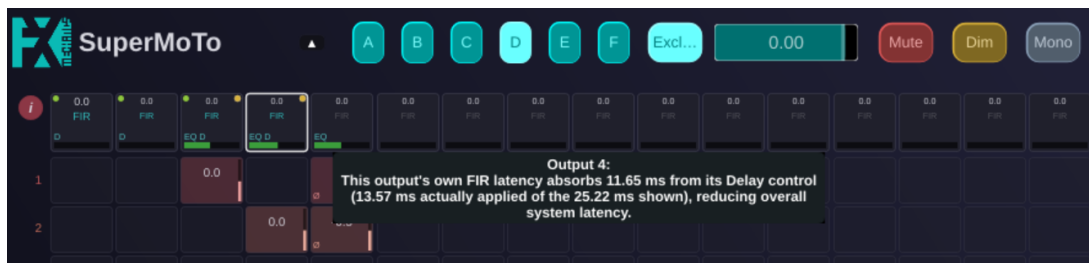


Figure 3.3: Output strip close-up: the gold LED marks an output whose alignment is automatic (compensation added, self-absorption, or both); the tooltip shows the amounts in samples / milliseconds.

3.4 The global spectrum analyzer

The analyzer can display any combination of individual matrix frames of output frames, up to 8 simultaneous traces. A clickable "avg/peak" badge switches the per-point aggregation. The analyzer also can be controlled with mouse wheel and ctrl-mouse wheel to change the axes limits and zooming.

3.5 Host-automatable parameters

The matrix settings themselves are deliberately not exposed as host parameters: $6 \times 32 \times 32$ frames with ~ 10 fields each would swamp any host. They are saved with the plugin state (Section 2.4). Only the master controls are real host parameters:

Parameter	Type / range	Meaning
Level	float, -60 to 6 dB	Master output level.
A...F	bool (6)	Engage configuration A-F (default: A on).
Exclusive	bool (default on)	Engaging one config releases the others; off = sum.
Mute	bool	Mute master output.
Dim	bool	Dim master by 6 dB (factor 0.5).
Mono	bool	Mono sum.

Table 3.2: The complete set of host-automatable parameters (APVTS).

3.6 Compact (collapsed) view

The collapse button ($\blacktriangle$) shrinks the window to a compact "output-strip-only" view, or a very compact view with only switches and gain slider, handy as a permanent monitor controller once the routing is set.

Chapter 4

The monitor target curve

The correction SuperMoTo designs aims the loudspeaker at a target. The ideal correction objective designed in the analysis and group analysis panes correspond to a flat and horizontal frequency response curve. The Target view lets you aim at something else, which almost always looks like a gentle downward tilt.

4.1 Why a flat in-room response is the wrong target

A loudspeaker whose direct sound is flat does not measure flat in a room. Its [directivity index](#) rises with frequency (ITU-R BS.1116-3 requires a monitor's to, smoothly, between 6 dB and 12 dB over 500 Hz to 10 000 Hz [27]) so the reverberant share of a steady-state, spatially averaged measurement falls as frequency rises, and absorption in a treated room takes more off the top. The ear follows the direct sound plus the first arrivals and the microphone average follows the steady state. Correct the steady state to flat and the direct sound has been lifted in the treble by the difference. This results in a "harsh" sound.

EBU Tech 3276's operational room response mask holds its upper limit flat at 3 dB but lets the lower one fall at 1 dB per octave above 2 kHz, reaching -6 dB at 16 kHz [26]. ITU-R BS.1116-3's widens downwards at 1.5 dB per octave over the same range [27]. Preference work puts the middle of the range lower still: the Harman in-room target falls about 1 dB per octave across the band with a low shelf under 105 Hz [29]. Toole's survey of the whole question is the long version [28].

A total of -3 dB to -6 dB from 20 Hz to 20 kHz is therefore a conservative reading of all three, and is where the factory curves sit.

Not the cinema X-curve. SMPTE ST 202 and ISO 2969 specify 3 dB per octave above 2 kHz for rooms over 125 m³. That is a very different target. Allen's account of where it came from also records what happened when small, dry mix rooms were tuned to it in the 1980s, and material sounded dull there, engineers compensated, so the mixes came out bright [31]. A small room needs less tilt than a cinema, and the useful idea to borrow is that the turnover frequency, not the slope, is what should move with room size and listening distance.

4.2 Choosing how much tilt

Section 4.1's argument is about a room in general. In a particular room, how much of what a measurement picks up is reverberant rather than direct sets how much tilt is actually owed, and that balance is set by the room and by where the listening position sits in it, not by the loudspeaker.

Rule of thumb. The drier and closer, the flatter; the livelier and further, the steeper. A small, well-treated room with the listening position close to the speakers, as near-field monitoring intends, keeps the direct sound dominant well into the treble, and there is little for a room to add beyond what Section 4.1 already accounts for from *directivity* alone. Flat or Gentle usually reads right. A larger or more reflective room, a listening position further back, or a wide correction area, lets the reverberant field's share grow, and its spectrum falls with frequency faster than the direct sound's, for the same reason. Strong or Harman-like tends to read right there. This is one reading of Olive, Welti and McMullin's finding that listeners preferred a downward-tilting in-room target to a flat one [29]: the more of the room is in what they heard, the more welcome the tilt.

A number to hang it on. Room acoustics has a standard quantity for exactly this balance: the *critical distance* d_c , the distance from the loudspeaker at which direct and reverberant sound pressure are equal. A common estimate, combining Sabine's reverberation-time formula with the classical room-constant approximation for a diffuse field, is

$$d_c \approx 0.057 \sqrt{\frac{QV}{T_{60}}} \quad (\text{metres, } V \text{ in m}^3, T_{60} \text{ in seconds}), \quad (4.1)$$

where Q is the loudspeaker's directivity factor in the band of interest (roughly 2–4 for a small monitor's tweeter) and T_{60} the room's reverberation time. At a listening distance r , the direct-to-reverberant ratio there is approximately $20 \log_{10}(d_c/r)$ dB: well positive favours Flat or Gentle, well negative favours Strong or Harman-like, and close to 0 dB is squarely Moderate, the default.

Two rooms of the same 50 m^3 and the same 1.2 m listening distance, differing only in treatment, illustrate the swing. Well treated, $T_{60} = 0.25 \text{ s}$, Equation 4.1 gives $d_c \approx 1.4 \text{ m}$, about 1.3 dB: direct sound leads, mildly. Untreated, $T_{60} = 0.6 \text{ s}$, $d_c \approx 0.9 \text{ m}$, about -2.5 dB : the same chair, in the same room, is now on the reverberant side. Nothing else moved; only how much of the room was absorbed.

The approximation folding the room constant into the Sabine absorption understates d_c once the average absorption coefficient is no longer small, so it errs, if anything, towards recommending more tilt in a live room rather than less. Q is frequency-dependent and hard to pin down without a polar measurement, and a small room's low end rarely reaches the diffuse-field assumption behind Equation 4.1 at all. So treat the result as an order of magnitude, one input among others, not a setting to compute and forget.

T_{60} is measured by SuperMoTo. Load a set of measurements in the Analysis pane with the swept-sine estimator, choose the raw impulse view, and the read-out beside it gives the room's reverberation time. You can get a tooltip showing the per octave band values by moving the mouse over it (Section 7.2.3). That is the number Equation 4.1 wants, from the room the correction was designed in, and the single figure it reports (the mean of the 500 Hz and 1 kHz octaves) is the one to put in it.

What is not measured is V and r : a tape measure settles both, and Q stays an assumption unless the loudspeaker came with a polar plot. So the estimate is still an order of magnitude arrived at deliberately rather than a setting computed from data, which is the right status for it.

What actually decides it. None of the above replaces the switch. A curve chosen from the room's numbers is a starting point to load and then judge as Section 4.4 describes, on programme material you know well, with Engage flicked in and out.

4.3 Curve parameters

The target curve is defined through four parameters:

- **Tilt:** the total fall from 20 Hz to 20 kHz, in dB. Negative falls, which is the useful direction; 0 aims at a flat in-room response, as before.
- **Turnover:** keep the target flat below this frequency and spread the whole fall over what is left. "off" runs the tilt straight across the band, which is the Harman shape; a turnover gives the EBU one.
- **Bass** and **Bass freq:** an optional first-order low shelf under the tilt, for the bass lift the Harman target carries near 105 Hz.

The curve is normalised so that its mean over 200 Hz to 2000 Hz is 0 dB (the same band the analysis normalises a measured response in section 7.2.1) so a target and a measurement are read against the same zero.

It is then applied as attenuation only. The whole curve is pulled down until its highest point reaches 0 dB, which is the dimmed second line on the plot, and the read-out next to the Engage switch says by how much. The layer can therefore never clip an output.

4.4 Usage

The target curve corrects nothing by itself. It shapes what a correction aims at. On a rig that has not been measured, corrected and time-aligned it is a tone control and nothing more. Measure (Chapter 6), correct and align (Chapters 7 and 8), and then come here to decide how the corrected system should tilt.

The button "Engage" controls a plugin parameter, so a host can automate it and it can be bound to a key, because the way to judge a target curve is to switch it in and out while listening to something familiar, and a curve that is obvious on paper is often inaudible in the room. The switch fades over 50 ms, so it does not click.

The curve is applied identically to every output, after that output's own EQ and before its delay. Applying it everywhere equally is what keeps it out of the main-to-subwoofer phase relationship: an identical minimum-phase filter on both sides of a crossover cancels out of the difference between them, where a tilt baked into the mains' correction filters alone would not. EBU Tech 3276 asks for the same thing: "All channels should be adjusted in the same way" [26].

A "Dry" measurement bypasses the target curve, as it bypasses everything else on the output; "Output + FIR" and "System" measurements include it (Section 6.1). So a verification measurement shows the tilt, and a measurement made to design a new correction does not.

4.5 The curve bank

Curves are files, one per curve, in a `target_curves` folder beside the preset folder, whose full path is shown under the buttons. Factory curves cannot be changed and are written again if the folder loses them. "Save as..." writes your own beside them, and "Save", "Rename" and "Delete" act on yours only. When the values match none of the stored curves the selector reads "(edited)".

The shape in force is part of the configuration: it is saved with the session and with presets, like the matrix and the outputs. The bank is machine-wide, like the preset folder.

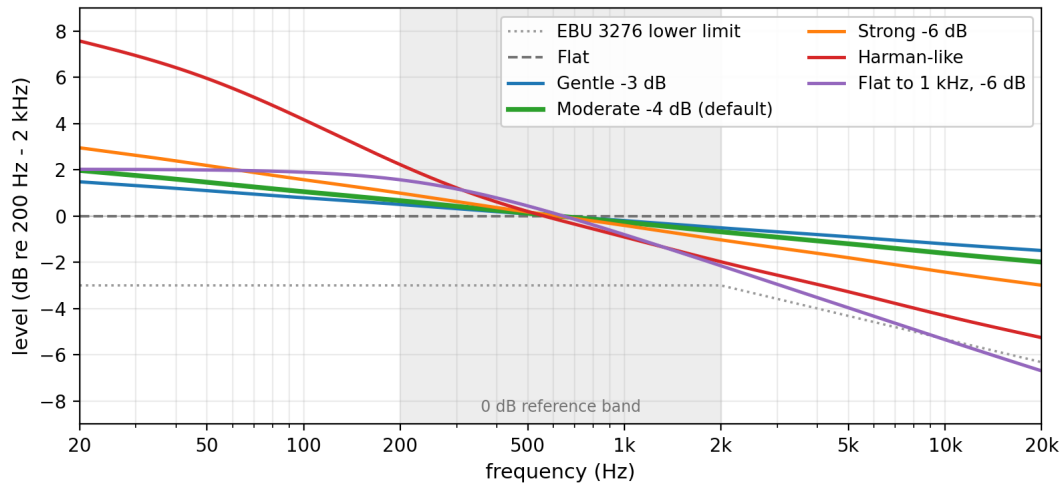


Figure 4.1: The factory target curves, all normalised to 0 dB on the 200 Hz to 2000 Hz band the analysis uses as its own reference. Flat is the behaviour before this view existed. The dotted line is EBU Tech 3276’s lower tolerance limit for a measured room response, -3 dB to 2 kHz then falling 1 dB per octave, drawn for scale: it is a tolerance on what a room does, not a rule about what to aim at, and only the steepest curve crosses it, by 0.4 dB in the last octave. Produced by `doc/figures/target_curves.py`, which reads the factory set out of the source so the figure cannot disagree with what the plugin ships.

Curve	Tilt	Where it comes from
Flat	0 dB	A flat in-room response.
Gentle	-3 dB	The cautious end of the range the standards allow.
Moderate	-4 dB	The default.
Strong	-6 dB	About EBU Tech 3276’s 1 dB per octave above 2 kHz, spread across the band.
Harman-like	-10 dB	Roughly 1 dB per octave with 3 dB of bass shelf at 105 Hz [29].
Flat to 1 kHz	-6 dB	The EBU shape: flat, then falling, with the fall spread over the top four octaves.

Chapter 5

The Configuration tool

The Configuration tool is a helper writing nothing that the matrix pane cannot express by hand, producing a base configuration of frames, gains, delays and crossovers (Chapter 3). It saves typing (a 5.1 layout is about a dozen crosspoints, a 16-speaker third-order ambisonic decoder up to 256).

For standard layouts the tool writes a whole matrix for you. Choose a layout (**2.0**, **2.1**, **4.0**, **4.1**, **5.1**, **7.1**), assign the input and output channel of each speaker, set the gains and the bass-management **crossover** frequency, then apply to a configuration (A..F). The mains are high-passed and their sum is sent, low-passed, to the subwoofer output.

The ambisonics case need a special attention and is exposed in the rest of this chapter.

5.1 Ambisonics decoding

The config tool also generates periphonic (3D) ambisonics decoders for orders 1-3 (Appendix G is a primer on the underlying theory). It follows the AmbiX convention (channels in ACN order with SN3D normalisation) and takes the B-format signal on the first $(\text{order} + 1)^2$ inputs (4, 9 or 16 channels). Instead of pairing one input with one speaker, each row now describes one loudspeaker: its direction (azimuth and elevation), its output channel, and its distance from the centre. Only the direction takes part in the decode. The distance is not an ambisonic quantity at all, and never reaches the decoder matrix. It drives the optional radius compensation described below, which is a per-output delay and trim for speakers that are not equidistant. The "Speakers" control chooses the number of outputs: at the canonical count the tool ships a hand-tuned periphonic rig (an 8-speaker cube, a 12-speaker icosahedron, a 16-speaker dome); at any other count it lays the speakers out on a near-uniform spherical-Fibonacci lattice. Either way you then edit the angles and radii to match the real installation (the minimum is $(\text{order} + 1)^2$ speakers, the number of harmonics).

Clicking "Apply" builds a max-rE sampling decoder: for each speaker s in direction Ω_s and each harmonic j the decode gain is

$$D[s][j] \propto a_{n(j)} Y_j^{\text{SN3D}}(\Omega_s), \quad (5.1)$$

where Y_j^{SN3D} is the real SN3D spherical harmonic of ACN index j , a_n the per-order max-rE gains (from the largest root of the Legendre polynomial $P_{\text{order}+1}$, improving localisation), and the matrix is scaled so an on-axis plane wave decodes to about unity. Each non-negligible $D[s][j]$ is written as a harmonic $\rightarrow$ speaker frame, its sign becoming a **polarity** flip. With bass management enabled, the omnidirectional W component (ACN 0) is low-passed to the subwoofer while the speaker feeds are high-passed, which, summed at each output, is exactly the high-passed decoded speaker because the filters are linear. The matrix grows to expose the B-format inputs and the speaker (plus sub) outputs. Refine gains, delays and per-output **FIR** corrections in the matrix afterwards.

Importing an IEM AllRADecoder. As an alternative to the built-in sampling decoder, "Load IEM decoder..." reads a `.json` configuration exported by the IEM [AllRADecoder](#) (part of the IEM Plug-in Suite). AllRAD (all-round ambisonic decoding) handles irregular and hemispherical rigs better than a plain sampling decoder. The file's `decode Matrix` replaces the computed one; SuperMoTo converts it to its SN3D / ACN feed on the fly:

- if `ExpectedInputNormalization` is `n3d`, each coefficient of ACN degree n is multiplied by $\sqrt{2n+1}$ (N3D $\rightarrow$ SN3D);
- if `Weights` is `maxrE` and `WeightsAlreadyApplied` is `false`, the per-degree max-rE gain is folded in;
- `Routing` maps each matrix row to a hardware output channel; `IsImaginary` loudspeakers (used only to regularise the decoder design) carry no row and are skipped;
- the loudspeaker layout supplies each speaker's azimuth / elevation, its **Radius** (used by the radius compensation below) and its per-speaker **Gain** (folded into the frame gain as a trim).

A gain matrix cannot encode time, so the imported decoder, exactly like the built-in one, carries no delay. The radius delay below is the only time alignment, and it is optional.

Radius compensation. Ambisonic decoding assumes the speakers are equidistant from the centre. When they are not, the tool eventually corrects for each radius r_s relative to the farthest speaker $r_{\max}$, writing a per-speaker output "delay" and "trim":

$$\Delta t_s = \frac{r_{\max} - r_s}{c}, \quad \Delta g_s = 20 \log_{10} \frac{r_s}{r_{\max}} \ (\leq 0), \quad (5.2)$$

with $c = 343 \text{ m s}^{-1}$. Closer speakers are thus delayed (so all wavefronts reach the centre at the same time) and attenuated, the reference being the farthest speaker. No channel is ever boosted or asked for a negative delay. The "Write radius gain" and "Write radius delay" are independently optionnal, and both apply equally to the built-in and the imported (IEM) decoders. Turning one off leaves that output property untouched, so an alignment or level already set (in the group analysis, or in the single-speaker analysis pane) survives an "Apply". The radius delay is independent of, and rides on top of, the inter-output FIR [latency compensation](#) (Section 3.3), which only equalises the FIR processing latency. The final arrival of a speaker is $\Delta t_s + l_{\max}$, so the radius alignment survives untouched.

Subwoofer polarity. The decoder's signs are frame polarity flips, part of the decode, and "Apply" rewrites them. A subwoofer that sums better inverted against the speakers is a property of the rig, not of the decode, so its place is the "Phase inv." switch of the subwoofer's output (Section 3.1.2). The tool never writes output polarity, so that setting survives rebuilding the decoder. Group analysis sets it from "Invert sub" (Section 8.7). Do not also invert the frames feeding the subwoofer, since the two flips multiply and cancel.

Measured vs. geometric alignment (do not double-count). The geometric radius delay above and a measured alignment are two routes to the same delay, not two delays to add. Both write the same per-output delay, and each simply overwrites the other, so the question is what method you want to determine that delay (estimation based on speaker positions or measurement). And there are two ways to measure it. First is the group analysis (Section 8.5), which aligns a whole rig on its most-distant driver and is the natural choice for an ambisonic array. The other way is for a single speaker, the subwoofer time-alignment (Section 7.5.2). Both reference the farthest driver exactly as the radius delay does, and both measure what the radius can only estimate. When you run the per-output sub analysis for a main speaker k at distance d_k from the mic centroid, the main set has d_k/c removed per file and the sub is anchored on

that same delay, so the reported "Mains delay" is $\approx (d_{\text{sub}} - d_k)/c$ plus the intrinsic **group delay**. Setting that on the output gives an acoustic-plus-electronic arrival of

$$\underbrace{d_k/c}_{\text{acoustic}} + \underbrace{(d_{\text{sub}} - d_k)/c}_{\text{electronic}} = d_{\text{sub}}/c, \quad (5.3)$$

independent of d_k : every measured main lands at the same time as the sub and as the other mains. So a per-speaker measured alignment already contains the path-length (radius) term, and more accurately than an entered radius, since it uses the true acoustic path, references the same centroid as the average, and folds in the speakers' intrinsic group delay. Therefore:

- **Measured route** (best): align the rig in the group analysis, which writes each output's delay for you. Alternatively, you can do one speaker at a time, by running the sub analysis per main and set each output delay to the reported value. To (re)build the decode afterwards without clobbering those delays, turn "Write radius delay" off before "Apply" so that the tool then writes the decode, the crossover and the radius trim (unless "Write radius gain" is also off) but leaves each output's delay untouched. (Otherwise set the radii equal so the geometric delay is zero.) Applying the decode with the toggle left on replaces a measured alignment with the geometric estimate, which is the one real way to lose work here.
- **Geometric route** (quick): enter the radii and let the tool write the delay and trim. Use this when there is no sub, or you have not measured each speaker.

Level matching is a separate question from alignment which the correction FIR does not answer, because each speaker's correction is normalised to that speaker's own mid-band, so inter-speaker amplitude differences survive it untouched. The group analysis does answer it by deriving a suggested level-matching trim per speaker from the corrected mid-band level (Section 8.6). Like the radius trim it is referenced to the quietest speaker, so it only ever attenuates. It is informational only, though: "Apply & export" never writes it, so you enter it yourself in that output's **Trim** in the matrix. Failing that, the radius trim Δg_s (or a calibrated-SPL measurement) still handles it.

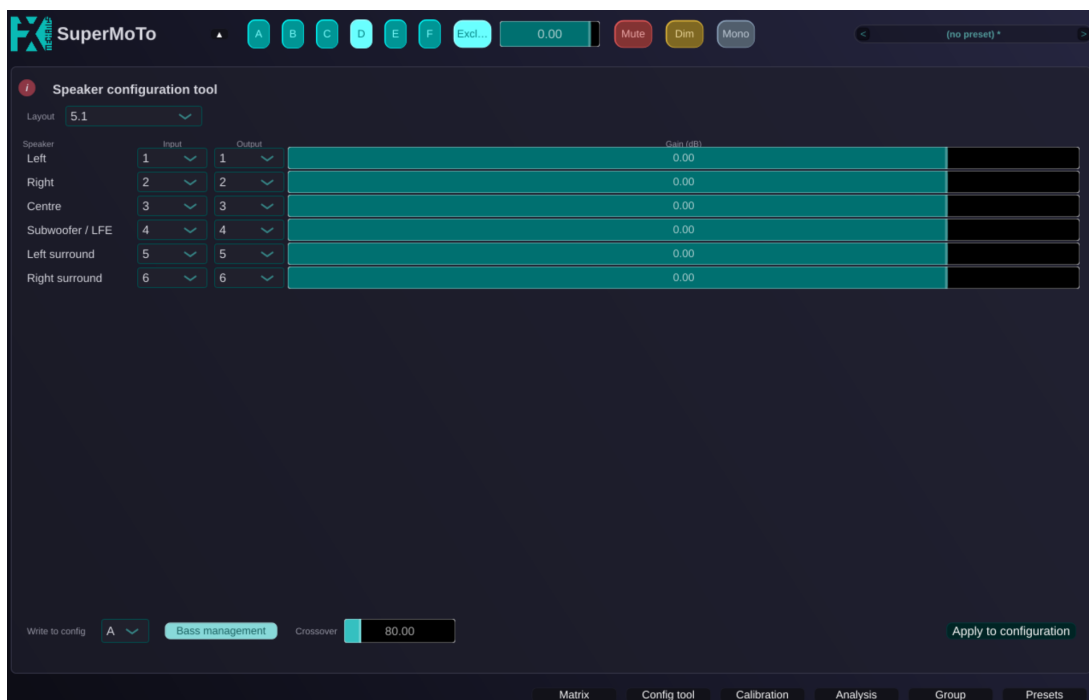


Figure 5.1: The Configuration tool: a standard layout (here 5.1) with per-speaker input/output assignment, gains and the bass-management crossover frequency, ready to Apply into one of the A–F configurations.

Chapter 6

Measurement (Calibration view)

The Calibration view is where SuperMoTo emits a test signal, records the measurement microphone and writes the result to disk. This chapter takes the three measurement modes first, since they decide what everything else means, then the controls of the pane, then what a run leaves on disk, and finally the practical recommendations. The reasoning behind the microphone-position recommendations is in [Appendix D](#).

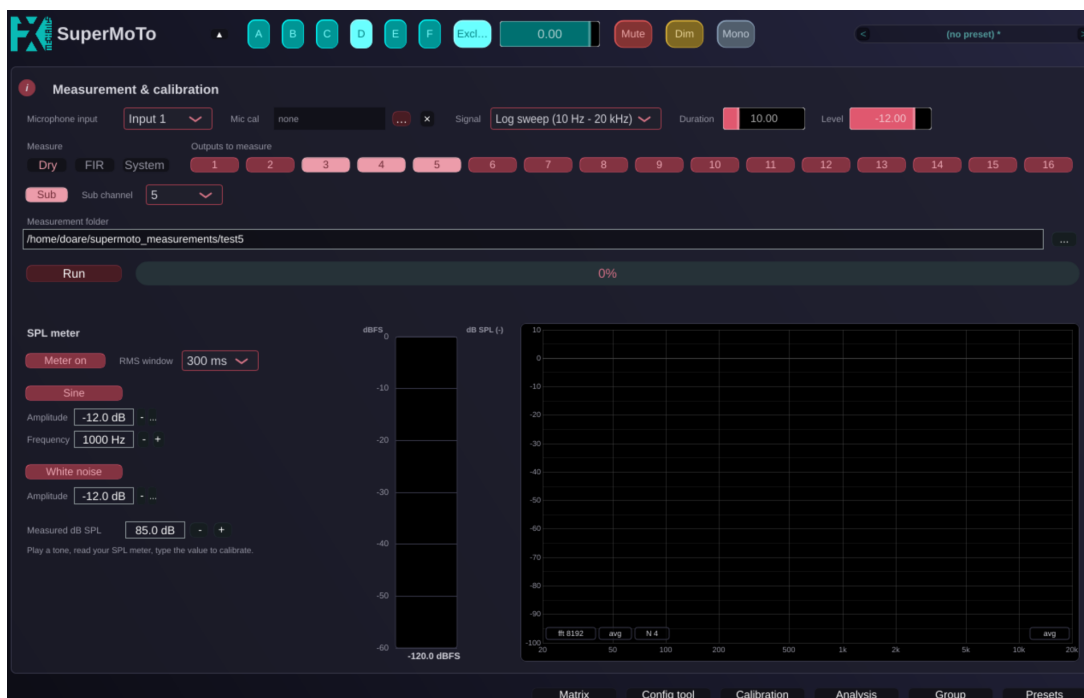


Figure 6.1: The Calibration view: microphone input, measurement type, channel selection, stimulus (noise / sweep), duration, level, the **Sub** switch and sub-channel combo, the measurement **folder**, the Run control, and the right-hand SPL-meter section with its dual dBFS / dB SPL bar and test-signal generators.

6.1 The three measurement modes

The measurement type selector decides where the stimulus is injected and how much of the engine it passes through before reaching the loudspeaker and the microphone. The three modes form a natural progression. The first characterises a bare speaker ("Dry"), the second verifies its correction ("FIR"), and the third validates the whole monitored chain ("System"). [Table 6.1](#)

summarises them and Figure 6.2 shows the injection point of each.

Mode	What is exercised
Dry	Stimulus written straight onto the measured <i>output</i> , with no trim, no polarity flip and no FIR. The raw loudspeaker, as wired. The first step before designing a FIR.
FIR	Stimulus to the measured <i>output</i> through its trim, polarity and FIR chain only, to verify a correction by re-measuring.
System	Stimulus injected into a plugin <i>input</i> and run through the whole engine (matrix routing, frame EQ / crossover, output FIRs, inter-output latency compensation), the complete system as heard. Channel selection means <i>inputs</i> in this mode; the "Sub" switch does not apply.

Table 6.1: Measurement modes. "Dry" characterises a bare speaker, "FIR" verifies its correction, "System" captures the complete monitored chain. What each mode writes to disk is in Section 6.4.

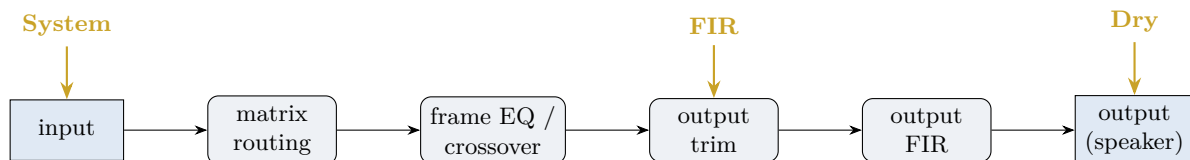


Figure 6.2: Where each mode injects the stimulus. "Dry" drives the output directly (bypassing trim and FIR); "FIR" enters at the output trim so the signal passes trim + FIR; "System" enters at a plugin input and traverses the entire engine.

6.1.1 The raw loudspeaker (Dry)

The stimulus is written directly onto the selected output channel without the output's trim, polarity or FIR applied. The capture therefore characterises the loudspeaker and room exactly as the speaker is physically driven, with no correction in the path. This is the measurement you feed to the analysis (Chapter 7) or group analysis (Chapter 8) to design a correction. Run it for each output you intend to correct, at several microphone positions around the listening spot.

6.1.2 Verifying the correction (FIR)

The stimulus is emitted to the selected output through that output's trim, polarity and FIR only (the matrix is otherwise silent). Use it after you have exported a correction IR and assigned it to the output. Re-measuring should show a flatter response than the corresponding dry capture. It is the direct before/after check on a single speaker's correction.

6.1.3 The complete monitored chain (System)

The stimulus is injected into the selected plugin input and run through the whole engine with the currently engaged configuration(s), including matrix routing, frame EQ and bass-management *crossover*, the output FIRs and the inter-output *latency compensation*. This is the final system you want to be optimised. The master gain is bypassed (unity) during the run, so the capture level is set by the stimulus alone. Because the source is an input, the channel selection chooses inputs. This is the end-to-end validation which, for a bass-managed rig, shows whether mains and subwoofer sum smoothly through the crossover at the listening position.

6.2 The Calibration pane, control by control

A measurement is set up by working across the pane, picking the microphone, the mode, the channels, the stimulus and the destination folder, then pressing "Run".

6.2.1 Microphone input and calibration

The "Microphone input" selector chooses which plugin input channel carries the measurement microphone. Next to it, the "Mic cal" control loads and clears the microphone calibration curve. Because there is normally only one physical microphone, this is a global setting remembered across sessions and projects. Section 6.3 covers the calibration in detail.

6.2.2 Measurement type and channel selection

The measurement type buttons choose Dry, FIR or System (Section 6.1). The channel toggles then choose what to measure, and their meaning follows the mode. They are physical outputs in Dry and FIR, and plugin inputs in System. Each selected channel is measured in turn within one run.

6.2.3 The Sub switch

The "Sub" switch turns subwoofer handling on for this measurement run. The "Sub channel" combo next to it then picks which of the measured channels is the subwoofer. Both are disabled in System mode, where the toggled channels are plugin inputs rather than physical outputs. Leaving the switch off (the default) measures every channel as an ordinary loudspeaker.

The "Sub" switch's only effect is to flag the corresponding channel under the `sub` name instead of its channel number (Section 6.4.1), and the manifest records which channel it was. That is what later lets Group analysis load a whole campaign and know which of the sets is the sub.

6.2.4 Stimulus, duration and level

The stimulus is either band-limited white noise (10 Hz to 20 000 Hz) or a logarithmic sweep, which runs from about 11 Hz up to Nyquist over a whole number of octaves. The sweep is generally a better choice. It also enables the synchronized swept-sine analysis of Section 7.1.5, which separates the harmonic-distortion curves. Its exact band is written into the manifest per run, so the analysis always knows what was emitted, and folders measured with an older version keep analysing with the band they were actually made with. In order to keep the deconvolved `impulse response` free of ringing around its peak, the sweep is performed all the way to the Nyquist frequency. Appendix E discusses the measurements behind that choice.

Longer is quieter, through more averaging, but slower. The level should be loud enough to sit well above the room noise and quiet enough not to drive the speaker into distortion.

The duration (5s to 30s) is not exact for sweeps, since in that case, the sweep rate is quantized (Section B.3.1).

6.2.5 The measurement folder

The folder chooser sets where a run organises the files of an entire measurement campaign into a single folder that Group analysis (Chapter 8) can later load in one step. Section 6.4 describes what ends up there.

6.2.6 Comments

Two free-text fields let you record what a set of measurements actually was. Both are written into the manifests, next to the files themselves, where they will still be legible later.

- **Folder comment...** opens a small editor for a general comment describing the whole campaign (the room, the rig, the date, the speaker positions). It is pre-filled from the folder's existing manifest when you point the path at a folder that already has one, so it accumulates all consecutive runs. Edit it between runs and the next run rewrites it.
- **Run comment** is a one-line note attached to the next run only, logged with that run's entry ("mic at pos xxx", "after moving the left speaker").

6.2.7 Running the measurement

"Run" measures each selected channel in turn. Each capture records the stimulus plus a short decay tail (1 s) so the room's late energy is included. The stimulus is faded in over 20 ms to avoid a click at the start. It is faded out over 20 ms as well for noise, but over only 0.5 ms for a sweep. A long fade-out would taper the top of the swept band and smear the deconvolved impulse response, while a very short one still removes the step at the end.

When measuring several channels in one run, the engine holds the monitor path silent while each finished capture is written to disk, so the speakers are not blasted between consecutive measurements.

Stopping. The button reads "Stop" while a run is going, and it cancels the run rather than merely ending it: the captures already written are deleted, and the status line says how many went. A run is all or nothing because half of one is worse than none — a folder holding some channels of a position, with nothing in the manifest to say which, since that is only written when a run completes. Runs that finished are never touched, and neither is anything else in the folder: each run writes the next free position number per channel, so it can only ever delete its own files, and once they are gone the next run takes the numbers this one was using. A run that dies on a write error (a full disk, a folder that went away) cleans up the same way.

6.2.8 The SPL meter

The right-hand SPL-meter section reuses the microphone / channel / measurement-type selections. It shows the microphone RMS on a dual dBFS / dB SPL bar and can emit a sine and/or white-noise test signal, routed exactly like the selected measurement mode. To calibrate the dB SPL scale, play a tone, read a real SPL meter at the listening position, and type that value into the plugin.

6.3 Microphone calibration

6.3.1 Why calibrate

Even a good measurement microphone is not perfectly flat. Each capsule has small deviations from flat, most noticeably a gentle rise or roll-off in the top octave (above 8 kHz to 10 kHz) and sometimes a few tenths of a dB elsewhere. The calibration file is the measured deviation of that microphone from flat. If it is not removed, the analysis attributes the microphone's own colouration to the loudspeaker and builds it into the correction. Dividing the calibration out first means the estimated [transfer function](#) reflects the loudspeaker and room alone. Note that the measurement microphone's defects are generally of a much lower order than that of the speakers in the room.

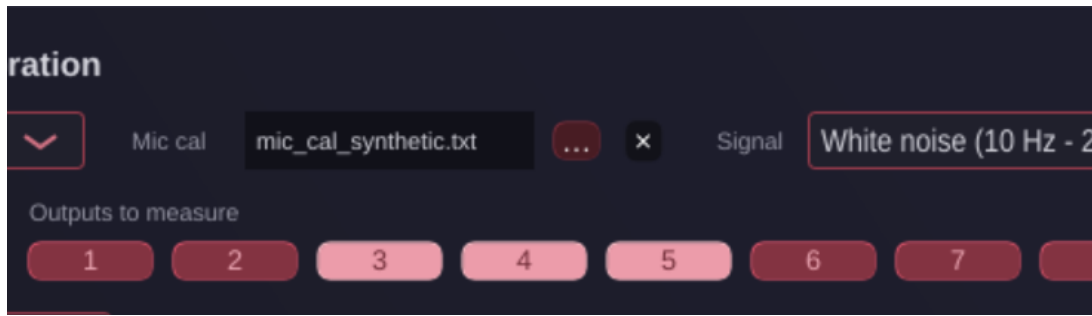


Figure 6.3: The **Mic cal** control on the Calibration view: the loaded file name, the load (...) and clear (x) buttons, beside the microphone-input selector. The same calibration is shown (read-only) in the Analysis view.

6.3.2 The calibration file format

There is no formal standard, but a single de-facto plain-text format is shared by REW, miniDSP, Dayton (OmniMic / iMM-6) and the FRD tools, and SuperMoTo reads it directly (.txt, .cal or .frd):

- comment lines start with *, # or ; (the miniDSP `Sens Factor =...` header is one such line);
- every other line is `frequency_Hz magnitude_dB [phase_deg]`, whitespace- or comma-separated, with the phase column optional.

The magnitude is the microphone's deviation from flat (mean ≈ 0 in the midband). Individually-measured mics (e.g. the UMIK series) have a file keyed to the unit's serial number, downloaded from the vendor. Cheaper mics may ship only a model-typical file, or none. The **Sens Factor** line encodes absolute sensitivity for dB SPL work and is ignored here.

6.3.3 Where the calibration is used

The calibration is a global setting, loaded with "Mic cal" ... on the Calibration view and remembered across sessions and projects, since there is normally one physical microphone. It is used in three places.

1. **The live spectrum display** (SPL-meter section) plots the corrected microphone spectrum, so what you see already has the microphone's own response taken out.
2. **The measurement folder.** Every run records the calibration in effect, its name and a verbatim copy of the file, in the folder's manifest (Section 6.4.4), so the folder carries its own microphone correction.
3. **The analysis views** divide it out of every measured transfer function, and can use the folder's copy instead (Section 7.1.2).

The captured waves stay raw. The calibration is applied at analysis time and never baked into the recordings, so a better file, or the file for another microphone orientation (Section 6.5.1), can be used later without measuring again.

6.4 What a measurement writes to disk

Every run leaves two kinds of thing in the measurement folder, the capture files themselves and two manifests describing them. Together they make the folder self-describing. Hand it to Group analysis, to a colleague, or to yourself in six months, and the analysis knows how the recordings were made without you having to remember.

6.4.1 The capture files

Each measured channel produces one stereo wav file. The naming follows the mode, so a System set can never collide with a Dry or FIR set in the same folder:

Mode	Channels are	File written
Dry	outputs	ch<output>_pos<N>.wav, or sub_pos<N>.wav for the channel flagged as the subwoofer
FIR	outputs	ch<output>_pos<N>.wav, or sub_pos<N>.wav for the channel flagged as the subwoofer
System	inputs	in<input>_pos<N>.wav; the Sub switch does not apply, so there is no sub file

The position number <N> is found automatically for each channel by scanning the folder for that channel's existing files and incrementing. Moving the microphone and running the same channel selection again therefore adds the next position without any manual renaming, and channels measured together in one run stay in step across runs.

6.4.2 The "sent" channel convention

In every mode the captured stereo file stores the raw stimulus (pre-everything) on channel 1 and the microphone on channel 2:

- channel 1 = the sent signal (the raw stimulus),
- channel 2 = the recorded microphone signal.

The [transfer function](#) computed in the analysis part is therefore always microphone / raw stimulus. The consequence is deliberate and important:

- in "Dry" mode that ratio is the bare speaker + room response;
- in "FIR" and "System" modes it already includes the correction (and, for System, the full chain), so a successful correction shows up directly as a flatter measured transfer function, with nothing to subtract by hand.

6.4.3 The two manifests

After every run SuperMoTo (re)writes two manifests in the folder, both regenerated from the folder's actual contents each time, so they stay correct even if files are added or removed by hand:

- `measurement.xml` is the machine-readable manifest, detailed below. This is what Group analysis's "Load measurement folder..." button actually reads (Section 8.3), so hand-editing or reformatting the readme can never break folder loading.
- `readme_measurement.md` holds the same information as human-readable documentation of the campaign (mode, stimulus, duration, level, mic input, files written per run). It is also the fallback for folders recorded before the XML manifest existed, and the copy you can read without any tooling.

6.4.4 What the manifest records

`measurement.xml` carries everything a later analysis needs to interpret them. At the folder level it holds:

- the **channels** measured and which one is the subwoofer, plus the highest position number reached and the time of the last run. A channel that the preset names (Section 3.1.2) carries that description with it, so the folder records which loudspeaker each capture came

from. Names are kept from run to run: a run that cannot name a channel, a System run whose channels are inputs for instance, leaves the names an earlier run wrote;

- the **general comment** (Section 6.2.6);
- the **SPL calibration** in effect, as the single offset relating full scale to sound pressure, $\text{dB SPL} = \text{dBFS} + \text{offset}$, which is what later allows the analysis to plot estimated dB SPL (Section 7.2.1);
- the **microphone calibration** in effect, stored as its display name plus a verbatim copy of the calibration file itself, so the folder carries its own microphone correction (Section 7.1.2).

and, per run, a log entry giving the time, the mode, the stimulus (type, duration and level), the microphone input, the files written, the comment for that run, and, for a sweep, the sweep identity f_1 , f_2 and L . That last item is what enables the synchronized swept-sine analysis of Section 7.1.5. Without a recorded L the deconvolution cannot reconstruct the stimulus, and the analysis falls back to [Welch](#).

6.5 Recommendations

The reasoning behind the position recommendations is in Appendix D, and what follows is the summary.

6.5.1 Choosing and orienting the microphone

What the analysis sees is the loudspeaker and room as sampled by the microphone. The microphone is therefore part of the measurement chain, and the kind of microphone, together with how well it is characterised, sets a ceiling on how trustworthy the resulting correction can be.

Ordinary recording microphones (cardioid vocal or instrument mics) are deliberately voiced, with a directional pickup pattern and a shaped, non-flat response designed to flatter sources. That is exactly wrong for measurement. Use a measurement microphone, which has the following interesting characteristics:

- Omnidirectional in almost all the audible frequency range
- Small diaphragm (typically quarter-inch or half-inch), for a wide, uniform high-frequency response with little directional colouration at high frequencies.
- Nominally flat magnitude response over the audio band, with only small residual deviations, which a calibration file then removes (Section 6.3).

Common, affordable choices include the USB miniDSP UMIK-1 / UMIK-2, the Dayton Audio EMM-6 (XLR) and IMM-6, the Behringer ECM8000, and any class 1/class 2 measurement capsule. USB models carry their own ADC and appear as a soundcard input, while XLR models need phantom power and an audio interface. Some are delivered with calibration data, some not. It is highly correlated to the price. Either works, since SuperMoTo only needs the microphone to arrive on one of its plugin input channels.

Because the microphone is omnidirectional only at low frequencies, its high-frequency response depends on the angle of arrival, so it is characterised for a specific orientation. The two orientations differ above a few kilohertz, and most in the top octave. Some vendors ship two calibration files for this reason:

- 0° (**on-axis**), with the capsule pointed at the loudspeaker.
- 90° (**grazing**), with the capsule pointed at the ceiling, so that sound from a loudspeaker at ear height arrives from the side.

A run measures every selected channel with the microphone in one place, so the capsule cannot point at each speaker in turn. Point it at the ceiling and load the 90° file. Every speaker in the horizontal plane then arrives at 90°, whatever its azimuth, so one file is correct for all of them, and the orientation is easy to reproduce at every position. Keep the 0° file for runs that measure a single speaker, with the capsule aimed at it.

Elevated speakers are the limit of this approach. A speaker 30° above ear height arrives at about 60° , and one overhead at 0° , so for a height layer or an Ambisonic dome the 90° file is only approximate for them. The error is confined to the treble, where a coarse high-frequency smoothing and a moderate "Range" already keep the correction from following fine detail (Section 7.3.2).

The calibration data matters most above a few kilohertz. Hence, in the bass the capsule is essentially flat and room modes dominate regardless. Measuring with no calibration at all is therefore acceptable for a broad bass correction, but any treble target then inherits the microphone's error, so a calibrated microphone is advised for full-range work.

6.5.2 How many positions, and how far apart

- **Seven to nine positions** for a single listening position. Five is already enough below 500 Hz. The extra positions only earn their keep higher up, where the return diminishes as $1/\sqrt{N}$.
- **Spread them over 30 cm to 50 cm**, about the volume a head occupies while working. Much wider and the mid band is averaged towards something true at no single position, without improving accuracy in the bass, where the wavelengths are metres. A subwoofer set argues for keeping the cluster tight for a second reason (Appendix D.3).
- **Use all three dimensions.** A flat cross in the horizontal plane misses the fastest-changing direction, because the desk and ceiling reflections vary more with height than with lateral position. Two or three points at ± 10 cm to 15 cm vertically are worth more than the same number added to the plane.
- **Measure every speaker from the same points.** Group analysis compares the speakers' arrival times and levels, which only means something if every speaker was measured from the same places. Measuring all the channels in each run guarantees it, and also lets a subwoofer be paired with each speaker by run (Section 8.3).
- If the system must serve a listening zone rather than a spot, microphone positions have to cover that area, and the low end then becomes an explicit compromise rather than a correction. No single filter can serve two positions a metre apart below a few hundred hertz.

The cluster's size determines what it averages out. A response changes noticeably over about half a wavelength, so a 30 cm cluster only averages what lies above roughly 570 Hz, where the **comb filtering** of reflections moves from one position to the next. Below that, every position sees much the same response, and the average keeps it as it is at the seat, whether it comes from a **room mode**, from **speaker-boundary interference** or from the low-frequency part of any other reflection. The correction is therefore valid for the area the cluster covers.

What the correction can do about a feature depends on another distinction, peaks against dips. A resonant peak can be cut safely. A dip, whether a modal null or a cancellation by a reflection, cannot be filled by any filter, and "Max boost" stops the correction from trying (Section 7.4). Appendix D works through the numbers.

6.5.3 Choosing the stimulus

- Prefer the log sweep. Both stimuli give magnitude and phase, since the **Welch** estimate is complex, but for the same duration the sweep gives a cleaner result. It spends as long in each octave as in any other, so the bass, where room noise is highest, gets far more energy than white noise gives it. It keeps harmonic distortion out of the linear response and shows it as separate curves. And it needs no averaging of random segments, whose variance a Welch estimate trades against frequency resolution.
- 10 s is a good default duration. Go longer in a noisy room rather than louder, and remember a sweep needs enough length for its harmonic orders to separate (Section B.3.3).

- Set the level loud enough to sit well clear of the room noise and no louder. Check the distortion curves after a first run. If the harmonic orders are close to the fundamental, the speaker is being driven too hard for a clean measurement.

Chapter 7

Analysis & correction design



Figure 7.1: The Analysis view: individual smoothed transfer functions (thin), their average (thick), the proposed correction and the corrected response, in magnitude and phase; with the n th-octave smoothing, correction-level, boost-ceiling and analysis-range controls, the subwoofer integration section (crossover, Invert, Mains delay), and the Export buttons.

The Analysis view turns the measurements of one loudspeaker into a correction filter. You load the measurement set, check the averaged response on the plot, tune the [smoothing](#) and the correction controls until the corrected curve looks right, optionally align a subwoofer, then export the filter and assign it to an output. This chapter describes what each step does and how to set it. The signal processing behind it, with the equations, is in [Appendix B](#).

7.1 Loading and estimating the response

Click "Load measurements. . ." and select the files of one loudspeaker, measured at several microphone positions around the listening point. For each file SuperMoTo estimates the loudspeaker-room [transfer function](#) from the stimulus and the microphone signal stored in it (Section [6.4](#)). If a microphone calibration is in effect, it is divided out at this stage (Section [7.1.2](#)), so the curves

describe the loudspeaker and the room and exclude the microphone.

7.1.1 Choosing the estimator

The "TF" selector chooses how each transfer function is estimated.

- **Welch** averages the spectra of many overlapping segments of the recording. It works with any stimulus and is always available.
- **Sweep (Farina)** deconvolves a logarithmic sweep. It is offered, and selected automatically, as soon as the loaded folder's `measurement.xml` records the sweep's parameters (Section 6.4.4). It estimates the whole band from a single sweep rather than from an average of segments, and it moves the harmonic distortion out of the linear response instead of leaving it folded in, which gives a cleaner estimate.

Both give magnitude and phase, and everything after the estimate, from averaging to export, is identical for both.

7.1.2 Microphone calibration

The microphone's own response is divided out of each measured transfer function as soon as it is estimated, so the curves describe the loudspeaker and the room alone. Section 6.3 covers the calibration file itself and Section 6.5.1 which orientation to use. At each frequency bin the measured response is multiplied by the inverse of the microphone curve,

$$H_{\text{corr}}(f) = H_{\text{meas}}(f) 10^{-c_{\text{dB}}(f)/20} e^{-j c_{\phi}(f)}, \quad (7.1)$$

where c_{dB} and c_{ϕ} are the calibration magnitude and phase, interpolated in log-frequency and held flat (at the endpoint value) outside the file's range. When the file has no phase column, $c_{\phi} \equiv 0$ and only the magnitude is corrected.

The right curve is not always the one currently loaded on the Calibration view, so a source selector chooses which one is divided out.

- **Global mic cal** is the curve loaded on the Calibration view. This is the default and is what you want while the measurements in front of you were made with the microphone you are using now.
- **Folder mic cal** is the curve embedded in the measurements' own `measurement.xml` (Section 6.4.4). Offered only when the loaded set actually carries one.
- **No mic cal** analyses the raw measurements, with no microphone correction at all. Useful to see how much the calibration is doing, or when you suspect the calibration file itself.

When a loaded set carries an embedded curve, "Folder" is selected automatically. Measurements made a year ago with a different microphone are then analysed with that microphone's calibration, although that can be overridden. The name of the curve in effect is shown next to the selector (*Mic cal: ...*) as a reminder. Because the captures stay raw, changing the source re-analyses the loaded set immediately, with no need to measure again.

7.1.3 The Welch window

The "Welch window" sets the length of the analysis segments, from 16384 to 262144 samples, 65536 by default. A longer window gives a finer frequency resolution, about 0.67 Hz at 44.1 kHz with the default, which is what resolves low-frequency **room modes**. A shorter window cuts the same recording into more segments to average, which lowers the noise of the estimate. With the swept-sine estimator the window also sets how much of the deconvolved response is kept, and a window too large for the sweep makes the analysis fall back to Welch (Section 7.1.5).

7.1.4 Delay removal and spatial averaging

Each microphone position is at a different distance from the loudspeaker, so each measurement arrives with a different delay. SuperMoTo finds that delay from the peak of each measurement's [impulse response](#) and removes it, so every measurement starts at the same instant. The responses are then averaged, taking the magnitude from a power mean and the phase from the complex one — and keeping only as much of that phase as the positions agree about, fading the rest toward the minimum-phase equivalent of the same magnitude. Appendix [B](#) gives the arithmetic and says why both halves of that are necessary. This [spatial averaging](#) is what makes several positions worthwhile. The [comb filtering](#) that reflections produce, narrow peaks and nulls that move with the microphone, tends to cancel in the average, leaving the response the loudspeaker reproduces consistently across the listening area [[19](#), [16](#)]. A correction designed from a single position would instead try to fill nulls that do not exist a few centimetres away. Appendix [D](#) discusses how many positions to take and how far apart.

7.1.5 The swept-sine estimator

The sweep method needs the sweep's parameters, which SuperMoTo reads from the measurement folder's manifest. Three practical consequences follow.

- The sweep rate is quantized, so the actual sweep is slightly longer or shorter than the "Duration" you asked for. It sounds the same, and the quantization is what lets the harmonic orders come out with their true phase (Section [B.3.3](#)).
- Harmonic distortion orders 2 to 5 appear on the plot as the H2...H5 traces. They follow the same [smoothing](#) and microphone calibration as the other curves, and are averaged across positions in power, since distortion phase is not coherent from one position to the next.
- An order is dropped when the sweep is too short to separate it cleanly, which is why a short measurement shows fewer distortion curves than a long one.

A file is estimated with Welch instead whenever

- the folder carries no sweep identity, as with files loaded by hand, a noise run, or a folder recorded before the manifest stored it;
- the recording is shorter than the sweep it claims to be;
- the Welch window is too large for the sweep, so the harmonic responses would not stay clear of the linear one.

The fallback is per file and silent. A set that mixes sweep and noise runs still analyses, each file by whichever estimator applies to it. Welch remains the right, and only, choice for a noise stimulus, and stays selectable for a sweep recording if you want to compare the two.

7.2 The plot component

The plot shows:

- the individual smoothed [transfer functions](#), one per position (thin lines);
- their complex average (thick line);
- the proposed correction;
- the predicted corrected response, the average with the correction applied;
- the subwoofer response, when a subwoofer set is loaded (Section [7.5](#));
- the harmonic-distortion magnitudes H2...H5, with the swept-sine estimator only.

By default, magnitudes are normalised to the 200 Hz to 2000 Hz mean of the average, so 0 dB is the mid-band level, independent of the absolute measurement gain. Phase is shown as principal value with the propagation delay already removed. The correction's phase follows the "Phase" setting, so with "Min phase" selected it shows the phase the exported filter will actually have. Two selectors change what the panel displays.

7.2.1 Level reference

The "Level" selector sets the vertical reference of the measured curves. It is a display choice only. The correction is designed from the same data either way, and the correction curve itself is always shown in absolute dB, since it is a gain and has no reference to choose.

- "Normalized" is the default described above, with 0 dB at the 200 Hz to 2000 Hz mean. Best for judging shape, and the only sensible choice when comparing speakers measured at different gains.
- "Absolute dB" shows the recorded level per unit of stimulus level, with no normalisation. Use it to compare the actual sensitivity of two speakers, or to see how much level a correction is giving away.
- "dB SPL" shows the estimated sound pressure during the measurement. This needs both an SPL calibration and the run's stimulus level, so it is offered only when the folder's manifest supplies them (Section 6.4.4), and greys out otherwise. It is exact for a sweep run and approximate for noise, the stimulus level being defined for a sine.

7.2.2 Frequency response or impulse response

The "View" selector swaps the frequency plot for a time-domain view of the same speaker. "Impulse response" renders the design at the current "FIR length":

- "measured" is the averaged response rendered back to the time domain;
- "corrected" is what the loudspeaker is predicted to do once the exported filter is loaded, following the "Phase" setting;
- "+ sub (as measured)", with a subwoofer loaded, is the corrected main summed with the subwoofer at the level it was measured, with its *polarity* and the "Mains delay" applied.

Time zero is placed at the centre of the buffer, where a *linear-phase* correction puts its main peak. This is the quickest way to see what the length and phase settings are really doing. It shows how much the corrected response rings, whether the FIR is long enough for the correction to decay inside it rather than being truncated, and how much pre-ringing a linear-phase filter introduces. Changing "FIR length" or "Phase" updates it immediately.

7.2.3 The raw impulse view, and the reverberation time

"Impulse response (raw)" shows something else: the measurements themselves, one trace per microphone position, the whole analysis window long, with no smoothing, no band limiting and no correction. The direct sound sits at time zero, the propagation delay having been taken out when the files were loaded (Section 7.1.4).

The "dB" button puts the amplitude on a decibel axis, which this view turns on for you, because what it is for is the part of the response a linear axis flattens against zero: the reverberant tail, the noise floor under it, and how far apart the positions drift once the room rather than the loudspeaker is doing the talking. The mouse wheel moves the floor.

T60. Beside the selector, the read-out gives the room's reverberation time, and hovering it lists the octave bands it was measured in. Each position is estimated on its own, from its own raw response band-limited to the octave, by Schroeder backward integration with the line fits of ISO 3382 (T30 where the decay reached -35 dB, T20 where it only reached -25 dB, both extrapolated to 60 dB); the positions are then averaged as ISO 3382 averages them, by taking the mean of the times. The single figure is the usual mid-frequency one, the mean of the 500 Hz and 1 kHz octaves. Section 4.2 is what it is there for.

Three things it will not do, each for the same reason — a decay can only be read from a response that still has one in it:

- It is not measured on the smoothed curves. Smoothing is a convolution in frequency, so it is a multiplication in time by a window of about $1/\Delta f$: 1/6 octave is 43 ms at 200 Hz and 4.3 ms at 2 kHz. On a synthetic 0.40 s room the smoothed response reads 0.28 s to 0.30 s broadband and gives no readable decay in any octave band at all. What would be left to measure is the smoothing.
- It is not measured on the average. The average takes its magnitude from a power mean and its phase from a complex mean (Section 7.1.4), which is right for what it is for and means the two no longer describe one response: transformed back, its energy does not even fall monotonically. Averaging the *times* instead is what the standard asks for anyway.
- It needs the swept-sine estimator (Section 7.1.5). The Welch estimate of a swept measurement smears the response in time: the same 0.40 s room reads 0.63 s to 0.66 s that way, and a longer sweep does not improve it, where the sweep deconvolution reads 0.399 s. In Welch mode the read-out says so rather than reporting a number it cannot stand behind.

A last caveat that is the room's rather than the plugin's: the estimate is only as long as the analysis window lets it be. At 48 kHz a 65 536-sample window holds 1.37 s, which is ample for a control room and short for a church.

7.3 Fractional-octave smoothing

Smoothing is applied both to the displayed curves and to the average the correction is derived from. It averages the complex response, magnitude and phase together, over a band a fixed fraction of an octave wide around each frequency, so it is narrow in the bass and wide in the treble. That matches the ear's roughly logarithmic frequency resolution and keeps the correction from chasing the dense, narrow interference structure of the room [14]. The available fractions are Off, 1/24, 1/12, 1/6, 1/3, 1/2 and 1 octave.

7.3.1 Frequency-dependent smoothing

The "Smooth LF/HF" controls set two fractions independently, 1/6 octave each by default. The low-frequency one applies at and below 100 Hz, the high-frequency one at and above 10 kHz, and the fraction is blended smoothly in between (Section B.4). Setting both to the same value gives uniform smoothing. This lets you correct the bass in detail and the treble only broadly, in a single correction.

7.3.2 Why smoothing matters, and a recommendation

Do not try to correct a raw or lightly-smoothed response. An unsmoothed in-room measurement is full of sharp, deep nulls and peaks produced by reflections and standing waves. These narrow features have three properties that make inverting them counter-productive:

- They are largely non-**minimum-phase**. These are true cancellations that no magnitude or phase filter can undo, and attempting it only demands enormous boosts that the boost ceiling (Section 7.4) then has to claw back, wasting headroom [17, 16].
- They are position-dependent. A null at the measurement point has moved or vanished a few centimetres away, so "filling" it corrects one point while degrading every other listening position.
- Inverting a narrow notch produces a long, high-*Q* filter whose ringing (pre-/post-echo in the rendered **FIR**) is itself audible.

The correction here is derived from the smoothed complex average, so the smoothing setting is your primary control over how much of this fine structure the inverse is even allowed to see. Combined with the **spatial averaging** of Section 7.1.4, which already cancels much of the position-dependent **comb filtering**, enough smoothing keeps the correction on the broad, reproducible

trends of the loudspeaker-room response, exactly the part that is stable across the listening area and safe to invert.

Practical guidance.

- Treat the corrected curve, not the raw one, as the target. It should look smooth, not a mirror image of every wiggle in the measurement. If the correction curve is spiky, increase the smoothing.
- Exploit the frequency-dependent smoothing (Section 7.3.1). It is generally safe to correct more aggressively at low frequencies, which are broad, modal and relatively position-stable, and only lightly at high frequencies. There the response is dominated by direct sound plus early reflections, so only broad trends should be corrected and narrow dips left alone [19]. A good starting point is a finer LF fraction (e.g. 1/12-1/6 octave) with a coarser HF fraction (e.g. 1/3-1 octave).
- For uniform smoothing, set both controls equal, with 1/6 octave on both as a sensible neutral baseline. If a correction sounds harsh, "phasey" or over-processed, move coarser and/or lower the "Correction level" (Section 7.4) rather than fighting individual peaks.
- Use the "Range" to keep the correction off the extreme top and bottom entirely, where even heavy smoothing cannot make the data trustworthy.
- Smoothing is your first line of [regularisation](#). The boost ceiling and the fade outside the analysis range are the safety nets, not a substitute for choosing a sensible smoothing.

7.4 Designing the correction

The correction compensates both magnitude and phase. It starts from the exact inverse of the smoothed average, normalised to its 200 Hz to 2000 Hz mean level, which would flatten the response completely. Exact inversion is only safe for the [minimum-phase](#) part of the response, because deep room nulls cannot be filled without enormous, ringing boosts [17, 16]. Four stages [regularise](#) that inverse into the correction that is actually applied, and Section B.5 gives their formulas.

- **Max boost** limits how much the correction may lift the response, from 0 to 24 dB with a default of 12 dB. The limit is approached through a soft knee rather than a hard clamp, so notches do not leave a kink. Cuts are never limited, because attenuating a peak is always safe and only filling a null is dangerous.
- **Low-frequency slope.** The correction rolls off below 30 Hz like a second-order high-pass, so it does not chase the woofer's natural roll-off by boosting subsonic content where the speaker produces only noise. This stage has no control.
- **Correction level** morphs from no correction at 0 to the full correction at 1, the default, scaling magnitude and phase together. Intermediate values are useful when full inversion sounds over-processed.
- **Range** sets the band the correction acts on, 20 Hz to 20 kHz by default, with editable pre-sets. Outside it the correction fades back to unity over half an octave, so noise and content the loudspeaker cannot reproduce are neither inverted nor written into the exported filter.

7.5 Subwoofer phase alignment

Where a main and a subwoofer overlap around the [crossover](#), their outputs add as complex pressures. The sum is only flat if the two are in phase through the crossover region, otherwise it shows a dip, or a deep cancellation if they are in anti-phase [20]. SuperMoTo can steer the corrected main's phase onto the subwoofer's so the pair sums coherently, without changing either magnitude.

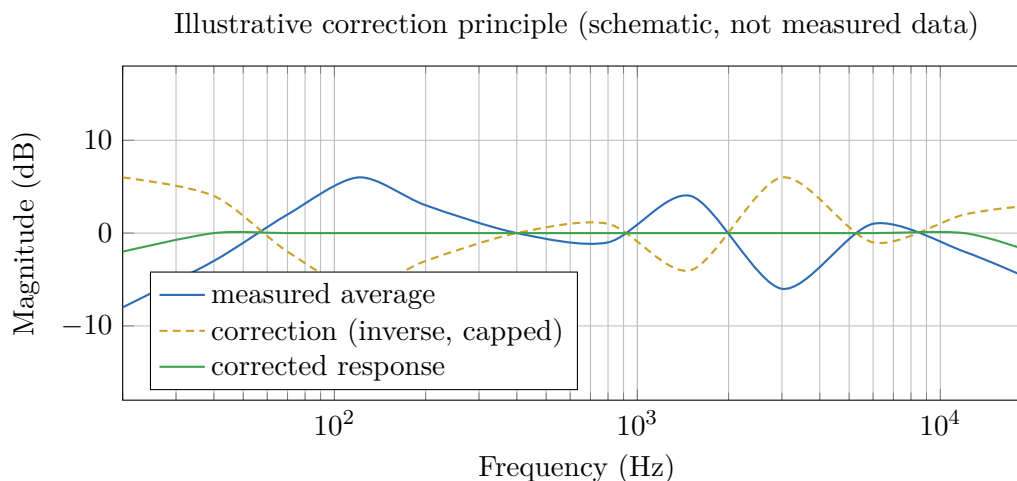


Figure 7.2: Schematic of the correction principle: the correction (gold, dashed) is the regularised inverse of the smoothed measured average (blue); applying it yields an approximately flat corrected response (green). The boost ceiling prevents the correction from fully inverting deep notches; the low-frequency slope rolls the correction off below the target corner. *This plot is illustrative, not real measurement data.*

7.5.1 Setting it up

1. Load the main's measurement set as usual.
2. Click "Load sub measurements..." and select the subwoofer set, measured at the same positions and loaded in the same order as the main's. Each subwoofer measurement is paired with the main measurement at the same position, which preserves their relative timing (Section B.6).
3. Set the "Crossover" to the crossover frequency of the system, 80 Hz by default. Below it the main's phase is steered onto the subwoofer's, and the steering is released over the octave above, so the main keeps its own flat-phase correction in its passband.
4. Tick "Invert sub" if the subwoofer is wired, or behaves, out of [polarity](#). The design then assumes an inverted subwoofer, so also turn on "Phase inv." on the subwoofer's output (Section 3.1.2). This pane does not know which output that is and leaves it to you.
5. Tune the "Mains delay", described below, and export.

7.5.2 The Mains delay control

The subwoofer usually lags the mains by several milliseconds, its own [group delay](#) added to any difference in path or processing latency. Folding all of that into the phase correction would demand a long, low-frequency-heavy [FIR](#). Instead the "Mains delay" slider, -40 to 40 ms, declares a bulk delay that you apply to the main output(s) with the per-output delays in the matrix, and the correction only carries the residual. A positive value delays the main(s) and a negative one delays the subwoofer. When it matches the real offset, the corrected main's phase is already flat through the crossover, which gives a short, well-behaved FIR. The corrected-phase curve visibly flattens as you approach the right value.

On loading the subwoofer set, SuperMoTo estimates the offset from the slope of the subwoofer's phase around the crossover and shows it as a recommendation. It is not applied automatically. You have to move the slider, and the on-screen message then states the exact delay to dial in, and on which output. Appendix F explains why the phase slope is the right measure for a subwoofer, not the arrival time.

Applying it with the FIR. To skip the manual matrix step, tick "Apply bulk delay". "Export correction IR" then writes the delay onto the assigned output at the same time it loads the FIR, so the bulk delay and the residual correction land together. Output delays cannot be negative, so a negative value is clamped to zero, and in that case you delay the subwoofer's output by hand. This is the per-speaker measured route of Section 5.1. Each main's exported delay equals its path-length term, so all mains, and the subwoofer, arrive together at the centre.

The alignment only exists in the [linear-phase](#) rendering. With "Min phase" selected the exported filter carries no phase steering, and "Crossover", "Invert sub" and "Mains delay" have no effect on it (Section 7.6).

7.6 Phase type and latency

Both exports turn a complex spectrum into a finite [impulse response](#) of the chosen "FIR length", from 256 to 65536 samples, 4096 by default. Longer filters reach lower frequencies, at the cost of more latency in linear phase. The "Phase" selector chooses how the correction is rendered:

- **Linear phase**, the default, keeps the correction's magnitude and phase, including the subwoofer alignment. How far up the phase is corrected is set by "Limit phase", below. Its energy is centred in the filter, which adds a latency of half the FIR length on that output. The matrix's [inter-output latency compensation](#) absorbs it automatically so all outputs stay time-aligned (Chapter 3).
- **Min phase** rebuilds the filter from the correction's magnitude alone, as a [minimum-phase](#) filter (Section B.9.2). The magnitude correction is identical, and the filter adds essentially no latency, which suits monitoring while tracking. It gives up the [excess-phase](#) flattening and the subwoofer alignment, since a minimum-phase filter cannot impose an arbitrary phase.

The latency compensation reads each FIR's actual peak position, so it recognises both renderings without any setting. It aligns every fed output to the longest engaged FIR, however, so the low-latency benefit only appears when all engaged outputs use minimum-phase FIRs or none. A single [linear-phase](#) FIR in the active configuration pulls the rest back up to its delay.

Limit phase. Next to the "Phase" selector, and on by default, "Limit phase" restricts the linear-phase correction to where the subwoofer alignment needs it. The designed phase is kept in full up to twice the "Crossover", then faded out over the octave above; from four times the crossover up, the filter is exactly the minimum-phase one. Measured through the whole system on a ten-position campaign, it took 14 dB to 24 dB of [pre-echo](#) out of every octave from 500 Hz to 8 kHz, bringing each to within 2 dB of minimum phase, and left the main-subwoofer summation unchanged (Section B.8).

Turning it off corrects the phase at every frequency where the positions agree. Above the crossover region that brings no timing improvement anyone can hear: on that campaign the median listening position was already time-coherent to 0.2 ms from 250 Hz up without any correction. It does put energy ahead of the direct sound, heard as a short pre-reverberation before impacts. The read-out beside "FIR length" turns amber and says so while the switch is off. "Limit phase" has no effect in "Min phase", which keeps no phase of its own. Without a subwoofer the band still follows the "Crossover" setting, which then has no other effect.

7.7 Exporting

- **Export IR** saves the measured response itself, the delay-aligned complex average, band-limited to the analysis range so out-of-band room and microphone noise does not leak into the file.

- **Export correction IR** renders the correction to a mono 32-bit float wav with the chosen "FIR length" and "Phase", and can load it directly onto the output selected in "Assign to". In linear phase it adds half the FIR length of delay on that output, which the [inter-output latency compensation](#) absorbs. In [minimum phase](#) it adds essentially none.

Chapter 8

Group analysis and multi-speaker alignment

8.1 Overview

Chapter 7 designs one speaker's correction at a time. Once a rig has more than a couple of drivers, two things become impractical to do by hand. One is comparing every speaker's measured arrival time to work out the delays that time-align them all. The other is repeating the whole load, tune and export cycle once per speaker with the same settings. Group analysis addresses both. It holds one group of speaker analyses, plus one shared subwoofer, side by side, shares the correction-design settings across all of them, computes the relative delays that align the whole group, and exports every speaker's correction filter in one step.

For each speaker, Group analysis runs exactly what the single-speaker Analysis pane does to one measurement set, namely **Welch** or Farina's method estimation (Section 7.1), fractional-octave **smoothing** (Section 7.3.1), correction design (Section 7.4) and **FIR** rendering (Section 7.6). Nothing about the underlying DSP differs. What Group analysis adds is the delay computation across the whole group and the batch apply/export step.

8.2 Loading measurements

- **Speakers** sets how many drivers to align (1 to 16). Each gets its own row with a "Load..." button, a file-count / measured-delay read-out, and an output assignment combo. "Load..." takes the same multi-position, stereo sent/recorded measurements as Section 7.1.4, one file per microphone position, at the same positions across every speaker. Whichever route the files come in by, the folder they sit in is read with them: if it holds a **measurement.xml** manifest, its embedded microphone calibration is adopted (Section 7.1.2) and a set measured with a sweep is analysed by deconvolution rather than by the Welch estimate (Section 7.1.5), exactly as "Load measurement folder..." and the Analysis view do. Since both are settings the whole group shares, adopting a transfer-function method that differs from the one in force re-analyses every set already loaded.
- A **Sub** switch (top row, next to "Speakers") turns subwoofer handling on or off for the whole group. When it is off, the Sub row is disabled and the subwoofer is left out of alignment and export entirely (Section 8.7). One dedicated "Sub" row takes the shared subwoofer measurement set, at the same microphone positions as the speakers.
- Past six speakers the rows lay out in two columns instead of one long scrolling list, so a large rig stays legible.



Figure 8.1: The Group analysis view: speaker count and Compute alignment / Apply & export controls; the shared correction-design row; the per-speaker rows (Load..., measured delay, output assignment) and the dedicated Sub row; the Preview selector and the shared transfer-function plot.

8.3 Loading a whole measurement folder in one step

Loading every speaker (and the sub) by hand with individual "Load..." clicks works, but is tedious for a large rig and depends on picking files in the right order in each dialog. "Load measurement folder..." does the whole thing in one step. Point it at a folder written by the Calibration view's folder-based capture (Section 6.4.1) and it reads that folder's `measurement.xml` manifest to find which channels were measured and which one is the subwoofer. It then sets "Speakers" to match and loads every speaker's files, and the sub's, sorted by position rather than by file-dialog order. Each channel's output assignment is set automatically to that same channel number. The "Sub" switch is set to match the folder, on if it contains a subwoofer measurement and off otherwise.

Each subwoofer file is paired with the speaker file of the same run, because every channel measured in one run was measured at the same microphone position. This holds even when a speaker missed a run, where the position numbers in the file names, counted per channel, drift apart. Files that share no run with the sub, a subwoofer measured in runs of its own for instance, are paired in the order they were measured.

8.3.1 Choosing the measurement runs

"Runs..." next to "Load measurement folder..." lists the runs of the loaded folder, one row per run, with its number, time, mode, signal, the channels it measured and its comment (Section 6.4.1). Unchecked runs are left out of every speaker and of the sub, and "OK" re-analyzes the group in the background. If "Compute alignment" had been run, it runs again once the analysis is done, so the delays and trims on the rows always match the runs in use. "Cancel" changes nothing, and "OK" is refused while a selection would leave a speaker or the sub with no measurement. When some runs are left out the button reads "Runs 5/7...", and the report lists them with their time and comment.

Runs measured in "FIR" mode start unchecked, since they record the speaker through its correction rather than the bare speaker a correction is designed from. Check one only to compare. Verification runs are often better kept in a folder of their own and opened in the single-speaker Analysis view. Files that no run lists, added to the folder by hand for instance, share one row of their own.

The selection belongs to the loaded folder. Loading another folder starts again from the default, and loading a row or the sub by hand withdraws "Runs..." until the next folder load, because applying a selection would replace those files. A folder without a `measurement.xml` has no run log, so "Runs..." stays disabled for it.

8.4 One set of correction settings for the whole group

The **Welch** window, the low/high-frequency **smoothing**, the correction level, the max boost, the **FIR** length, the phase type and "Limit phase", the analysis range, the **crossover** and the sub **polarity** are shared. They form one "tone" applied to every speaker in the group, and to the sub apart from the analysis range (Section 8.7), using the same controls described in Sections 7.1-7.6. A "Preview" selector picks which speaker's curves, or the sub's, the single plot at the bottom shows while these are tuned. Switching it never changes what is computed, only what is displayed. The "TF" estimator (Section 7.1.5), the mic calibration source (Section 7.1.2), and the "View" and "Level" display selectors (Sections 7.2.2 and 7.2.1) behave exactly as in the single-speaker Analysis view, the first two applying to every engine in the group and the last two following the previewed speaker.

8.5 Computing the alignment

"Compute alignment" reads the propagation delay of every loaded speaker and of the sub, using the same impulse-response-peak estimate Section B.2 describes for a single speaker, and proposes the delay that time-aligns the whole group. The most-distant driver is left at 0ms and every other driver is pushed back by the difference, so every proposed delay is non-negative and everyone arrives together at the reference position. The result is shown next to each row and is what "Apply & export" later writes onto the corresponding output.

For any speaker that also has a subwoofer measurement loaded (i.e. whose own row was analyzed together with the Sub row's files), the newly computed delay is fed back into that speaker's own subwoofer phase alignment (Section 7.5.2) so the correction's **crossover all-pass** is designed for the delay that will actually be applied, not the raw, unaligned one. What is fed back is the speaker's delay relative to the subwoofer. The Sub row also carries a trim that offsets the subwoofer against the whole group, and a "Use x-over" button that sets it from the crossover-band phase slope. Appendix F explains both, and why the default trim of zero means pure arrival-time alignment.

8.6 Level matching between speakers

Speakers of different models (or the same model driven by different amplifier gains) rarely play at the same level for the same input. "Compute alignment" therefore also derives a suggested level-matching trim per speaker, shown next to each row's delay and recorded in the report.

The level estimate is the mean power of each speaker's corrected response, over the band 500Hz to 2000Hz. The corrected response is the **spatially averaged transfer function** (Section 7.1.4) multiplied by its designed correction, i.e. the level that speaker will actually play at once its **FIR** is applied. That mid-band restriction follows SMPTE ST 2095-1 which specifies band-limited pink noise over exactly 500Hz to 2000Hz as the calibration signal for setting channel levels [6], and CTA-2034-A rates loudspeaker sensitivity over a similar mid-band average

(300 Hz to 3000 Hz) [7]. The reasoning is that below a few hundred hertz the reading is dominated by the room's position-dependent modal field, and by how far each model's bass extends, rather than by channel gain, while above a few kilohertz rising loudspeaker directivity makes the level a function of toe-in and microphone orientation [19]. In between, the estimate is robust.

The quietest speaker is the reference, so every suggested trim is ≤ 0 dB and attenuates the louder channels down to it, which preserves headroom. This is the same convention as the delays, where the most-distant driver is the one left untouched. The gain suggestion is informational only. "Apply & export" never writes it, so an intentional level offset, such as a rear-channel shelf you set by ear, is never silently overwritten. To apply one, enter it in that output's "Trim" in the matrix view (Section 3.1.2).

The subwoofer gets no trim suggestion, because the matching band sits above its passband and a sub's level against the mains is a [crossover](#)-balance decision made by ear or by an SPL measurement, not a sensitivity-matching one.

8.7 The subwoofer gets no correction filter

Group analysis never designs a correction [FIR](#) for the subwoofer. Above a subwoofer's real passband, a broadband measurement (sent full-band, recorded through a driver that reproduces only the low end) is just noise. The sub's own analysis range is therefore automatically capped at 300 Hz regardless of the shared "Range" control, so even its diagnostic-only, never-exported preview curve stays meaningful. Only two settings are ever written to its assigned output, the computed time-alignment delay (Section 8.5) and the [polarity](#).

The polarity follows "Invert sub" in both directions, since every speaker's correction was designed with the subwoofer at that polarity against its "Dry" measurement, and "Dry" bypasses the output settings (Section 6.1). It is written on the output, so it holds for every configuration that feeds the subwoofer and survives rebuilding a decoder with the Config tool. Frame and output flips multiply, so leave the frames feeding the subwoofer at normal polarity, or an inverted frame turns it back (Section 3.1.2).

Not every rig has a subwoofer, so handling it at all is optional. While off, the "Sub" switch (Section 8.3) leaves any already-loaded sub measurement in place but excludes it from "Compute alignment" and "Apply & export", so turning it back on later resumes using whatever was loaded before.

8.8 Applying and exporting

"Apply & export..." asks for a destination folder. For every speaker assigned to an output, it then renders and writes the correction IR (Section 7.6) into that folder, and writes the computed delay together with the [FIR](#) path onto that output's settings. Those settings are global, exactly as with the single-speaker Analysis pane, because an output's processing is shared by every configuration A-F (Section 3.1.2). The sub, if assigned to an output, gets its delay and its polarity written (Section 8.7).

A `report.md` is written alongside the exported [impulse responses](#). It records the group's shared settings and, for each speaker and the sub, the full path of every measurement file used, the measured and applied delay, the corrected in-band level and suggested level-matching trim (Section 8.6), which output it was assigned to, the exported file name or, for the sub, the polarity written, or why nothing was written, for instance because it was not measured or not assigned to an output.

8.8.1 The export prefix

Everything a run writes is named from the "Prefix" field, as in `<prefix>_speaker1_correction.wav`, `<prefix>_report.md` and so on. Leave it empty and the names have no prefix at all.

The point is that several alignments, for different rooms, different speaker sets or successive attempts at the same rig, can then share one folder instead of overwriting each other. That matters more than tidiness. A second run into the same folder with no prefix replaces the first run's impulse responses, including ones already assigned to outputs and in use. The field is pre-filled from the measurement folder's name when you load one, and never overwrites a prefix you have typed yourself.

8.8.2 Figures in the report

With "Figures" enabled, the report embeds two plots per measured speaker. The first is its frequency response, with the measurements, average, correction, corrected response, the sub curve and, for a sweep measurement, the distortion traces. The second is its impulse response, measured and correction, as the "View" selector shows them (Section 7.2.2). They are rendered from the same display components the pane itself uses, so a figure looks exactly like what you saw on screen.

The PNGs go into `<prefix>_figs/` beside the report and are referenced relatively, so the report folder can be moved, copied or zipped and the markdown still resolves. Rendering adds a few seconds to a large group, which is why it can be switched off.

8.9 Background processing

Loading a speaker's measurement set, re-analyzing the whole group after changing the [Welch](#) window, and "Apply & export" all run on background threads rather than blocking the editor. A progress bar appears, and every control that could otherwise touch an analysis engine mid-computation is disabled until the batch finishes. This keeps the editor responsive even with long, multi-position measurement sets across many speakers.

Chapter 9

Building from source

SuperMoTo is CMake based and mirrors the MechanOdd project layout. JUCE and the FxmeJuceTools user module are expected as siblings of the repository:

```
../JUCE/                               juce-framework/JUCE checkout
../JUCE/usermodules/FxmeJuceTools/      symlink to FxmeJuceTools/module/FxmeJuceTools

git submodule update --init --recursive # FxmeFX + WDL
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
```

The build produces VST3, AU and Standalone formats, all with a fixed 32 in/32 out discrete bus. The active matrix size, 2 in / 8 out by default, is independent of the bus and set with the plugin's own "Inputs" and "Outputs" selectors. Continuous-integration builds for Linux, macOS (universal) and Windows are provided by the GitHub Actions workflow [.github/workflows/release.yml](#), triggered by pushing a version tag (vX.Y.Z).

9.1 License

SuperMoTo is distributed under the GNU Affero General Public License version 3 or later, or under commercial terms from the author for holders of a commercial JUCE licence:

SPDX-License-Identifier: AGPL-3.0-or-later OR LicenseRef-FXME-Commercial

JUCE 8 is itself dual-licensed (AGPLv3, or a commercial JUCE licence) and the plugin mirrors that shape: whichever route you are already on for JUCE, the same route works here. A commercial grant from the author covers this code only and is not a JUCE licence, which must be obtained separately from Raw Material Software.

A few parts are framework-free and stay under the GNU Lesser General Public License version 3 or later, so they can be used without JUCE: the Ambisonics decoding helper (`Source/Dsp/AmbisonicsDeco`), the microphone calibration script (`Tools/mic_cal.py`), this manual and the paper. The shared FX-Mechanics code in the `lib/FxmeTools` submodule is split the same way, its framework-free `core/` half being LGPL as well, and the WDL convolution engine it carries is zlib-licensed (Cockos Incorporated). The repository's `LICENSE.md` states which licence applies to which file, and every file names its own in an SPDX header.

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Appendix A

Worked examples

This appendix teaches SuperMoTo by building real setups, from one corrected pair of speakers to an Ambisonic rig. Each example adds an idea to the ones before it, so Examples 1 and 2 are worth reading in order. The later ones can be read on their own and point back to the steps they reuse. Table A.1 lists them with what each one teaches.

Setup	What it teaches
1 A stereo pair	The measure, correct and verify loop, one speaker at a time
2 A stereo pair and a subwoofer	Bass management, subwoofer phase alignment, checking the whole chain
3 Several speakers at once	Group analysis, time alignment across a rig, level matching
4 Two pairs, a sub and headphones	Several configurations over the same corrected outputs
5 5.1, switchable to 2.1, 5.0, 2.0	Listening modes as configurations, frames added by hand
6 Eight monitors and a sub in Ambisonics	The decoder, radius compensation and measured alignment together

Table A.1: The worked examples.

A.1 Three things to know first

Routing and loudspeakers are kept apart. A **configuration** (A-F) stores the routing, meaning which inputs feed which outputs, with a gain, a polarity flip, a small 2-band EQ and an active state per frame (Chapter 3). Everything that characterises a physical loudspeaker (its trim, polarity, 2-band EQ, time-alignment delay and **FIR** correction) lives on the output instead, and is shared by every configuration that uses that output (Section 3.1.2). This split is what makes multi-configuration monitoring cheap. You correct and align each speaker once, then describe each way of listening as a different routing over the same corrected outputs.

One consequence is easy to trip over. The bass-management **crossover** is an output filter too, a highpass on a main and a lowpass on a subwoofer. When the Config tool applies a layout, it rewrites those filters on the layout's outputs, whichever configuration it writes the routing to. Applying a layout with "Bass management" off therefore removes the highpass from those outputs for every configuration.

Every correction follows the same loop. The first measurement sees the bare speaker, the analysis turns it into a correction on the output, and two further measurements check the result, first on the speaker alone and then through everything you actually listen to (Table A.2 and Section 6.1).

Step	Where	What is recorded or written
Measure	Calibration, "Dry"	The speaker and the room, with no processing in the path
Correct	Analysis or Group analysis	A correction and a delay, written onto the output
Check the speaker	Calibration, "FIR"	The speaker through its own trim, polarity, EQ and FIR
Check the system	Calibration, "System"	A plugin input through the whole engaged configuration

Table A.2: The loop every example goes through.

How a measurement session goes. Put the measurement microphone on a plugin input and select it in "Microphone input". Point the capsule at the ceiling and load the microphone's 90° calibration file with "Mic cal". A run measures several speakers from the same place, and this orientation is the one that is the same for all of them (Section 6.5.1). Plan seven to nine microphone positions spread over 30 cm to 50 cm around the listening seat, a few of them at different heights (Section 6.5.2). One "Run" measures every selected channel in turn with the microphone in one place. The rhythm of a session is therefore to place the microphone, click "Run", move it and click "Run" again, until every position is done. Keep one "Measurement folder" per rig, and note where the microphone was in "Run comment" (Section 6.2.6).

A.2 Example 1: a corrected stereo pair

Two monitors sit on outputs 1 and 2 and a stereo source arrives on inputs 1 and 2. This example goes once around the whole loop of Table A.2, correcting one speaker at a time.

1. **Route.** In the Config tool choose "Stereo 2.0", turn "Bass management" off (there is no subwoofer to take the low end), leave "Write to config" on A and click "Apply to configuration". The matrix view now shows two active frames, in1→1 and in2→2. Routing by hand works just as well, by double-clicking the two frames in the matrix view.
2. **Measure.** In the Calibration view choose "Dry", tick outputs 1 and 2, keep the log sweep and pick a measurement folder. Measure every microphone position as described above. Each run writes one file per output, `ch01_pos001.wav` and `ch02_pos001.wav` for the first position, and so on (Section 6.4.1).
3. **Look before correcting.** In the Analysis view click "Load measurements..." and select all the `ch01` files. The thin curves are the individual positions and the thick one their average (Chapter 7). Their spread tells you how far to trust the data. Where the positions agree the average is a property of the speaker in the room, and where they scatter it is a compromise that no filter can make right at every position.
4. **Design the correction.** The defaults are a sensible start, with 1/6-octave smoothing on both "Smooth LF/HF" controls, a "Correction level" of 1 and a "Max boost" of 12 dB. Judge the predicted corrected response rather than the correction itself. If the correction looks spiky, smooth more, typically coarser in the treble than in the bass, and if the result sounds over-processed, lower "Correction level" (Section 7.3.2). Keep "Phase" on "Linear

phase", with "Limit phase" on, unless you monitor while tracking, where "Min phase" avoids the latency (Section 7.6).

5. **Write it onto the output.** Choose "Output 1" in "Assign to" and click "Export correction IR...". The file is saved and loaded as output 1's FIR in one step. Load the `ch02` files and do the same for output 2.
6. **Check the speakers.** Back in the Calibration view, choose "FIR", select a folder of its own (verification captures are then never mixed with the "Dry" ones) and measure outputs 1 and 2 again at a few positions. Load them in the Analysis view. Their average should now look like the predicted corrected response of step 4.

With only two speakers there is nothing to align, since the matrix's [latency compensation](#) keeps both outputs in step even if their FIRs differ (Section 3.3). If the two speakers are at clearly different distances from the seat, Example 3 measures and applies the delay that brings them back together.

A.3 Example 2: adding a subwoofer

The same pair gets a subwoofer on output 3. Below the [crossover](#) the mains are rolled off and their sum goes to the sub. The new difficulty is the crossover region, where main and sub play together and only add up smoothly if they are in phase (Section 7.5).

1. **Route with bass management.** In the Config tool choose "Stereo 2.1". The "Subwoofer" row takes "Sum of mains" on output 3. Keep "Bass management" on, set "Crossover" (80 Hz is typical) and apply to A. The tool writes a highpass on outputs 1 and 2, a lowpass on output 3, and routes both inputs to the sub.
2. **Measure the three speakers together.** In the Calibration view choose "Dry", tick outputs 1 to 3, turn "Sub" on and set "Sub channel" to 3, so the sub's files are named `sub_pos001.wav` and so on. Measuring all three in each run matters, because every channel of a run was measured with the microphone at the same place, and that is what lets the analysis compare main and sub timing.
3. **Correct each main against the sub.** In the Analysis view load the `ch01` files, then click "Load sub measurements..." and select the `sub` files in the same order. Set "Crossover" to the same frequency as the Config tool, and tick "Invert sub" if the subwoofer is wired, or behaves, out of [polarity](#). Design the correction as in Example 1.
4. **Find the delay.** A message next to "Mains delay" recommends a main/sub offset, estimated from the sub's phase around the crossover. Move "Mains delay" towards it and watch the corrected phase flatten through the crossover. The message then says how much delay to add and on which output (Section 7.5.2). A delay is used rather than a longer correction because it costs nothing, while folding several milliseconds into the FIR would make it long and bass-heavy.
5. **Write the correction and the delay together.** Tick "Apply bulk delay", assign output 1 and click "Export correction IR...". The FIR and the delay land on output 1 at once. Repeat steps 3 to 5 for output 2 with the `ch02` files. If "Mains delay" had to go negative, it is the sub that needs delaying. An output delay cannot be negative, so enter that amount as output 3's delay in the matrix view instead.
6. **Match the sub's polarity.** If "Invert sub" is ticked, click output 3 in the matrix view and turn on "Phase inv.". The corrections were designed for an inverted sub, and the Analysis view does not know which output the sub is on.
7. **Check the whole chain.** In the Calibration view choose "System", which measures plugin inputs rather than outputs, tick input 1 and measure at the seat. The stimulus now runs through configuration A exactly as you listen to it, with the routing, the crossover filters, the FIRs, the delays and the latency compensation (Section 6.1). In the Analysis view, the response should pass smoothly through the crossover. A dip there means the delay or

the polarity is wrong.
 The alignment lives in the phase of the mains' corrections, so it only exists with "Linear phase". "Min phase" keeps the magnitude correction but gives the subwoofer alignment up (Section 7.6). SuperMoTo delays the sub output by itself to match the latency of the mains' FIRs, which the gold dot on output 3 shows (Section 3.3).

A.4 Example 3: several speakers at once

Examples 1 and 2 repeat the same analysis once per speaker and never compare the speakers with each other. As soon as a rig has more than a couple of drivers, or speakers at different distances that play together, Group analysis does it all in one pass (Chapter 8). It designs every correction with the same settings, measures each speaker's arrival time, and writes the delays that make them all arrive together.

1. **Measure the whole rig into one folder.** In the Calibration view choose "Dry", tick every speaker's output, and if there is a subwoofer turn "Sub" on with its "Sub channel". Measure every position.
2. **Load the folder.** In Group analysis click "Load measurement folder..." and pick that folder. "Speakers", the output of each row and the "Sub" switch are set from the folder's manifest, and each speaker's files are loaded (Section 8.3). If the folder also holds runs you do not want, a verification in "FIR" mode or a position that went wrong, "Runs..." leaves them out. "FIR" runs already start unchecked (Section 8.3.1).
3. **Tune one set of settings for everyone.** The correction controls work as in the Analysis view (Welch window, *smoothing*, "Correction level", "Max boost", "FIR length", "Phase", "Range", "Crossover", "Invert sub"), shared by the whole group. Use "Preview" to look at each speaker's curves in turn.
4. **Compute the alignment.** "Compute alignment" measures when each speaker's sound arrives at the microphone. The farthest speaker gets 0 ms and every other one is delayed by the difference, so all delays are positive and the whole rig arrives together (Section 8.5). Each row also gets a suggested trim that would bring its level down to the quietest speaker's (Section 8.6).
5. **Apply.** Check that each row's output is the one that speaker is on, and click "Apply & export...". Every speaker's correction is exported and loaded with its delay, and the subwoofer gets its delay and its polarity from "Invert sub", so no step 6 of Example 2 is needed here (Section 8.7). A `report.md` next to the *impulse responses* records what was done.
6. **Apply the level trims yourself.** The suggested trims are never written, because a level offset may be deliberate. Enter the ones you want in each output's "Trim" in the matrix view.
7. **Check the system** as in step 7 of Example 2.

The subwoofer gets no correction filter, only its delay and polarity (Section 8.7). The Sub row's trim, and "Use x-over", adjust its timing against the group when arrival-time alignment is not what sounds best through the crossover (Appendix F).

A.5 Example 4: two monitor pairs, a subwoofer and headphones

A typical control room has two stereo pairs, mains on outputs 1-2 and nearfields on 3-4, a subwoofer on output 5 and headphones on 7-8. The source is stereo on inputs 1-2. Each speaker is corrected once, and it is the configurations that describe the ways to listen (Table A.3).

1. **Correct the speakers.** Correct outputs 1-4 and align the sub to each pair as in Example 2. Each pair's delay goes on its own outputs, and output 5's lowpass and delay are shared by both. The default matrix size, 2 in / 8 out, already fits this rig.

Config	Listen on	Routing
A	Mains + sub	in1→1, in2→2, (in1+in2)→5
B	Nearfields + sub	in1→3, in2→4, (in1+in2)→5
C	Mains alone	in1→1, in2→2
D	Nearfields alone	in1→3, in2→4
E	Headphones	in1→7, in2→8

Table A.3: Five configurations over the same corrected outputs.

2. **Config A.** In the Config tool choose "Stereo 2.1", set the "Subwoofer" row's output to 5, and apply to A with "Bass management" on.
3. **Config B.** Still "Stereo 2.1", with the Left and Right rows on outputs 3 and 4, the sub on 5, and the same "Bass management" and "Crossover" as for A. Apply to B.
4. **Configs C and D.** Choose "Stereo 2.0" and apply to C with outputs 1-2, then to D with outputs 3-4, keeping "Bass management" and "Crossover" as for A. With it off, the tool would remove the mains' highpass, and configuration A would lose it too.
5. **Config E.** In the matrix view click "Edit: E" and double-click frames in1→7 and in2→8. Routing by hand touches no output filter.
6. **Use it.** Keep "Exclusive" on, so engaging one configuration from the top bar releases the others and switching is an instant A/B. With "Exclusive" off, engaged configurations sum, headphones together with the mains for instance. Once the setup is done, the collapse button (▲) shrinks the window to the output strip, or further to a mini controller with the configuration buttons, the level, "Mute", "Dim", "Mono" and output meters (Section 3.6).

Full range or bass management. The highpass that bass management puts on the mains is an output filter, so configurations C and D play the mains highpassed too, and "mains alone" loses its bottom octave. There is no separate full-range version of output 1. If a room really switches between with-sub and without-sub on the same speakers, run the mains full range and let the subwoofer only extend them below the crossover. Apply the layouts with "Bass management" off. The tool still sets the sub output's first EQ band as a lowpass at the crossover, only switched off, so click that output in the matrix view and switch the band on (Section 3.1.2). Configurations C and D are then genuinely full range. Keep the highpass form of bass management for small satellites that must never see deep bass, where "mains alone" is a fallback you rarely engage.

There is a second way, which keeps a real crossover in A and B while leaving C and D full range. A frame carries the same 2-band EQ as an output, applied to that one route before it sums into the output (Section 3.1.2), and a frame belongs to one configuration. Put the highpass on the frames feeding the mains in A and B, leave outputs 1-4 flat, and the same speakers play highpassed in the bass-managed configurations and full range in the others. The Config tool always writes the crossover onto the outputs, so this is a hand edit in the matrix view, and every frame feeding a main in a bass-managed configuration needs its own copy of the filter. The subwoofer's lowpass can stay on output 5, which only ever plays in A and B.

A.6 Example 5: a 5.1 system switchable to 2.1, 5.0 and 2.0

A 5.1 feed arrives on inputs 1-6 in the usual order, L, R, C, LFE, L_s , R_s , which is also the Config tool's order. The speakers are wired the same way, with the subwoofer on output 4. Four configurations cover the common listening modes (Table A.4).

1. **Correct and align the rig.** The five satellites play together, so their arrival times matter as much as their responses. Measure the six speakers into one folder with "Sub channel" set to 4, and correct and align them with Group analysis (Example 3).

Config	Mode	Routing
A	5.1	in1→1 ... in6→6, (in1-3 + in5-6)→4
B	2.1	in1→1, in2→2, (in1+in2+in4)→4
C	5.0	in1-3→1-3, in5-6→5-6
D	2.0	in1→1, in2→2

Table A.4: One corrected 5.1 rig, four listening modes.

2. **Config A.** In the Config tool choose "5.1" and apply to A with "Bass management" on. The tool grows the matrix to 6 in / 6 out, highpasses the satellites and lowpasses the sub. The "Subwoofer / LFE" row feeds the sub from the LFE input only. To send the satellites' bass there as well, which bass management is for, click "Edit: A" in the matrix view and activate the frames from inputs 1, 2, 3, 5 and 6 to output 4, lowering their gain to balance the sum by ear or with the SPL meter.
3. **Config B.** Choose "Stereo 2.1" with the sub on output 4, same "Bass management" and "Crossover", and apply to B. Then activate frame in4→4 by hand, so the LFE still reaches the sub.
4. **Configs C and D.** The tool has no 5.0 layout, so apply "5.1" to C and deactivate its frame into output 4 with a double-click. For D, apply "Stereo 2.0" to D. Keep the same "Bass management" and "Crossover" for both.
5. **Use it.** Switch modes from the top bar with "Exclusive" on, and check each mode in "System" mode as in Example 2, measuring the inputs that mode uses.

The paragraph on full range and bass management in Example 4 applies unchanged. With full-range satellites, run them full range and let the sub extend them, and 5.0 and 2.0 keep their low end. With small satellites keep them highpassed, and treat 5.0 and 2.0 as reduced-bass fallbacks.

A.7 Example 6: an Ambisonic rig with eight monitors and a subwoofer

Eight monitors surround the listener, for instance as a cube with four at ear height and four above, and a subwoofer completes the rig. The Ambisonic (AmbiX) feed arrives on inputs 1-4, the monitors are on outputs 1-8 and the sub on output 9. Eight speakers decode first order robustly, since it needs at least four. Second order needs nine, so this rig stays at first order.

1. **Describe the rig.** In the Config tool choose "Ambisonics", with "Order" 1 and "Speakers" 8. Enter each speaker's azimuth, elevation, radius and output, starting from the default cube (Section 5.1).
2. **Build the decoder.** Keep "Bass management" on, set "Crossover" and apply to A. The tool writes the harmonic→speaker decode frames, feeds the omnidirectional W channel to the sub on the next free output (9 here), highpasses the eight monitors and lowpasses the sub. It also compensates the radii on the outputs, delaying and attenuating the closer speakers so everything arrives together at the centre, and grows the matrix to fit.
3. **Correct and align by measurement.** The radius compensation is a geometric estimate, and a measurement does better. Measure the eight monitors and the sub into one folder ("Sub channel" 9) and use Group analysis as in Example 3. With the capsule pointing at the ceiling, the 90° calibration is only approximate in the treble for the four upper monitors, so keep the high-frequency smoothing coarse (Section 6.5.1). "Apply & export. . ." replaces the geometric delays with the measured ones and adds the corrections. The decode, the corrections, the delays and the [latency compensation](#) all run in the same engine, so nothing is counted twice.

4. **Rebuild without losing the measurement.** If you later edit the decoder (a speaker moved a few degrees, say), turn "Write radius delay" off before applying again, so the measured delays stay on the outputs. "Write radius gain" does the same for the output trims.
5. **Irregular rigs.** When radii or angles are uneven, prefer "Load IEM decoder..." to import an AllRAD matrix designed for that layout in the IEM AllRADecoder, instead of the built-in sampling decoder. The radius toggles still apply.
6. **A quick check configuration.** Optionally keep configuration B routing W to a front pair, for a quick non-periphonic comparison from the top bar.

Larger rigs. Higher orders need more speakers and more channels. Third order takes 16 B-format inputs and at least 16 speakers, so a 16-speaker dome with a sub uses 16 inputs and 17 outputs, well within the fixed 32 in / 32 out bus (Section 1.4). The steps are the same, with "Order" 3 and "Speakers" 16. The 32 channels of the plugin could theoretically allow up to order 4. Extending config tool to manage higher orders is considered for future versions. The background behind the decoder (spherical harmonics, orders, the AmbiX convention and the $\max-r_E$ weighting) is in Appendix G.

Appendix B

Analysis and correction theory

This appendix gives the signal processing behind the Analysis view of Chapter 7, in the order the data flows. Each **transfer function** is estimated, the positions are aligned and averaged, the average is smoothed, the correction is designed and optionally aligned to a subwoofer, and the result is rendered as an **FIR**.

B.1 Welch estimation

For each file the plugin estimates the loudspeaker-room **transfer function** $H(f) = Y(f)/X(f)$, where x is the emitted stimulus (channel 1 of the file) and y the captured microphone signal (channel 2). If a microphone calibration is in effect, it is divided out of H at this stage (eq. (7.1)).

B.1.1 The H_1 estimator

A direct bin-by-bin division Y/X is unusable in a room. It is dominated by measurement noise wherever X is small, and a single FFT gives an estimate with 100 % standard error. SuperMoTo instead uses the classical **Welch** averaged-periodogram approach [1]. The recording is cut into overlapping segments, each is windowed and transformed, and the auto- and cross-spectra are averaged over the L segments,

$$\hat{P}_{xx}(f) = \frac{1}{L} \sum_{\ell=1}^L |X_{\ell}(f)|^2, \quad \hat{P}_{xy}(f) = \frac{1}{L} \sum_{\ell=1}^L X_{\ell}^*(f) Y_{\ell}(f). \quad (\text{B.1})$$

The transfer function is then formed as the H_1 estimator

$$\hat{H}(f) = \frac{\hat{P}_{xy}(f)}{\hat{P}_{xx}(f)}. \quad (\text{B.2})$$

H_1 is the least-squares estimator that is optimal when the noise is on the output (the microphone path), the realistic case for an acoustic measurement, and it rejects that noise as more segments are averaged. (Its companion $H_2 = P_{yy}/P_{yx}$ assumes input noise. Both bracket the true response, and they coincide only when the **coherence** is unity.) The derivation and noise properties of H_1/H_2 are standard material in [2, 3, 4].

B.1.2 Windowing, overlap and resolution

Each segment is weighted by a Hann window before the FFT, to control spectral leakage [5]. Consecutive segments overlap by 50 %, the standard choice that makes the Hann-windowed segments use the recording almost uniformly (Welch's WOSA scheme). The window length W , from 16384 to 262144 samples, sets the frequency resolution

$$\Delta f = \frac{f_s}{W}, \quad (\text{B.3})$$

about 0.67 Hz at $f_s = 44.1$ kHz with the default $W = 65536$. Bins where $\hat{P}_{xx}$ is essentially zero are **regularised** by a small term $\varepsilon = 10^{-10} \max_k \hat{P}_{xx} + 10^{-30}$ added to the denominator of (B.2), so silent bins do not blow up.

B.2 Delay removal and complex averaging

Each microphone position is at a different distance from the speaker, so each $\hat{H}$ carries a different bulk propagation delay, a linear phase term $e^{-j2\pi f\tau}$. Averaging the complex responses without removing it would let the position-dependent phase ramps cancel each other and destroy the result. The plugin therefore estimates the delay τ of each measurement from its **impulse response** (inverse FFT of $\hat{H}$), taken as the position of the peak of $|h(t)|$. A peak in the second half of the buffer is interpreted as a small negative delay that wrapped around. The linear phase is then removed,

$$\hat{H}'(f) = \hat{H}(f) e^{+j2\pi f\tau}, \quad (\text{B.4})$$

which time-shifts every measurement so its main arrival sits at $t = 0$. This is the excess-delay normalisation used in swept-sine room measurement practice [8]. Appendix F discusses what this arrival-time estimate measures, and where it fails.

The delay-aligned responses of all N positions are then averaged, and the magnitude and the phase of that average come from different means, for reasons that are worth separating.

The **magnitude** is a power mean,

$$M(f) = \sqrt{\frac{1}{N} \sum_{n=1}^N |\hat{H}'_n(f)|^2}, \quad (\text{B.5})$$

because $|\overline{H}|$ is not $\overline{|H|}$. Above the room's transition frequency the positions no longer agree on phase, and the modulus of a complex mean of N near-random phasors reads about $1/\sqrt{N}$ low. For instance, on a ten-position set, one gets 0.2 dB at 200 Hz but 12 dB at 13 kHz, at bins where every individual curve is flat. Inverting that would boost a cancellation that exists at no microphone position. Equation B.5 is the spatial average of the sound field, and averaging out the position-dependent **comb filtering** caused by reflections is what makes several positions worthwhile in the first place [19, 16].

The **phase** comes from the complex sum $S(f) = \sum_n \hat{H}'_n(f)$, provided that this sum is statistically meaningful. The validity and reliability of this sum are quantified by the vector coherence:

$$R(f) = \frac{|S(f)|}{N M(f)} \in [0, 1], \quad (\text{B.6})$$

which is unity where every position agrees and zero where they are independent. N unit phasors that agree on nothing still sum to $\sqrt{N}$, so R has a floor of $1/\sqrt{N}$ that means nothing. Removing it gives the weight actually used,

$$w(f) = \text{clamp}\left(\frac{NR^2(f) - 1}{N - 1}, 0, 1\right), \quad (\text{B.7})$$

so $w = 0$ reads "no better agreement than chance". The average is then the power-mean magnitude carried by a direction blended between the one the positions voted for and the minimum-phase direction of that same magnitude,

$$\bar{H}(f) = M(f) \frac{b(f)}{|b(f)|}, \quad b(f) = w \frac{S}{|S|} + (1 - w) e^{j\varphi_{\min}(f)}, \quad (\text{B.8})$$

with $\varphi_{\min}$ the minimum-phase argument of M , obtained from the real cepstrum as in Section B.9.2. The two are blended as phasors on the short arc, as the subwoofer alignment blends its own, so no phase is ever unwrapped and nothing can jump.

Why the phase needs a weight at all. Where the positions disagree, S nearly cancels, and its argument is then the direction of a residual between near-random phasors. The phase can shift by 180° between adjacent frequency bins even as the magnitude M varies smoothly. Because this abrupt phase flip leaves the magnitude unaffected, it cannot be detected by analyzing the magnitude spectrum alone. However, a [linear-phase](#) rendering exposes this transition by forcing the complex trajectory to pass through a near-zero point on the unit circle. Measured on a ten-position campaign whose right loudspeaker had $R = 0.07$ near 670 Hz, the exported filter carried a -25 dB notch of $Q \approx 500$ at 697 Hz, audible as a ringing tone, where the minimum-phase rendering of the same design (using the magnitude alone) was 2.3 dB at that frequency and showed nothing. Applying (B.8) leaves the magnitude correction completely unchanged at 0.000 dB RMS over the 200 Hz to 10 000 Hz range. Meanwhile, the crossover delay estimate shifts by merely 0.02 ms and the summation efficiency by 0.02 dB to 0.04 dB, while the notch is entirely eliminated: no bin between 650 Hz and 750 Hz falls below -0.4 dB, as in the minimum-phase rendering. This approach is also fundamentally justified: spatial variations in phase do not reflect the acoustic properties of the overall listening area. Attempting to correct these position-dependent phase differences might optimize the response for a single seat, but it will inevitably degrade performance for the rest of the audience.

What a value of R is worth. R is the mean resultant length of circular statistics, and for a wrapped-normal or von Mises spread of phases it fixes the scatter directly: the circular standard deviation is $\sigma = \sqrt{-2 \ln R}$. That is what Table B.1 reads out, beside the weight (B.7) gives at $N = 10$. The two agree where they should: at $R = 1/\sqrt{10} = 0.316$, the value N independent positions produce on their own, the scatter is 87° (phases spread almost uniformly around the circle) and the weight has fallen to 0.003. The bias correction in (B.7) is the standard small-sample correction for the squared mean resultant length [30].

R	0.99	0.95	0.90	0.80	0.70	0.50	0.32
circular sd σ (deg)	8	18	26	38	48	68	87
weight w , $N = 10$	0.98	0.89	0.79	0.60	0.43	0.17	0.00

Table B.1: What the positions’ agreement is worth. R is the vector coherence of (B.6), $\sigma = \sqrt{-2 \ln R}$ the phase scatter it implies, and w the weight (B.7) gives that phase in the average. Independent positions put a floor of $1/\sqrt{N}$ under R — 0.58 at three positions, 0.32 at ten, 0.22 at twenty — which is what the correction removes.

What it looks like on a real set. Table B.2 is R measured per octave on a ten-position campaign, both main loudspeakers. It follows the shape Appendix D predicts from the correlation length: near unity through the bass, where a 30 cm cluster samples a single correlation cell, and falling through the midrange as the positions begin to sample independently. The weight therefore lets the average’s phase through in full below a few hundred hertz and fades it out above, without anyone choosing a frequency. That settles which phase is information. Whether inverting it is worth what it costs is a separate question, and Section B.8 answers it with a frequency after all.

How rare the failure is, and why that is the problem. On that same campaign the defect occupied *one bin*. Taking the largest phase step between neighbouring bins anywhere in the analysis band, the left loudspeaker reached 11.9° and the right 154.9° , the latter at 697.8 Hz and at exactly one of the 29 632 bins in band. With (B.8) in place they fall to 8.3° and 9.7° , both at the crossover where the subwoofer alignment turns the phase on purpose, and no bin anywhere exceeds 90° . A defect that occupies one bin in thirty thousand, leaves the magnitude

octave from (Hz)	31	63	125	250	500	1k	2k	4k	8k
left main	0.98	0.99	0.90	0.89	0.79	0.67	0.50	0.67	0.46
right main	0.97	0.99	0.90	0.88	0.78	0.68	0.59	0.59	0.34

Table B.2: Vector coherence per octave band, labelled by the band’s lower edge, on ten positions spanning 45 cm by 34 cm with four of them 6 cm higher. The floor for ten independent positions is 0.32.

untouched and is plainly audible is exactly the kind that a curve on a screen will not show and a listening test will.

B.3 Synchronized swept-sine deconvolution

With the sweep stimulus, the [transfer function](#) can be estimated by deconvolution instead of by [Welch](#) averaging. The implementation follows the synchronized swept-sine formulation of Novak, Lotton and Simon [9].

B.3.1 The synchronized sweep

The stimulus is the exponential (logarithmic) sweep

$$x(t) = \sin\left(2\pi f_1 L \left(e^{t/L} - 1\right)\right), \quad (\text{B.9})$$

from f_1 to f_2 , where f_2 is the Nyquist frequency and f_1 lies a whole number of octaves below it, near 11 Hz at every sample rate (Appendix E explains why). Its instantaneous frequency is $f(t) = f_1 e^{t/L}$, so the sweep rate L is the time the stimulus takes to multiply its frequency by e , and the sweep reaches frequency f at

$$t(f) = L \ln \frac{f}{f_1}. \quad (\text{B.10})$$

What makes the sweep synchronized is that L is not taken directly from the requested duration T , but quantized so that $f_1 L$ is an integer:

$$L = \frac{1}{f_1} \text{round}\left(\frac{f_1 T}{\ln(f_2/f_1)}\right). \quad (\text{B.11})$$

The actual sweep duration is then $L \ln(f_2/f_1)$, slightly different from the T requested.

B.3.2 Deconvolution by the analytic inverse

The recorded response is deconvolved by the sweep’s analytic spectral inverse, the closed form obtained from the stationary-phase approximation of the sweep’s spectrum [9],

$$X^{-1}(f) = 2\sqrt{\frac{f}{L}} \exp\left(-j 2\pi f L \left(1 - \ln \frac{f}{f_1}\right) + j \frac{\pi}{4}\right). \quad (\text{B.12})$$

The recording is transformed, multiplied by (B.12), and transformed back. Because the inverse is known in closed form, nothing is divided by a measured spectrum. There is no ill-conditioned denominator to [regularise](#) (contrast the ε term in (B.2)), and the whole band, magnitude and phase, comes from a single sweep rather than from an average over segments.

B.3.3 Why the harmonic orders separate

A distortion product of order m at frequency f is generated while the sweep is passing f/m . By (B.10) that happens

$$\Delta t_m = L \ln m \quad (\text{B.13})$$

seconds before the linear response at the same frequency. After deconvolution, therefore, the **impulse responses** of the different orders sit at different places on the time axis. The linear impulse response lands at the system's propagation delay, and the order- m impulse response lands $L \ln m$ seconds before it, wrapping circularly towards the end of the buffer.

This is where the quantization (B.11) earns its place. With $f_1 L$ an integer, the sweep's phase at every arrival time Δt_m is an exact multiple of 2π , so the harmonic impulse responses emerge not merely separated in time but with their true phase. Deconvolving a recording whose L was not quantized (a file from another tool, say) still separates the orders in time, but their phase relationship is lost. Under the Welch estimator, by contrast, the distortion products are simply part of the measured output and end up folded into the linear estimate.

B.3.4 Extracting the linear and harmonic responses

The deconvolution uses an FFT the next power of two at least as long as the recording, and never shorter than the sweep, so the wrapped harmonic responses land far from the linear one. The first W samples of the deconvolved response hold the linear impulse response. They are transformed onto exactly the same $W/2 + 1$ bin grid the Welch path produces, so everything downstream is method-agnostic.

Harmonic orders 2 to 5 are extracted, each centred $L \ln m$ seconds before the linear peak. The extraction half-width is bounded by 0.45 times the spacing to the next order, which shrinks as $L \ln(m/(m-1))$, so neighbouring orders cannot leak into one another. An order is dropped when the sweep is too short to separate it cleanly. Across microphone positions the harmonic magnitudes are averaged in power, because distortion phase is not coherent from one position to the next and the vector average of Section B.2 would cancel the distortion instead of measuring it.

B.4 Fractional-octave smoothing

The $1/n$ -octave **smoothing** is a complex moving average, magnitude and phase together, over a logarithmic band $[k/r, kr]$ around each bin k , with $r = 2^{\text{fraction}/2}$, evaluated in $O(n)$ with a prefix sum. Working on a constant- Q grid follows Hatziantoniou and Mourjopoulos [14]. The $1/2$ -octave setting fills the large perceptual gap between $1/3$ and 1 octave.

With frequency-dependent smoothing, the octave fraction used at bin k is interpolated logarithmically in frequency between the low- and high-frequency settings,

$$\text{frac}(f) = \text{frac}_{\text{LF}} + t(f) (\text{frac}_{\text{HF}} - \text{frac}_{\text{LF}}), \quad t(f) = \text{clamp}\left(\frac{\log_2 f - \log_2 f_{\text{lo}}}{\log_2 f_{\text{hi}} - \log_2 f_{\text{lo}}}, 0, 1\right), \quad (\text{B.14})$$

with anchors $f_{\text{lo}} = 100$ Hz and $f_{\text{hi}} = 10$ kHz. This is distinct from the constant- Q widening that any fractional-octave window already has. That widening grows the window in Hz as frequency rises but keeps the octave fraction constant, whereas here the octave fraction itself varies with frequency.

B.5 Correction design

Starting from the normalised smoothed average $h = \bar{H}_s/g_{\text{ref}}$, with g_{ref} the 200 Hz to 2000 Hz mean magnitude, the ideal inverse is the complex reciprocal

$$C_0(f) = \frac{1}{h(f)} = \frac{\overline{h(f)}}{|h(f)|^2}, \quad (\text{B.15})$$

which flattens magnitude and zeroes phase at once. Four mechanisms regularise and shape C_0 into the correction that is actually applied.

1. Soft-knee boost ceiling. Rather than hard-clamping the boost, which leaves a kink at every notch, the boost portion of the correction magnitude in dB is bent towards a ceiling L (default +12 dB) with a tanh knee of width $\kappa = \min(6, L)$ dB:

$$C_{\text{dB}} \mapsto \begin{cases} C_{\text{dB}}, & C_{\text{dB}} \leq L - \kappa, \\ (L - \kappa) + \kappa \tanh \frac{C_{\text{dB}} - (L - \kappa)}{\kappa}, & C_{\text{dB}} > L - \kappa. \end{cases} \quad (\text{B.16})$$

The boost passes untouched up to $L - \kappa$, then saturates smoothly and asymptotes to L . Cuts ($|C| < 1$) are left alone. This is the practical form of regularised inversion discussed in [18].

2. Low-frequency slope. C is multiplied by a 2nd-order Butterworth high-pass target at $f_c = 30$ Hz,

$$H_{\text{hp}}(f) = \frac{(j\omega)^2}{(j\omega)^2 + \sqrt{2}(j\omega) + 1}, \quad \omega = f/f_c, \quad (\text{B.17})$$

so below f_c the correction rolls off rather than chasing the woofer's natural roll-off to infinity.

3. Correction level. The correction level λ morphs continuously from bypass to full correction by raising the correction to a power in the log/phase domain,

$$C \mapsto |C|^\lambda e^{j\lambda \arg C}, \quad \lambda \in [0, 1], \quad (\text{B.18})$$

so $\lambda = 0$ gives unity and $\lambda = 1$ the full inverse.

4. Analysis band. Outside the user range $[f_{\text{lo}}, f_{\text{hi}}]$ the correction is faded back to unity through a half-octave raised-cosine skirt $w(f)$,

$$C \mapsto (1 - w(f)) + w(f) C. \quad (\text{B.19})$$

B.6 Subwoofer phase alignment

Preserving the relative timing. The subwoofer set is taken at the same positions as the main and paired by load order. Each subwoofer measurement is delay-anchored on the corresponding main measurement's delay rather than having its own delay estimated and removed. If each file were independently zeroed, the physically meaningful main-vs-sub arrival-time difference at each position would be discarded. Anchoring keeps it, so the averaged subwoofer response $\bar{S}(f)$ retains its true phase relative to $\bar{H}$.

The all-pass alignment term. From the unit-magnitude phasor of the subwoofer, $\hat{s}$, an all-pass factor is built and blended towards "no change" above the crossover,

$$A(f) = \frac{(1 - w_x) + w_x \hat{s}(f)}{|(1 - w_x) + w_x \hat{s}(f)|}, \quad C \mapsto C \cdot A(f), \quad (\text{B.20})$$

where the alignment weight $w_x(f)$ is 1 at and below the crossover and is released to 0 over one octave above it (raised cosine). The renormalisation makes A strictly phase-only, leaving the main's magnitude correction unchanged. Blending the phasors, rather than scaling an unwrapped phase angle, keeps the rotation on the short arc between 1 and $\hat{s}$, so A applies only the minimal phase shift and never unwinds the subwoofer's large accumulated low-frequency phase. The polarity inversion negates $\bar{S}$.

Bulk time alignment. The subwoofer phasor entering the blend is advanced by the Mains delay t_a before its phase is taken,

$$\hat{s}(f) = \frac{\hat{S}(f)}{|\hat{S}(f)|}, \quad \hat{S}(f) = \pm \bar{S}_s(f) e^{j2\pi f t_a}, \quad (\text{B.21})$$

so when t_a matches the measured offset the corrected main's phase is already flat through the crossover and $A \rightarrow 1$. The recommended t_a comes from a power-weighted least-squares fit of the unwrapped, delay-anchored subwoofer phase over $[f_c/2, 2f_c]$, the local group delay $\hat{\tau} = -d\varphi/d\omega$. Appendix F develops this estimator and the condition the applied delays must satisfy.

B.7 The complete correction filter

Collecting the stages of Sections B.5 and B.6, the correction applied at each frequency bin is the single complex expression

$$C(f) = (1 - w(f)) + w(f) A(f) \mathcal{P}_\lambda \left[H_{\text{hp}}(f) \mathcal{B}[1/h(f)] \right] \quad (\text{B.22})$$

evaluated on the Welch bin grid ($f_k = k f_s/W$). Section B.9 resamples and inverse-transforms this spectrum $C(f)$, not an analytic filter, into the exported FIR. Reading (B.22) from the inside

out, the operators are exactly the design steps above:

$$h(f) = \frac{\bar{H}_s(f)}{g_{\text{ref}}} \quad \text{normalised smoothed average, (B.23)}$$

$$\frac{1}{h(f)} = \frac{\overline{h(f)}}{|h(f)|^2} \quad \text{exact complex inverse, eq. (B.15), (B.24)}$$

$$\mathcal{B}[z] = z 10^{(\ell(z_{\text{dB}}) - z_{\text{dB}})/20}, \quad z_{\text{dB}} = 20 \log_{10} |z| \quad \text{soft-knee boost limit (phase kept), (B.25)}$$

$$\ell(g) = \begin{cases} g, & g \leq L - \kappa, \\ (L - \kappa) + \kappa \tanh \frac{g - (L - \kappa)}{\kappa}, & g > L - \kappa, \end{cases} \quad \kappa = \min(6, L), \quad L = \text{max boost (dB), (B.26)}$$

$$H_{\text{hp}}(f) = \frac{(j\omega)^2}{(j\omega)^2 + \sqrt{2}(j\omega) + 1}, \quad \omega = \frac{f}{f_c} \quad \text{LF slope, } f_c = 30 \text{ Hz, (B.27)}$$

$$\mathcal{P}_\lambda[z] = |z|^\lambda e^{j\lambda \arg z} \quad \text{correction level } \lambda \in [0, 1], \quad \text{(B.28)}$$

$$\hat{s}(f) = \frac{\pm \bar{S}_s(f) e^{j2\pi f t_a}}{|\bar{S}_s(f)|} \quad \text{time-aligned sub phasor, } t_a = \text{Mains delay, (B.29)}$$

$$A(f) = \frac{(1 - w_x) + w_x \hat{s}(f)}{|(1 - w_x) + w_x \hat{s}(f)|} \quad \text{sub phase alignment (} A \equiv 1 \text{ if no sub), (B.30)}$$

$$w(f) \in [0, 1] \quad \text{analysis-band weight (raised-cosine skirts). (B.31)}$$

Two boundary cases complete the definition. Where $|h(f)|$ is negligible, the inverse is replaced by the boost ceiling rather than diverging. Cuts ($z_{\text{dB}} \leq 0$) pass $\mathcal{B}$ unchanged since $L - \kappa \geq 0$, so only boosts are limited. Setting $\lambda = 1$, no subwoofer ($A \equiv 1$) and a fully open analysis band ($w \equiv 1$) reduces (B.22) to the boost-limited, LF-sloped exact inverse $H_{\text{hp}} \mathcal{B}[1/h]$. The linear-phase export then applies one more operator to C , the phase limit of Section B.8.

The ceiling limits a magnitude, and only a magnitude. $\mathcal{B}$ bounds $|1/h|$. It does nothing to $\arg(1/h) = -\arg h$, which turns fastest exactly where $|h|$ is smallest. A capped magnitude carried by an uncapped phase is no longer the inverse of anything, and the mismatch grows with how hard the ceiling is working. Two things keep it in proportion. The phase weight of (B.7) has already dealt with the case that matters most, a null the positions disagree about, where the phase was never information to begin with, and a null they all agree about is a property of the listening area whose phase is worth inverting as far as the ceiling lets the magnitude follow it, within the crossover region where the phase is kept at all (Section B.8). In the specific configuration analyzed above, the boost ceiling is never reached. The unconstrained inverse filter requires a maximum boost of only 5.1 dB and 6.4 dB for the two main channels, well below the 12 dB limit. Consequently, the bounding operator $\mathcal{B}$ acts as an identity function across all in-band frequency bins. Conversely, when a system does encounter its boost ceiling, the calculated correction phase becomes increasingly unreliable inside deep notches where the filter declines to apply further boost. The appropriate remedy for such deep nulls is not to increase the filter length (which merely implements an unrealistic target with higher precision) but rather to apply broader frequency smoothing or to lower the boost ceiling itself.

B.8 Limiting the phase to the crossover region

Equation (B.22) inverts the average's phase wherever the weight (B.7) lets it through. With "Limit phase" on, the default, the linear-phase export keeps that designed phase only where the subwoofer alignment needs it and hands it over to the minimum-phase phase of the same correction above,

$$\Lambda[C](f) = |C(f)| \frac{w_p(f) \hat{c}(f) + (1 - w_p(f)) \hat{m}(f)}{|w_p(f) \hat{c}(f) + (1 - w_p(f)) \hat{m}(f)|}, \quad \hat{c} = \frac{C}{|C|}, \quad (\text{B.32})$$

where $\hat{m}(f)$ is the unit phasor of the minimum-phase response with magnitude $|C|$, built by the cepstrum of Section B.9.2, and

$$w_p(f) = \begin{cases} 1, & f \leq 2f_x, \\ \frac{1}{2} + \frac{1}{2} \cos(\pi \log_2 \frac{f}{2f_x}), & 2f_x < f < 4f_x, \\ 0, & f \geq 4f_x, \end{cases} \quad (\text{B.33})$$

with f_x the crossover. The magnitude does not move, and from $4f_x$ up $\Lambda[C]$ is exactly the filter the minimum-phase rendering exports. The blend runs on the short arc, like those of (B.8) and the alignment term.

Why stop at twice the crossover. The alignment weight w_x has fallen to zero at $2f_x$, so above it the all-pass A is the identity and the only phase left to invert is the room's. Below it, the room's phase has to stay. A steers the main onto the subwoofer on the assumption that the main's own phase is already flat there. Replacing the average's phase with minimum phase at every frequency cost 1 dB of summation efficiency in the offline model of a ten-position campaign (-1.6 against -0.6). Limiting it at $2f_x$ cost nothing: below $2f_x$ the limited and unlimited exports are the same filter, and measured through the crossover band they differ by 0.01 dB.

Why above it at all. Only the part of the average's phase that every seat shares can be inverted everywhere [17, 16]. The rest stays at each seat, and a linear-phase filter puts half of whatever it does not cancel *ahead* of the direct sound, where the ear masks little of it: a *pre-echo*. At $R \approx 0.8$ – 0.9 , the coherence of the 250 Hz to 1000 Hz octaves in Table B.2, the weight still passes most of the phase while the positions scatter by 26° to 38° . That residual is exactly what was heard, as a short pre-reverberation before impacts, on a campaign re-exported after the fix of Section B.2. Making the weight stricter does not cure it: raising w to the fourth power still leaves the 500 Hz band 10 dB above minimum phase. Positions that agree this well can still leave each seat enough to hear.

Table B.3 measures it through the whole system: from 500 Hz to 8000 Hz the unlimited export put 14 dB to 24 dB more energy than the limited one in the 30 ms before the direct sound, and 14 dB to 30 dB more in the last 3 ms. The limited one is within 2 dB of minimum phase over the longer window. At 250 Hz, inside the fade, it removes 3 dB to 10 dB.

Nor did the inversion buy anything there. The arrival of each octave's envelope was measured against the direct sound at every seat, through the whole system. From 500 Hz to 4000 Hz the median seat was time-coherent to within 0.2 ms in every octave with no correction at all and through every export alike, and the worst seat to within 0.3 ms through any export. Against minimum phase, the unlimited export moved none of the octaves from 250 Hz up by more than 0.11 ms. Only below $2f_x$, where the alignment works, did linear phase move an arrival: the 125 Hz octave 0.4 ms to 0.5 ms earlier and the 63 Hz one 0.9 ms later, the main being steered onto the subwoofer.

	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz
<i>30 ms to 3 ms before the direct sound</i>						
minimum phase	-49/-50	-57/-57	-64/-66	-68/-67	-68/-68	-67/-67
linear, limited	-35/-34	-56/-56	-64/-64	-67/-67	-68/-68	-67/-67
linear, not limited	-32/-24	-38/-34	-45/-42	-46/-47	-44/-48	-50/-53
noise floor	-68	-71	-72	-71	-68	-67
<i>3 ms to 1 ms before</i>						
minimum phase	-62/-65	-56/-59	-64/-63	-59/-56	-74/-69	-77/-61
linear, limited	-42/-44	-56/-58	-64/-63	-59/-56	-69/-69	-73/-61
linear, not limited	-39/-26	-28/-28	-36/-33	-40/-42	-39/-39	-43/-47

Table B.3: What limiting the phase does, measured through the whole system at ten positions on a campaign crossed at 70 Hz, 8192 taps, left/right main. Energy arriving ahead of the direct sound, per octave, relative to that octave’s energy in the 300 ms after it, in dB, through causal band filters so the analysis cannot move anything earlier; power mean over the positions. The noise floor is the same 27 ms window taken 400 ms to 150 ms ahead of the direct sound, beyond the reach of any of the filters. Every response is deconvolved from its sweep whole: Welch averaging of the same sweeps reads some 20 dB more ahead of the direct sound in the 250 Hz to 1000 Hz octaves, in every row alike.

Why the limit is applied to the finished correction. The minimum-phase phasor $\hat{m}$ in (B.32) is that of $|C|$, not of the average. The average’s minimum phase also carries what the correction never inverts: the band edges of the measurement, and the loudspeaker’s roll-off below the analysis range, where w returns C to unity. Inverting their phase without their magnitude is a pre-echo of its own. On a synthetic set of three positions agreeing on a 1 kHz all-pass, a limit built from the average left -5.7 dB of the render’s energy ahead of its peak, more than no limit at all (-13). Built from $|C|$, as (B.32) does, it leaves -38 dB and matches the minimum-phase design to 0.001° above $4f_x$. The test suite checks both: the match above $4f_x$, and a pre-echo at least 20 dB below the unlimited design’s.

What it leaves. Below $2f_x$ the linear-phase filter still puts energy ahead of the direct sound. On the campaign of Table B.3 the 63 Hz octave carries -14 dB and the 125 Hz one -16 dB to -17 dB within 30 ms, against -20 dB and -26 dB in minimum phase. That is the alignment itself, and the design delay sets its size: crossed at 100 Hz, the 125 Hz octave carried -4 dB with the estimated delay and -13 dB to -16 dB with one 14 ms shorter, the summation being the same at both to 0.03 dB. What it buys is a summation that does not depend on that delay: measured, 0.2 dB to 0.5 dB more level through the crossover band than minimum phase at the estimated delay and 2.7 dB more at the shorter one; predicted, between -0.24 dB and -0.97 dB of summation efficiency for any delay from 10 ms to 40 ms, a range over which minimum phase falls as low as -8.4 dB. Whether that low-frequency energy is audible is a listening question. 30 ms is less than two periods at 63 Hz.

B.9 Rendering the FIR

Both exports turn a complex spectrum into a finite impulse response on the FIR’s frequency grid, with $N = \text{nextPow2}(\text{firLength})$ bins.

B.9.1 Linear- and mixed-phase rendering

The resampled spectrum is given a half-length linear phase shift, multiplying bin k by $e^{-j\pi k} = (-1)^k$, before the inverse FFT. Because the correction is a mixed-phase filter, deliberately carrying phase and **all-pass** content including the subwoofer alignment, it is not causal on its own, and centring its energy at sample $N/2$ is what makes it realisable as an ordinary FIR. A Tukey window (10 % taper) centred on $N/2$ then cleans up the edges. FIR design by frequency sampling and windowing is standard [4].

The cost of centring is a constant latency of $\approx N/2$ samples on that output. SuperMoTo's per-output convolution uses the WDL zero-latency (uniformly partitioned) convolution engine [21], so the filter adds no algorithmic block latency beyond that fixed $N/2$ offset.

B.9.2 Minimum-phase rendering

As defined in Appendix C, a **minimum-phase** response is the unique causal, stable response whose log-magnitude and phase form a **Hilbert-transform** pair (eq. (C.2)), so it is fully determined by $|C(f)|$ [4]. SuperMoTo computes it by the **real-cepstrum** method. With $c[n] = \mathcal{F}^{-1}\{\log |C|\}$ the real cepstrum, the minimum-phase log-spectrum is recovered by keeping its causal part,

$$\hat{c}[n] = c[n] \ell[n], \quad \ell[n] = \begin{cases} 1, & n = 0, \ n = N/2, \\ 2, & 1 \leq n < N/2, \\ 0, & N/2 < n \leq N-1, \end{cases} \quad (\text{B.34})$$

and the impulse response is $\mathcal{F}^{-1}\{\exp \mathcal{F}\{\hat{c}\}\}$, tapered only at the tail since its energy is front-loaded near $n = 0$ [4]. Only $|C|$ enters this construction, and the all-pass term A has unit magnitude, so the subwoofer alignment has no effect on a minimum-phase export.

Appendix C

Minimum phase, excess phase and invertibility

Whether an error can be undone by a filter hinges on a single property of the response, namely whether it is [minimum-phase](#). Section 1.2.5 says which phenomena fall on each side of that line. This appendix explains why, first physically and then in terms of poles and zeros.

C.1 Physically

A response is [minimum-phase](#) when, for its given magnitude, its energy arrives as early as possible. It has the least phase lag and the least [group delay](#) (the frequency-derivative of phase, i.e. the delay the response imposes near each frequency) consistent with that magnitude. Equivalently, both the response and its inverse are causal and stable, which is exactly what makes it correctable, because undoing it does not require predicting the future. A single loudspeaker resonance, a driver roll-off and an isolated low-frequency [room mode](#) are essentially minimum-phase, since the magnitude bump or dip and the phase rotation around it are two views of the same event. A pure propagation delay, a discrete reflection arriving after the direct sound, and the deep cancellation notch where two out-of-phase sources sum are not minimum-phase. They carry [excess phase](#), a lag beyond the minimum that no stable causal filter can remove.

C.2 Mathematically

For a discrete linear time-invariant system with rational [transfer function](#) $H(z)$, [minimum-phase](#) means every pole and every zero lies strictly inside the unit circle. Poles inside make H causal and stable. Zeros inside make the inverse $1/H(z)$, whose poles are H 's zeros, causal and stable too, so H can be inverted by a realisable filter. Any stable response factors uniquely into a minimum-phase and an [all-pass](#) part,

$$H(z) = H_{\min}(z) H_{\text{ap}}(z), \quad |H_{\text{ap}}(e^{j\omega})| = 1, \quad (\text{C.1})$$

where the all-pass factor H_{ap} (unit magnitude, so it changes only the phase) carries all the excess phase, obtained by reflecting each zero z_0 outside the unit circle to $1/\bar{z}_0$ inside. The minimum-phase factor alone sets the magnitude, $|H| = |H_{\min}|$, and for it alone magnitude and phase are not independent but a [Hilbert-transform](#) pair:

$$\arg H_{\min}(e^{j\omega}) = \mathcal{H}\{\log |H_{\min}(e^{j\omega})|\}, \quad (\text{C.2})$$

so the phase is completely determined by the magnitude. SuperMoTo uses this identity both ways, to design a flat-phase correction by conjugating the measured minimum-phase phase, and to render a low-latency filter from the magnitude alone (Section B.9.2).

C.3 What it means for the correction

The practical consequence governs the whole correction design. SuperMoTo can invert the magnitude exactly and flatten the **minimum-phase** phase. The **all-pass** / excess part and any pure delay can only be pre-compensated, with a **linear-phase** filter that is acausal and therefore adds latency (Section 7.6). A genuine non-minimum-phase cancellation, a zero on or outside the unit circle, has no stable inverse at all, so boosting into it merely wastes headroom. This one distinction is why the **spatial average**, the boost ceiling and the frequency-dependent **smoothing** exist (Chapter 7). They keep the design on the minimum-phase part of the response that is actually invertible.

Appendix D

Spatial averaging and microphone positions

This appendix gives the reasoning behind the microphone-position recommendations of Section 6.5.2. The analysis averages the measurements of one loudspeaker across microphone positions as complex quantities, after removing each one’s own arrival delay (Section B.2). How many positions to take, and how widely to spread them, then decides what that average actually represents. The two questions have different answers, and the second matters more than the first.

D.1 Spacing and the correlation length

Two microphones a distance d apart in a diffuse field do not measure independent pressures. The spatial correlation coefficient follows [15]

$$R(d) = \frac{\sin kd}{kd}, \quad k = \frac{2\pi f}{c}, \quad (\text{D.1})$$

whose first zero is at $d = \lambda/2$. Below that separation the two measurements are largely the same measurement, while above it they are effectively independent samples. The separation needed is therefore frequency dependent, and at low frequencies it is large:

Frequency	40 Hz	100 Hz	250 Hz	500 Hz	1 kHz	4 kHz
$\lambda/2$	4.3 m	1.7 m	69 cm	34 cm	17 cm	4.3 cm

A cluster of positions spanning 30 cm around the seat, which is the usual choice, therefore reaches $\lambda/2$ only above about 570 Hz. Everything below that is sampled inside a single correlation cell. Measured on a real five-position set in a small studio, the vector coherence $|\overline{H}|/|\overline{H}|$ (unity when every position agrees, zero when they are independent) bears this out:

Frequency	63 Hz	160 Hz	250 Hz	630 Hz	2 kHz	4 kHz
Coherence	0.997	0.98	0.98	0.73	0.75	0.36

The consequence is worth stating plainly, because it is easy to expect otherwise. What a cluster averages out is set by its size, not by the physical cause of a feature. Below about 570 Hz every position of a 30 cm cluster sees much the same response, so the average keeps it as it is at the seat. That is true of [room modes](#), which averaging is most often expected to remove (decorrelating 100 Hz would take moving the microphone 1.7 m, which is no longer the same listening position). It is just as true of [speaker-boundary interference](#) and of the low-frequency part of every other reflection. Above that frequency the [comb filtering](#) of reflections changes

from one position to the next and is averaged. The correction is therefore valid for the area the cluster covers, whatever the cause of each feature it corrects. That is a property of the physics rather than a shortcoming of the method.

What the correction can do about a feature depends on another distinction, peaks against dips. A resonant peak is safe to cut. A dip, whether a modal null or a cancellation by a reflection, cannot be filled by boosting, which is what the "Max boost" ceiling (Section 7.4) is there to prevent.

What the [spatial averaging](#) does buy, in the range where it works, is protection against over-fitting. The narrow peaks and nulls of a single-point measurement move with the microphone, and correcting them would make the response worse everywhere else.

D.2 How many positions are enough

The useful test is not a rule of thumb but whether the result still changes when a position is added or removed. Taking the same five-position set and dropping each position in turn, the designed correction moves by:

Band	RMS deviation	Worst case
30 Hz to 80 Hz	0.22 dB	0.57 dB
80 Hz to 200 Hz	0.36 dB	1.52 dB
200 Hz to 500 Hz	0.25 dB	1.27 dB
500 Hz to 2 kHz	0.50 dB	1.82 dB
2 kHz to 8 kHz	1.11 dB	6.90 dB

Below 500 Hz the answer is settled to a quarter of a decibel, which follows directly from the coherence above. Those positions are not independent, so adding more of them changes nothing. The only band that has not converged is above 2 kHz, and that is also the band where a detailed correction is least useful, since it will not hold a few centimetres away. The effective remedy there is broader high-frequency [smoothing](#) (Section 7.3), not more measurements. Reducing the set from five positions to three roughly doubles these figures, consistent with the $1/\sqrt{N}$ one would expect from averaging independent samples.

D.3 Why a subwoofer argues for a tight cluster

When a subwoofer set is loaded, each of its measurements is anchored on the delay of the main measurement taken at the same position (Section B.6), which is what preserves the main-to-sub timing. The residual offsets $\tau_{\text{sub}}^{(n)} - \tau_{\text{main}}^{(n)}$ differ from position to position, and the anchored average is only well conditioned while that spread is small compared with a period at the [crossover](#). On the five-position set above the spread is 0.57 ms, about 35° at 170 Hz, which is harmless. Spreading the same positions over a metre or more would grow it several-fold, the anchored average would partially cancel, and the crossover-region estimates (Section B.6) would degrade. Widening the array to improve the subwoofer integration makes it worse.

Appendix E

Why the sweep runs to Nyquist

The [impulse response](#) recovered by deconvolving a logarithmic sweep (Section [B.3](#)) is surrounded by ringing, and the level of that ringing is set almost entirely by where the sweep stops. On an ideal system, a sweep that stops at 20 kHz leaves the ringing only 15 to 21 dB below the peak, while the same sweep run up to Nyquist over a whole number of octaves pushes it 53 to 62 dB down. This appendix explains where the ringing comes from, gives the simulation data that separates its possible causes, and derives the band and fades SuperMoTo uses for the sweep stimulus.

E.1 The inverse filter and its group delay

The recording is deconvolved by the analytic inverse ([B.12](#)). Write $\varphi(f)$ for its phase. Since $\frac{d}{df}[f(1 - \ln(f/f_1))] = -\ln(f/f_1)$, the phase slope is $d\varphi/df = 2\pi L \ln(f/f_1)$, and the [group delay](#) of the inverse is

$$\tau(f) = -\frac{1}{2\pi} \frac{d\varphi}{df} = -L \ln \frac{f}{f_1} = -t(f), \quad (\text{E.1})$$

with $t(f)$ the arrival time ([B.10](#)). The inverse undoes exactly the dispersion the sweep introduced. Content that the sweep placed at frequency f at time $t(f)$ is returned to time zero, which is what makes the method work.

Two properties of the inverse matter for what follows. First, ([E.1](#)) is a property of the filter, not of the recording, so it applies to everything in the recording, including content at frequencies the sweep never visited. Second, the magnitude of ([B.12](#)) grows as $\sqrt{f}$ without bound, and the inverse is applied to every FFT bin from f_s/N up to Nyquist with no [regularisation](#) and no band limit. Only the DC bin is zeroed.

E.2 Where the ringing can come from

Take as a reference a 10 s sweep from 10 Hz to 20 kHz at 48 kHz, with a 20 ms linear fade at each end. The quantization ([B.11](#)) gives $L = 1.30$ s. Three mechanisms can plausibly produce ringing around the peak of its deconvolved [impulse response](#).

The fade-out. The literature on the exponential sweep attributes pre-ringing mainly to the fade-out applied to the end of the stimulus [[10](#)]. Its effect is easy to bound. The instantaneous frequency at the end of the sweep advances at $df/dt = f/L$, so a fade of duration Δt tapers a band of width $\Delta f \approx f_2 \Delta t/L$, here

$$\Delta f \approx \frac{20000 \times 0.02}{1.30} \approx 308 \text{ Hz}, \quad (\text{E.2})$$

that is from 19 695 Hz to 20 kHz, a span of 0.022 octave at the very top of an eleven-octave stimulus. The fade-in, by the same argument, tapers 10 Hz to 10.15 Hz. The emphasis on the fade-out comes from sweeps that run essentially to Nyquist, 22 kHz at 44.1 kHz in [10], where the fade-out is the only thing shaping the top edge of the stimulus band. A sweep that stops 4 kHz short of Nyquist has a hard band edge at 20 kHz whatever the fade does, and the fade merely rounds its last 0.022 octave.

Content above the stimulus band. Content in the recording above f_2 has no counterpart in the stimulus. It includes harmonic distortion of the sweep’s upper region, converter and preamplifier noise, and whatever splatter the end of the stimulus produces. By (E.1) the inverse assigns it a **group delay** $\tau(f) < -L \ln(f_2/f_1) = -T$, and amplifies it by $\sqrt{f}$ on the way. A displacement beyond $-T$ wraps circularly in the deconvolution’s FFT buffer, so where it lands depends on the transform length. With a 1 s silent tail the recording is 10.88 s long and the FFT is $N = 2^{19}$ samples, or 10.923 s. Table E.1 evaluates (E.1) just above f_2 .

Table E.1: Where content above f_2 lands after deconvolution. $f_s = 48$ kHz, $f_1 = 10$ Hz, $f_2 = 20$ kHz, $L = 1.30$ s, $N = 10.923$ s. The analysis keeps the first $W = 65\,536$ samples, that is 1.365 s.

f (Hz)	$\tau(f)$ (s)	wraps to (s)	
20000	−9.881	1.041	inside the analysis window
21000	−9.945	0.978	inside
22000	−10.005	0.918	inside
24000	−10.118	0.804	inside

All of it lands inside the part of the deconvolved response the analysis keeps. It does not appear as a recognisable pre-echo at a particular time, but as junk distributed through the retained impulse response, which after the forward transform contaminates the **transfer function** at every frequency, the region around the peak included.

The band edge. A stimulus occupying $[f_1, f_2]$ identifies the system only over that band. The recovered impulse response is the true one convolved with the impulse response of an ideal band-pass, whose symmetric sinc rings on both sides of the peak with a decay set by the sharpness of the edges. This is not an artefact in the sense the other two are. It is the resolution limit of the measurement, and the only way to reduce it is to widen the band.

E.3 How the mechanisms are separated

The three are separated by simulating SuperMoTo’s signal path on a system whose true **impulse response** is known exactly. It is a pure delay of 480 samples, standing for a loopback cable with a microphone about 3.4 m away. Deconvolving a stimulus that has passed through nothing should return a Dirac pulse [10], so everything else in the result is attributable to the method. The stimulus follows (B.9) with L quantized by (B.11), a 1 s silent tail is appended, and the deconvolution applies (B.12) over the full spectrum with only the DC bin zeroed.

The reported figure is the largest sample magnitude in the 20 ms preceding the peak, in dB relative to the peak, excluding the three samples adjacent to it. Both sides of the peak were measured throughout and agree to within a few tenths of a dB. That symmetry is itself evidence for the band edge, since a fade artefact would appear on one side only.

The simulation contains no distortion, no room and no converter, so it measures the floor the method imposes, not what a real measurement achieves. Its purpose is comparative. Every

configuration sees exactly the same idealised system, so differences between them are attributable to the stimulus and the deconvolution alone.

E.4 Results

The band dominates. Table E.2 varies only the stimulus band. The second row runs to Nyquist over a whole number of octaves, eleven at 44.1 and 48 kHz and twelve at 96 kHz.

Table E.2: Worst ringing within 20 ms of the peak, relative to the peak, for a 10 s sweep on a pure delay.

Stimulus band	44.1 kHz	48 kHz	96 kHz
10 Hz to 20 kHz	−21.0	−17.9	−14.9
f_1 to Nyquist, whole number of octaves	−52.9	−61.0	−62.2

Running to Nyquist gains about 43 dB at 48 kHz. The band stopping at 20 kHz also degrades as the sample rate rises, from −21.0 dB to −14.9 dB, because the gap between 20 kHz and Nyquist grows and the content above the band has more room to work in. A fixed band is therefore worst exactly where the interface offers the most bandwidth. Figure E.1 shows both *impulse responses* around the peak, and the symmetry of the 20 kHz trace is the signature of a band edge.

A low-frequency defect in the chain does not change this picture. Adding a second-order high-pass at 15 Hz, as an AC-coupled input would, moves the 20 kHz result from −17.9 dB to −17.8 dB, which is indistinguishable from the band edge alone.

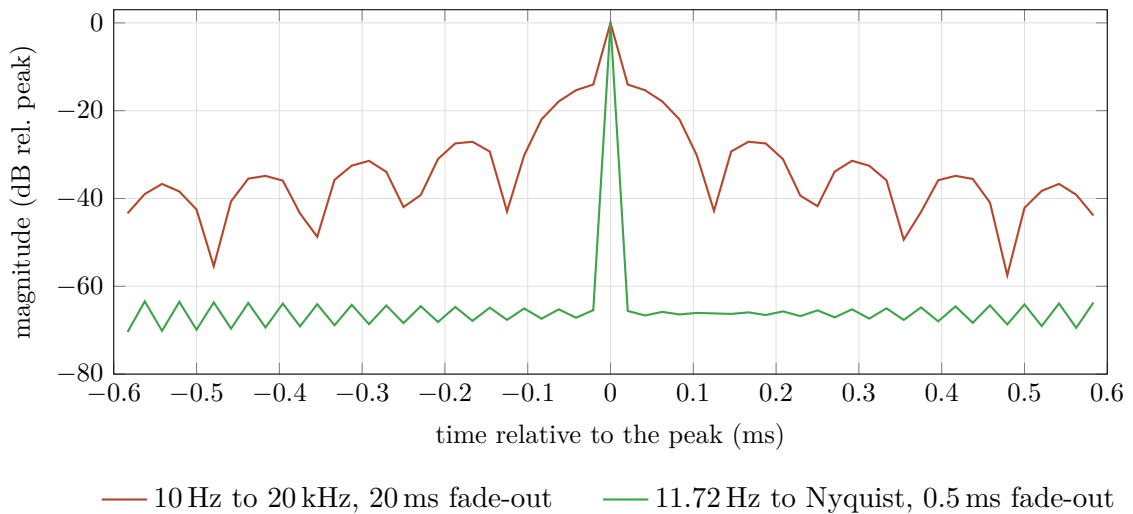


Figure E.1: Deconvolved impulse response of a pure delay, 48 kHz, 10 s sweep. A band stopping at 20 kHz shows the slowly decaying symmetric sinc of a band edge four kilohertz below Nyquist. A band running to Nyquist falls to the floor within one sample.

The fade-out matters only once the band reaches Nyquist. Removing the 20 ms fade-out from the 20 kHz band changes the result by less than 0.5 dB, consistent with the 0.022 octave estimate above, and cutting the sweep at its last zero crossing on top of that changes nothing further. Once the sweep runs to Nyquist the situation inverts, because the fade-out becomes the only thing shaping the top edge. Figure E.2 scans its length at three sample rates. The curves

are flat to the left and steep to the right. Anything up to about 1 ms sits within 1 dB of the optimum, beyond 4 ms the cost is real, and 20 ms costs 15 to 25 dB.

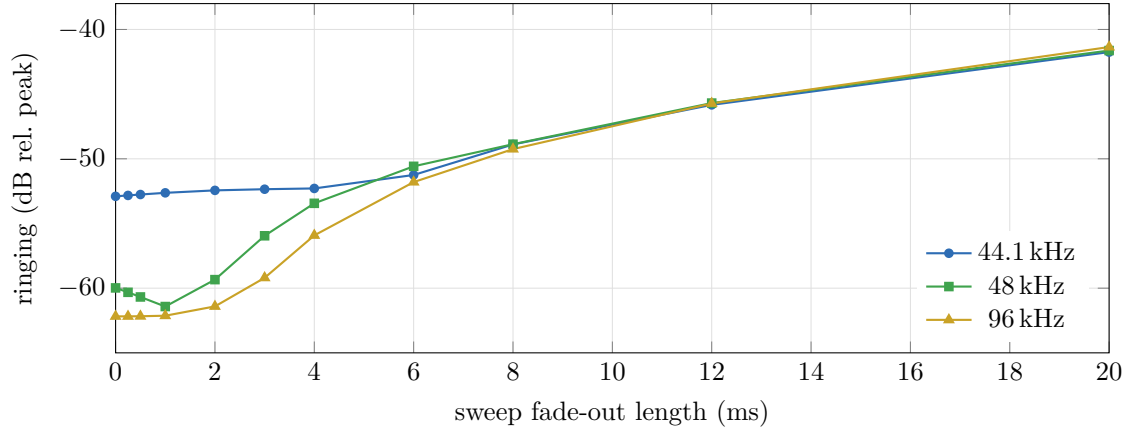


Figure E.2: Ringing against sweep fade-out length, 10 s sweep to Nyquist over a whole number of octaves. Everything up to about 1 ms sits within 1 dB of the optimum; beyond 4 ms the cost is real, and a 20 ms fade-out costs 15 to 25 dB.

E.5 Why the sweep cannot be cut at a zero crossing

Cutting the sweep at its last zero crossing is the remedy usually proposed alongside sweeping to Nyquist [10], but the two are incompatible. The synchronization condition already places the end of the sweep at a zero crossing in continuous time whenever the frequency ratio is an integer. From (B.9), with $k = f_1 L$ an integer, the total phase is

$$\Phi(T) = 2\pi f_1 L \left(\frac{f_2}{f_1} - 1 \right) = 2\pi k \left(\frac{f_2}{f_1} - 1 \right), \quad (\text{E.3})$$

so if $f_2/f_1 = 2^P$ then $\Phi(T)/2\pi = k(2^P - 1)$ is an integer and $x(T) = 0$ exactly. This is the phase-controlled chirp of Vetter and di Rosario [11], obtained here from the quantization (B.11).

That does not carry over to discrete time. The last emitted sample sits at index $n - 1$ with $n = \text{round}(T f_s)$, a fraction $\delta = T f_s - (n - 1)$ of a sample before the true end. At the end of the sweep the instantaneous frequency is Nyquist, so the phase advances by $2\pi f_2/f_s = \pi$ radians per sample, and the residual amplitude is

$$|x[n - 1]| = |\sin(\pi\delta)|. \quad (\text{E.4})$$

Table E.3 checks (E.4) against the generated signal.

Table E.3: Residual amplitude of the last emitted sample, predicted by (E.4) and measured on the generated sweep.

f_s	T	δ	$ \sin(\pi\delta) $	measured
48 kHz	10 s	0.5905	0.9598	0.9598
48 kHz	5 s	0.5149	0.9989	0.9989
44.1 kHz	10 s	1.1511	0.4572	0.4572
96 kHz	10 s	0.9661	0.1062	0.1062

The consequence is structural, not numerical. Near Nyquist the sampled sweep is $x[n] \approx (-1)^n \sin \varphi_0$, so consecutive samples differ only in sign and all share the magnitude $|\sin \varphi_0|$. No

sample is closer to a zero crossing than any other, so searching backwards for one is meaningless. Truncating anyway leaves both the last sample at 0.960 and the ringing at -61.0 dB, entirely unchanged.

At 48 kHz and 10 s the stimulus therefore ends by stepping from 0.96 to silence. Figure E.2 shows the deconvolution tolerates this, since a zero-length fade-out is within a decibel of the best result. The converter’s reconstruction filter, however, turns that step into a wideband transient delivered to the tweeter at measurement level, which is the reason to keep a short fade-out rather than none.

E.6 The band and fades SuperMoTo uses

The sweep band is derived from the sample rate. SuperMoTo sets $f_2 = f_s/2$ and $f_1 = f_2/2^P$, choosing P to place f_1 as near as possible to 10.5 Hz,

$$P = \text{round}(\log_2(f_2/10.5 \text{ Hz})), \quad (\text{E.5})$$

clamped to $[4, 16]$. Table E.4 lists the resulting bands.

Table E.4: Sweep band at common sample rates.

f_s (Hz)	P	f_1 (Hz)	f_2 (Hz)
44 100	11	10.767	22050
48 000	11	11.719	24000
88 200	12	10.767	44100
96 000	12	11.719	48000
192 000	13	11.719	96000

Two constraints are met at once. An integer P makes f_2/f_1 a power of two, so (E.3) holds and the sweep ends in sine phase. And f_1 stays near 11 Hz at every rate, below the half-octave skirt under the default 20 Hz lower edge of the analysis band (Section B.5), which reaches down to 14.1 Hz.

The fade-in is 20 ms. It tapers only the bottom 0.02 octave, and a fade-in, unlike a fade-out, does not raise the deconvolved coefficients around the central pulse [11], which the simulations here agree with. The sweep’s fade-out is 0.5 ms, the shortest value that removes the step of (E.4) while staying on the flat part of every curve in Figure E.2. Its cost against the best available value is at most 0.6 dB, at 48 kHz and 5 s, and is not measurable at 96 kHz.

The white-noise stimulus keeps a fixed 10 Hz to 20 kHz band with 20 ms fades at both ends. It is analysed by the Welch estimator (Section B.1), which never deconvolves anything, so none of the above applies to it.

E.7 What the figures mean for a real measurement

The figures describe an idealised chain. In a real measurement the floor is set by room noise, by the loudspeaker’s own distortion and by time variance, long before the -61 dB of Table E.2. What sweeping to Nyquist buys is that the method’s own contribution sits far enough below those floors to be ignored, which is not the case at -17.9 dB.

Two side effects of the wider band are worth knowing.

- **Driver loading.** An exponential sweep spends the same time in every octave, $L \ln 2$, which is about 0.9 s per octave in a 10 s sweep. Running to Nyquist adds the band above 20 kHz at that same energy per octave, and the converter’s reconstruction filter attenuates its extreme top. A first run at reduced level remains sensible on a delicate tweeter.

- **Harmonic curves.** The order- m harmonic of any fundamental above $f_s/(2m)$ lies above Nyquist and aliases. This holds whatever the band, but a sweep that reaches Nyquist carries more fundamentals in that region, so the H2 to H5 curves (Section [B.3.3](#)) are less reliable in the top octaves.

Appendix F

Delay estimation and subwoofer alignment

The analysis measures how much later a loudspeaker arrives in two independent ways. One is the peak of the [impulse response](#), used to align several loudspeakers to one listening position. The other is the slope of the subwoofer’s phase around the [crossover](#), used to integrate a subwoofer with the mains. They are easily taken for two measurements of the same thing, but they measure different physical quantities, can differ by several milliseconds, and each is the wrong tool for the other’s job. This appendix defines both, shows why they disagree, and derives the condition the output delays must satisfy for the subwoofer phase alignment of Section 7.5 to be correct.

Throughout, a pure delay τ contributes $e^{-j2\pi f\tau}$ to a [transfer function](#), and the [group delay](#) of a response with phase $\varphi(f)$ is $\tau_g = -\frac{1}{2\pi} d\varphi/df$. The subscripts m and s denote a main loudspeaker and the subwoofer.

F.1 Two estimators

Time of arrival. For each measurement file the analysis takes the largest sample of the [impulse response](#), reads a peak in the second half of the buffer as a small negative delay, and removes that delay before [spatial averaging](#) (Section B.2). The arrival time of a loudspeaker is the median of these per-position delays $\delta^{(p)}$,

$$d = \frac{1}{f_s} \text{median}_p \delta^{(p)}, \quad (\text{F.1})$$

so a single position that latches onto a reflection or a mode does not move the result. The sample grid gives a resolution of $20.8\mu\text{s}$ at 48 kHz, but the real limit is the width of the impulse, about $1/f_{\text{max}}$ for a signal band-limited to f_{max} . That is $50\mu\text{s}$ for a full-range loudspeaker measured to 20 kHz, and 10 ms for a subwoofer passing nothing above 100 Hz, where room modes and the subwoofer’s own filtering also shift the peak. The accuracy of any delay estimate is bounded by the signal’s bandwidth in this way [12], and no sample rate improves it. This estimator is excellent for full-range sources and weak for a subwoofer.

Crossover-band group delay. When a subwoofer set is loaded into a main’s analysis, each subwoofer file is anchored on the delay of the main file taken at the same position instead of its own (Section B.6). The averaged subwoofer response S therefore keeps the main-to-sub timing as a residual phase. The analysis unwraps $\varphi_S = \arg S$ at 48 logarithmically spaced frequencies f_n over $[f_x/2, 2f_x]$, with f_x the [crossover](#), and fits a least-squares slope weighted by the measured power $w_n = |S(f_n)|^2$,

$$\tau_{\text{grp}} = -\frac{1}{2\pi} \frac{\sum_n w_n (f_n - \bar{f})(\varphi_n - \bar{\varphi})}{\sum_n w_n (f_n - \bar{f})^2}, \quad (\text{F.2})$$

where $\bar{f}$ and $\bar{\varphi}$ are the weighted means. The result decomposes as

$$\tau_{\text{grp}} = \underbrace{(d_s - d_m)}_{\text{path difference}} + \underbrace{\langle \tau_g[\text{subwoofer}] \rangle_{[f_x/2, 2f_x]}}_{\text{own group delay in the band}}, \quad (\text{F.3})$$

the total effective lateness of the subwoofer's energy in the crossover region, geometry and filtering together. That is exactly what decides whether the two sources add or cancel there. It is a band average of a frequency-dependent quantity, not the time at which anything arrives.

The weighting is essential. The fit band reaches an octave above the crossover, where a subwoofer is already far down. On a real five-position measurement crossed at 85 Hz, $|S|$ falls from -6 dB at 60 Hz to -50 dB at 170 Hz, and the position-to-position coherence drops from 0.99 to 0.60 over the same span. An unweighted fit gives that noise the same say as the passband, and the estimate walks with the crossover setting:

f_x (Hz)	60	85	100	120	150
τ_{grp} , unweighted (ms)	27.6	21.8	17.7	13.7	11.3
τ_{grp} , weighted (ms)	26.4	31.3	32.3	34.8	30.7

The weighted figure is stable and agrees with the arrival time measured on the same set, 30.6 ms.

The estimate also depends on **smoothing**, because it is read off a complex-smoothed average. A window of β octaves at frequency f rotates the phasor of a residual delay τ by about

$$\Delta\varphi \approx 2\pi \tau \beta f \ln 2 \quad (\text{F.4})$$

across its width. As this approaches π the complex average collapses and the phase slope stops meaning anything. At 160 Hz with one-octave smoothing, that happens for $\tau \approx 4.5$ ms, well within the range of offsets being measured.

F.2 Why the two disagree

Beyond geometry, the dominant term in (F.3) is the **crossover** filter itself. A fourth-order Linkwitz-Riley low-pass [20] with $\Omega = f/f_x$ and $\omega_x = 2\pi f_x$ has the **group delay**

$$\tau_{\text{LR4}}(\Omega) = \frac{2\sqrt{2}}{\omega_x} \frac{1 + \Omega^2}{1 + \Omega^4}, \quad (\text{F.5})$$

equal to $2\sqrt{2}/\omega_x$ both at DC and at the corner, with a maximum about 21% higher at $\Omega \approx 0.64$. At $f_x = 80$ Hz this is 5.63 ms, about 1.9 m of apparent extra path from the filter alone (Figure F.1). A loudspeaker's low-frequency acoustic centre is also frequency-dependent [13], which adds a radiation term not treated here.

Both estimators contain the path difference. The arrival time adds wherever the subwoofer's band-limited impulse happens to peak, while the crossover-band estimate adds the average slope of its phase over an octave either side of the crossover, and there is no general relation between the two. Their difference $\tau_{\text{grp}} - (d_s - d_m)$ is therefore diagnostic. Close to zero means the subwoofer behaves like a delayed copy of the main and the offset is pure geometry, so moving the box moves the number. Several milliseconds means the offset is dominated by the crossover and driver alignment, and moving the box will not fix it.

F.3 Which estimator for which problem

Between two full-range loudspeakers only the arrival time exists. Each loudspeaker's own per-file delay is removed before its positions are averaged, so the two averaged responses carry no

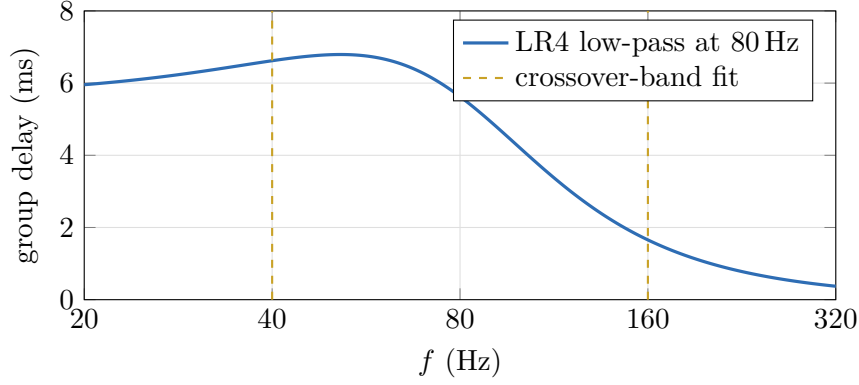


Figure F.1: Group delay of a fourth-order Linkwitz-Riley low-pass at 80 Hz, equation (F.5), with the band over which the crossover-band estimate fits its slope.

information about the difference between their delays, and there is no phase slope left to fit. The arrival time is the right answer anyway, because two full-range sources share essentially the same filtering, its contribution cancels in the difference, and what remains is the path difference that makes transients arrive together.

Between a main and the subwoofer the filtering does not cancel, and what governs summation at the [crossover](#) is the [group delay](#) there. The crossover-band estimate measures that directly, and it only exists for this pair because the anchoring deliberately preserves their relative timing.

	speaker to speaker	main to subwoofer
Arrival time	correct and robust	available but weak
Crossover-band group delay	not available	correct quantity
Goal	transients arrive together	sources sum coherently at f_x

F.4 The alignment condition

The correction of a main is designed assuming a delay T between the main and the subwoofer, the "Mains delay" of Section 7.5.2. Its [all-pass](#) term steers the main's phase onto the subwoofer advanced by that amount, $S e^{+j2\pi f T}$. When the outputs actually receive bulk delays a_m on the main and a_s on the subwoofer, the phase difference between the two acoustic paths in the [crossover](#) region is $2\pi f [T - (a_m - a_s)]$. It vanishes at every frequency only if

$$T_m = a_m - a_s. \quad (\text{F.6})$$

T is therefore not the delay applied to the main, but the main's delay relative to the subwoofer. In the single-speaker Analysis view the subwoofer output receives no delay from the analysis, so the condition reduces to $T = a_m$, which is what the "Mains delay" control means.

The condition only matters in [linear-phase](#) rendering. The [minimum-phase](#) rendering of Section B.9.2 rebuilds the filter from the correction's magnitude alone, and the all-pass has unit magnitude, so in that mode T , the crossover setting and "Invert sub" have no effect at all on the exported [FIR](#). The bulk delays are still written to the outputs, so the loudspeakers stay time-aligned on arrival, but there is no phase steering through the crossover. The subwoofer phase curve on the plot is drawn as the correction sees it, with the polarity and the assumed delay applied, so the corrected main and the subwoofer can be read against each other.

F.5 Group alignment and the subwoofer trim

"Compute alignment" in Group analysis (Section 8.5) aligns the group on arrival times. With D the largest arrival time d among the loaded entries and Δ the subwoofer trim set on the Sub row, the output delays are

$$a_i = D - d_i + \Sigma, \quad a_s = D - d_s + \Delta + \Sigma, \quad (\text{F.7})$$

where the group shift Σ slides every delay so that the earliest entry sits at zero, which gives the least overall latency. Each speaker's correction is then designed with

$$T_i = a_i - a_s = (d_s - d_i) - \Delta, \quad (\text{F.8})$$

so (F.6) holds for every trim value and the group shift cancels. A positive trim delays the subwoofer against the group, and because of the shift it shows up as less delay on the mains rather than more on the subwoofer. With mains at 30.6 ms, a 9 ms trim gives mains at 21.4 ms and the subwoofer at zero, the same alignment with 9.2 ms less latency. Output delays are limited to 0 ms to 100 ms, and a very spread group that reaches that limit no longer satisfies the condition for the clamped output.

The default trim of zero is pure arrival-time alignment. Setting $T_i = \tau_{\text{grp},i}$ in (F.8) instead gives the trim

$$\Delta_i^* = (d_s - d_i) - \tau_{\text{grp},i}, \quad (\text{F.9})$$

which is exactly the disagreement between the two estimators. The geometric term cancels, so Δ_i^* should be nearly the same for every speaker, and a wide spread is itself a sign that one of the estimates is unreliable. The Sub row shows the median over the loaded speakers, and "Use x-over" writes it into the trim. It is usually negative, since the subwoofer's low-pass has already made it late and it needs less delay than arrival-time alignment would give it.

The suggestion is withdrawn when the [smoothing](#) makes it untrustworthy. Evaluating (F.4) at the top of the fit band, with the suggested trim as the offset and the smoothing fraction in force at $2f_x$, the estimate counts as usable only while the rotation stays below $\pi/2$, half the point at which the complex average collapses. Past that, "Use x-over" is disabled and the read-out says the estimate is unreliable at this smoothing. The trim slider itself, ± 40 ms, stays available for setting by ear.

Appendix G

Ambisonics primer

This appendix gives the background behind the Ambisonics decoders the configuration tool generates (Section 5.1). It covers the spherical-harmonic representation, the orders, the AmbiX (ACN/SN3D) convention, how a decoder is computed, and the $\text{max-}r_E$ weighting.

G.1 Encoding the field and decoding to the rig

Ambisonics represents a sound field around a central point as a set of spherical-harmonic (SH) components, independently of any loudspeaker layout [22, 23]. A unit-amplitude plane wave arriving from direction $\Omega = (\theta, \varphi)$ (azimuth θ , elevation φ) carrying signal s is encoded into channels

$$B_j = Y_j(\Omega) s, \quad (\text{G.1})$$

where the Y_j are the spherical harmonics. A whole field is the superposition over all incident directions. Because the encoding does not mention speakers, one B-format signal can be decoded to any rig. Reproduction truncates the expansion at a finite order N , which sets the angular resolution and the size of the region around the centre where the field is faithfully reconstructed.

G.2 Spherical harmonics

The real spherical harmonics Y_n^m have a degree $n \geq 0$ and an order $m \in [-n, n]$. They form an orthogonal basis on the sphere, and degree n carries the angular detail of that order. Writing the unit direction as

$$x = \cos \varphi \cos \theta, \quad y = \cos \varphi \sin \theta, \quad z = \sin \varphi, \quad (\text{G.2})$$

the SN3D-normalised real harmonics used by SuperMoTo are, up to third degree,

$$\begin{aligned} n = 0 : Y_0 &= 1, \\ n = 1 : Y_1 &= y, \quad Y_2 = z, \quad Y_3 = x, \\ n = 2 : Y_4 &= \sqrt{3}xy, \quad Y_5 = \sqrt{3}yz, \quad Y_6 = \frac{1}{2}(3z^2 - 1), \quad Y_7 = \sqrt{3}xz, \quad Y_8 = \frac{\sqrt{3}}{2}(x^2 - y^2), \\ n = 3 : Y_9 &= \sqrt{\frac{5}{8}}y(3x^2 - y^2), \quad Y_{10} = \sqrt{15}xyz, \quad Y_{11} = \sqrt{\frac{3}{8}}y(5z^2 - 1), \quad Y_{12} = \frac{1}{2}z(5z^2 - 3), \\ Y_{13} &= \sqrt{\frac{3}{8}}x(5z^2 - 1), \quad Y_{14} = \frac{\sqrt{15}}{2}z(x^2 - y^2), \quad Y_{15} = \sqrt{\frac{5}{8}}x(x^2 - 3y^2). \end{aligned}$$

Each is scaled so its maximum magnitude over the sphere is 1 (the SN3D convention, below).

G.3 Ambisonic order

An order- N signal keeps every degree $n \leq N$. In 3D (periphonic) that is

$$K = (N + 1)^2 \quad \text{channels}, \quad (\text{G.3})$$

so orders 1/2/3 use 4/9/16 channels and need at least that many loudspeakers to decode. A horizontal-only (pantophonic) signal keeps just $m = \pm n$, giving $2N + 1$ channels. Higher order narrows the directional "beam" of each harmonic, improving localisation and enlarging the usable listening area, at the cost of more channels and speakers (Table G.1).

Order N	3D channels $(N + 1)^2$	2D channels $2N + 1$	default rig in the tool
1	4	3	8 (cube)
2	9	5	12 (icosahedron)
3	16	7	16 (dome)

Table G.1: Ambisonic order vs. channel count and the configuration tool's default periphonic rig.

G.4 The AmbiX convention (ACN and SN3D)

Two choices must be fixed to exchange B-format, the channel ordering and the normalisation. The modern standard is AmbiX [24]:

- **ACN** (Ambisonic Channel Number) orders the channels by the single index

$$j = n^2 + n + m, \quad (\text{G.4})$$

so $j = 0, 1, 2, 3, \dots$ are $W, Y, Z, X, \dots$. Note Y ($m = -1$) before Z ($m = 0$) before X ($m = +1$).

- **SN3D** (Schmidt semi-normalisation) scales each harmonic so its peak magnitude is 1. It relates to the fully orthonormal N3D form by $Y_n^{\text{SN3D}} = Y_n^{\text{N3D}} / \sqrt{2n + 1}$.

This supersedes the legacy FuMa (Furse-Malham) ordering $W, X, Y, Z, \dots$ with its $1/\sqrt{2}$ weighting on W , which was limited to third order. SuperMoTo expects AmbiX on the first K plugin inputs.

G.5 Decoding

A decoder maps the K B-format channels to L loudspeaker feeds. Collect the speaker directions in the $K \times L$ re-encoding matrix $\mathbf{C}$ with $C_{j,s} = Y_j(\Omega_s)$, so that driving the speakers with gains $\mathbf{g}$ re-creates the components $\mathbf{C}\mathbf{g}$. A good decode $\mathbf{D}$ ($\mathbf{g} = \mathbf{D}\mathbf{b}$) makes that match the input $\mathbf{b}$, i.e. $\mathbf{CD} \approx \mathbf{I}$. The general solution is the mode-matching pseudo-inverse $\mathbf{D} = \mathbf{C}^+$. For a near-uniform layout the harmonics stay orthogonal when sampled at the speakers, $\mathbf{CC}^T \approx \frac{L}{K}\mathbf{I}$, and the pseudo-inverse collapses to the scaled transpose, the sampling (projection) decoder

$$D_{s,j} = \frac{1}{L} Y_j(\Omega_s), \quad (\text{G.5})$$

which is what the tool builds. It is exact for ideal layouts and degrades gracefully for mildly irregular ones. Strongly irregular rigs are better served by an explicit pseudo-inverse or AllRAD decoder, which is why the tool also lets you trim each speaker's gain/angle/radius by hand.

G.6 The max- r_E weighting

At the centre, localisation is governed by two cues, the velocity vector $\mathbf{r}_V$ (low frequency, from the summed particle velocity) and the energy vector $\mathbf{r}_E$ (above ~ 700 Hz, from the summed

energy) [23, 25]. The basic decoder of (G.5) makes $|\mathbf{r}_V| = 1$ (correct low-frequency cue) but spreads energy, so $|\mathbf{r}_E| < 1$. The max- r_E decoder applies a per-degree gain a_n to maximise $|\mathbf{r}_E|$,

$$D_{s,j} = \frac{1}{L} a_{n(j)} Y_j(\Omega_s), \quad (\text{G.6})$$

where the optimal weights are samples of the Legendre polynomials,

$$a_n = P_n(r_E^{\max}), \quad (\text{G.7})$$

with $r_E^{\max}$ the largest root of $P_{N+1}(x)$ (the order- N energy limit). Normalised to $a_0 = 1$, the values used by SuperMoTo are listed in Table G.2. Max- r_E slightly widens phantom sources but markedly improves directional stability and timbre across the listening area, so it is the default for monitoring.

Order N	$r_E^{\max}$	a_0	a_1	a_2	a_3
1	0.5774	1	0.5774	–	–
2	0.7746	1	0.7746	0.4000	–
3	0.8611	1	0.8611	0.6123	0.3047

Table G.2: max- r_E per-degree gains (normalised $a_0 = 1$), from the largest root of P_{N+1} .

In SuperMoTo the resulting matrix is finally scaled so an on-axis plane wave decodes to about unity at the nearest speaker, written as harmonic $\rightarrow$ speaker frames, and combined with the radius compensation and bass management of Section 5.1. For deeper treatments see Zotter & Frank [25] and Daniel [23].

Appendix H

Glossary

Cross-references point to where each term is developed in the text. Each entry is also a link target. Wherever a term first appears in a section of the manual, it is set in [this colour](#) and clicking it jumps here. Readers using Acrobat Reader, and other viewers that display annotation tool-tips, also get the one-line summary in a pop-up on hover.

All-pass / excess phase

A factor with unit magnitude that changes only phase (eq. (C.1)). It carries the "excess" phase a response has beyond the minimum-phase minimum, such as pure delay, reflections and out-of-phase summation. It cannot be removed by a stable causal filter. SuperMoTo deliberately adds a short all-pass to steer a main onto a subwoofer's phase (Section B.6).

Cepstrum (real)

The inverse Fourier transform of the log-magnitude spectrum, $c[n] = \mathcal{F}^{-1}\{\log |C|\}$. Keeping its causal part reconstructs the minimum-phase response used for low-latency rendering (Section B.9.2).

Coherence

The fraction of the measured output linearly explained by the input at each frequency, from 0 to 1. It quantifies how much a Welch estimate can be trusted (Section B.1).

Comb filtering

Regularly spaced peaks and notches produced when the direct sound sums with a delayed copy such as a reflection. Wide-band and tied to geometry (Chapter 7).

Crossover

The frequency region where two drivers (e.g. main and subwoofer) hand over. A flat sum there is a phase/timing problem as much as a level one (Section 7.5.2).

FIR (finite impulse response) filter

A filter that convolves the signal with a finite list of coefficients (the impulse response). The exported correction is an FIR loaded per output. Its length N sets the frequency resolution f_s/N and, if linear-phase, the latency $N/2$.

Group delay

The delay a response imposes near each frequency, $-d(\arg H)/d\omega$. Used to auto-detect the main/sub offset (Section 7.5.2).

H_1 estimator

The ratio S_{xy}/S_{xx} of cross-spectrum to input auto-spectrum, averaged over Welch segments. It is the noise-robust transfer function SuperMoTo measures (Section B.1).

Hilbert transform

The operator linking log-magnitude and phase of a minimum-phase response (eq. (C.2)). It lets phase be recovered from magnitude (Appendix C).

Impulse response (IR)

A system's output to a unit impulse, the time-domain twin of the transfer function. Exported as a wav for the FIR engine and the measured-response file (Section 7.6).

Inter-output latency compensation

Automatic per-output delay so every engaged output stays time-aligned with the longest FIR's bulk latency (Chapter 3). An output's own manual delay first absorbs its own FIR's latency where it has slack, so only the unabsorbed remainder is added elsewhere. Shown by a gold LED on the output.

Linear-phase

A filter whose phase is exactly linear in frequency, i.e. a pure delay across the band. It can impose an arbitrary phase correction but adds $N/2$ samples of latency (Section 7.6).

Minimum-phase

A causal, stable response whose inverse is also causal and stable (all poles and zeros inside the unit circle), the only kind that is fully invertible by a realisable filter. Defined physically and mathematically in Appendix C.

Polarity (Invert)

A frequency-independent 180° sign flip, as opposed to a delay (which is frequency-dependent). Frames and outputs each have a "Phase invert" switch, and the two multiply (Section 3.1.2). "Invert sub" designs the alignment for a subwoofer wired out of polarity (Section 7.5.2), and Group analysis writes it onto the subwoofer output (Section 8.7).

Pre-echo

Energy a filter places before the sound it filters. A linear-phase correction puts half of whatever it does not cancel at the listener ahead of the direct sound, where the ear masks very little of it, so it is heard as a short pre-reverberation before transients. What "Limit phase" prevents (Section 7.6, Section B.8).

Regularization

Bounding the inverse where the measured magnitude is tiny, so deep notches do not demand infinite boost. SuperMoTo uses a soft-knee ($\tanh$) ceiling on the correction gain (Section B.5).

Room mode

A standing-wave resonance of the room producing large, strongly position-dependent peaks and dips at low frequency, with audible ringing (Chapter 7).

SBIR

Speaker-boundary interference response, the cancellation dip in the upper bass set by the speaker-to-wall distance (Chapter 7).

Smoothing (fractional-octave)

Averaging the complex response over a proportional bandwidth (e.g. 1/6 octave), finer in the bass and coarser in the treble, so the correction follows trends rather than single-seat interference (Chapter 7).

Spatial averaging

Averaging the (delay-aligned) responses of several microphone positions, so the correction targets what is consistent across the listening area rather than one seat (Section B.2).

Transfer function

The complex ratio $H(f) = Y(f)/X(f)$ of output to input spectrum (magnitude and phase together) that describes a linear system (Section [B.1](#)).

Welch method

Power and cross-spectra estimated by averaging windowed, overlapping FFT segments. It trades resolution for variance to give a usable in-room transfer function (Section [B.1](#)).

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